

Section B

Geometry and trigonometry

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Introduction to trigonometry

Why it is important to understand: Introduction to trigonometry

There are an enormous number of uses of trigonometry and trigonometric functions. Fields that use trigonometry or trigonometric functions include astronomy (especially for locating apparent positions of celestial objects, in which spherical trigonometry is essential) and hence navigation (on the oceans, in aircraft, and in space), music theory, acoustics, optics, analysis of financial markets, electronics, probability theory, statistics, biology, medical imaging (CAT scans and ultrasound), pharmacy, chemistry, number theory (and hence cryptology), seismology, meteorology, oceanography, many physical sciences, land surveying and geodesy (a branch of earth sciences), architecture, phonetics, economics, electrical engineering, mechanical engineering, civil engineering, computer graphics, cartography, crystallography and game development. It is clear that a good knowledge of trigonometry is essential in many fields of engineering.

At the end of this chapter, you should be able to:

- state the theorem of Pythagoras and use it to find the unknown side of a right-angled triangle
- define sine, cosine, tangent, secant, cosecant and cotangent of an angle in a right-angled triangle
- evaluate trigonometric ratios of angles
- solve right-angled triangles
- understand angles of elevation and depression
- use the sine and cosine rules to solve non-right-angled triangles
- calculate the area of any triangle
- solve practical problems involving trigonometry

12.1 Trigonometry

Trigonometry is the branch of mathematics which deals with the measurement of sides and angles of triangles, and their relationship with each other. There are many applications in engineering where knowledge of trigonometry is needed.

12.2 The theorem of Pythagoras

With reference to Fig. 12.1, the side opposite the right angle (i.e. side b) is called the **hypotenuse**. The **theorem of Pythagoras** states:

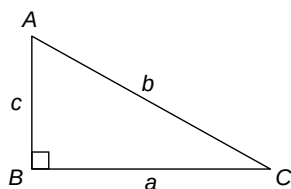
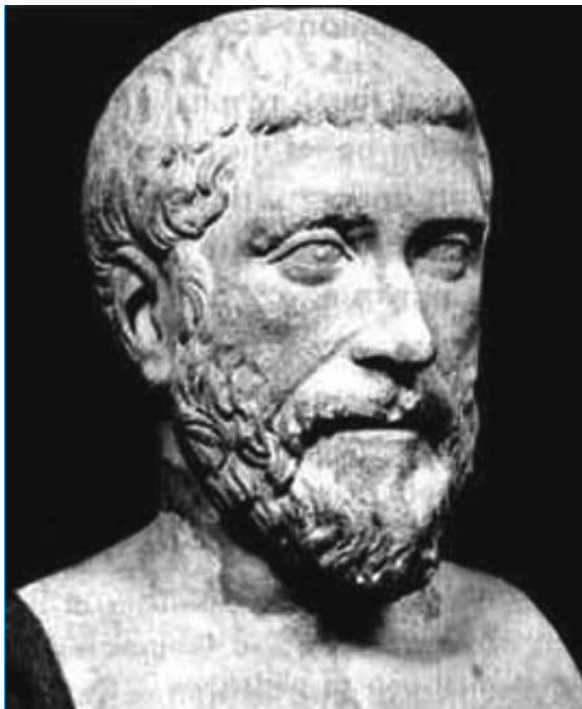


Figure 12.1



* **Who was Pythagoras?** **Pythagoras of Samos** (Born c. 570 BC and died about 495 BC) was an Ionian Greek philosopher and mathematician. He is best known for the Pythagorean theorem, which states that in a right-angled triangle $a^2 + b^2 = c^2$. To find out more go to www.routledge.com/cw/bird

‘In any right-angled triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides.’

Hence $b^2 = a^2 + c^2$

Problem 1. In Fig. 12.2, find the length of EF .

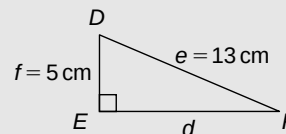


Figure 12.2

By Pythagoras' theorem:

$$e^2 = d^2 + f^2$$

Hence $13^2 = d^2 + 5^2$

$$169 = d^2 + 25$$

$$d^2 = 169 - 25 = 144$$

Thus $d = \sqrt{144} = 12 \text{ cm}$
i.e. **$EF = 12 \text{ cm}$**

Problem 2. Two aircraft leave an airfield at the same time. One travels due north at an average speed of 300 km/h and the other due west at an average speed of 220 km/h. Calculate their distance apart after four hours.

After four hours, the first aircraft has travelled $4 \times 300 = 1200 \text{ km}$, due north, and the second aircraft has travelled $4 \times 220 = 880 \text{ km}$ due west, as shown in Fig. 12.3. Distance apart after four hours = BC .

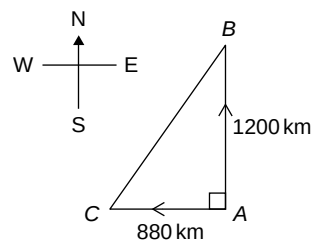


Figure 12.3

From Pythagoras' theorem:

$$BC^2 = 1200^2 + 880^2 = 1\,440\,000 + 774\,400$$

and $BC = \sqrt{(2\,214\,400)}$

Hence distance apart after four hours = 1488 km

Now try the following Practice Exercise

Practice Exercise 51 The theorem of Pythagoras (Answers on page 837)

1. In a triangle CDE , $D=90^\circ$, $CD=14.83$ mm and $CE=28.31$ mm. Determine the length of DE .
2. Triangle PQR is isosceles, Q being a right angle. If the hypotenuse is 38.47 cm find (a) the lengths of sides PQ and QR , and (b) the value of $\angle QPR$.
3. A man cycles 24 km due south and then 20 km due east. Another man, starting at the same time as the first man, cycles 32 km due east and then 7 km due south. Find the distance between the two men.
4. A ladder 3.5 m long is placed against a perpendicular wall with its foot 1.0 m from the wall. How far up the wall (to the nearest centimetre) does the ladder reach? If the foot of the ladder is now moved 30 cm further away from the wall, how far does the top of the ladder fall?
5. Two ships leave a port at the same time. One travels due west at 18.4 km/h and the other due south at 27.6 km/h. Calculate how far apart the two ships are after four hours.
6. Figure 12.4 shows a bolt rounded off at one end. Determine the dimension h .

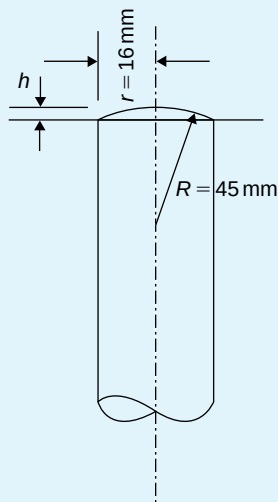


Figure 12.4

7. Figure 12.5 shows a cross-section of a component that is to be made from a round bar. If the diameter of the bar is 74 mm, calculate the dimension x .

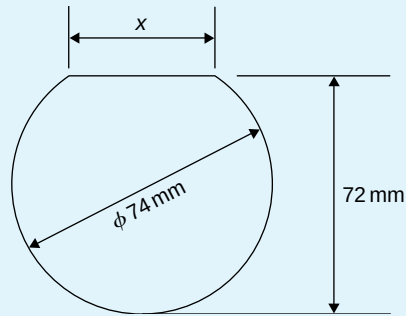


Figure 12.5

12.3 Trigonometric ratios of acute angles

- (a) With reference to the right-angled triangle shown in Fig. 12.6:

$$(i) \quad \text{sine } \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\text{i.e.} \quad \sin \theta = \frac{b}{c}$$

$$(ii) \quad \text{cosine } \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\text{i.e.} \quad \cos \theta = \frac{a}{c}$$

$$(iii) \quad \text{tangent } \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\text{i.e.} \quad \tan \theta = \frac{b}{a}$$

$$(iv) \quad \text{secant } \theta = \frac{\text{hypotenuse}}{\text{adjacent side}}$$

$$\text{i.e.} \quad \sec \theta = \frac{c}{a}$$

$$(v) \quad \text{cosecant } \theta = \frac{\text{hypotenuse}}{\text{opposite side}}$$

$$\text{i.e.} \quad \text{cosec } \theta = \frac{c}{b}$$

$$(vi) \quad \cotangent \theta = \frac{\text{adjacent side}}{\text{opposite side}}$$

$$\text{i.e.} \quad \cot \theta = \frac{a}{b}$$

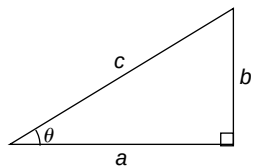


Figure 12.6

(b) From above,

$$(i) \quad \frac{\sin \theta}{\cos \theta} = \frac{\frac{b}{c}}{\frac{a}{c}} = \frac{b}{a} = \tan \theta,$$

$$\text{i.e.} \quad \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$(ii) \quad \frac{\cos \theta}{\sin \theta} = \frac{\frac{a}{c}}{\frac{b}{c}} = \frac{a}{b} = \cot \theta,$$

$$\text{i.e.} \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$(iii) \quad \sec \theta = \frac{1}{\cos \theta}$$

$$(iv) \quad \operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

(Note, the first letter of each ratio, 's' and 'c' go together)

$$(v) \quad \cot \theta = \frac{1}{\tan \theta}$$

Secants, cosecants and cotangents are called the **reciprocal ratios**.

Problem 3. If $\cos X = \frac{9}{41}$ determine the value of the other five trigonometry ratios.

Fig. 12.7 shows a right-angled triangle XYZ.

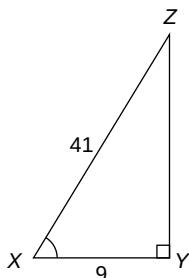


Figure 12.7

Since $\cos X = \frac{9}{41}$, then $XY = 9$ units and $XZ = 41$ units.

Using Pythagoras' theorem: $41^2 = 9^2 + YZ^2$ from which $YZ = \sqrt{(41^2 - 9^2)} = 40$ units.

Thus

$$\sin X = \frac{40}{41}, \tan X = \frac{40}{9} = 4\frac{4}{9},$$

$$\operatorname{cosec} X = \frac{41}{40} = 1\frac{1}{40},$$

$$\sec X = \frac{41}{9} = 4\frac{5}{9} \text{ and } \cot X = \frac{9}{40}$$

Problem 4. If $\sin \theta = 0.625$ and $\cos \theta = 0.500$ determine, without using trigonometric tables or calculators, the values of $\operatorname{cosec} \theta$, $\sec \theta$, $\tan \theta$ and $\cot \theta$.

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{1}{0.625} = 1.60$$

$$\sec \theta = \frac{1}{\cos \theta} = \frac{1}{0.500} = 2.00$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{0.625}{0.500} = 1.25$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{0.500}{0.625} = 0.80$$

Problem 5. Point A lies at co-ordinate (2, 3) and point B at (8, 7). Determine (a) the distance AB, (b) the gradient of the straight line AB, and (c) the angle AB makes with the horizontal.

(a) Points A and B are shown in Fig. 12.8(a).

In Fig. 12.8(b), the horizontal and vertical lines AC and BC are constructed.

Since ABC is a right-angled triangle, and $AC = (8 - 2) = 6$ and $BC = (7 - 3) = 4$, then by Pythagoras' theorem

$$AB^2 = AC^2 + BC^2 = 6^2 + 4^2$$

$$\text{and } AB = \sqrt{(6^2 + 4^2)} = \sqrt{52} = 7.211,$$

correct to 3 decimal places.

(b) The gradient of AB is given by $\tan A$,

$$\text{i.e. gradient} = \tan A = \frac{BC}{AC} = \frac{4}{6} = \frac{2}{3}$$

(c) The angle AB makes with the horizontal is given by $\tan^{-1} \frac{2}{3} = 33.69^\circ$

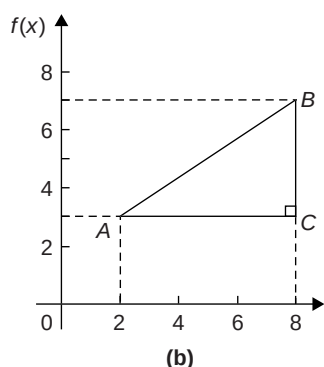
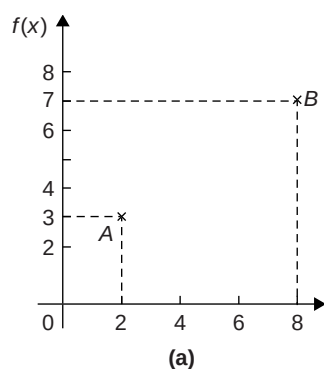


Figure 12.8

Now try the following Practice Exercise

Practice Exercise 52 Trigonometric ratios of acute angles (Answers on page 837)

1. In triangle ABC shown in Fig. 12.9, find $\sin A$, $\cos A$, $\tan A$, $\sin B$, $\cos B$ and $\tan B$.

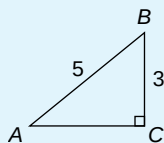


Figure 12.9

2. If $\cos A = \frac{15}{17}$ find $\sin A$ and $\tan A$, in fraction form.
3. For the right-angled triangle shown in Fig. 12.10, find:
- (a) $\sin \alpha$ (b) $\cos \theta$ (c) $\tan \theta$

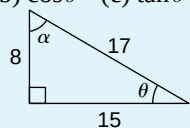


Figure 12.10

4. Point P lies at co-ordinate $(-3, 1)$ and point Q at $(5, -4)$. Determine
- (a) the distance PQ
- (b) the gradient of the straight line PQ and
- (c) the angle PQ makes with the horizontal.

12.4 Evaluating trigonometric ratios

The easiest method of evaluating trigonometric functions of any angle is by using a **calculator**.

The following values, correct to 4 decimal places, may be checked:

$$\begin{aligned} \sin 18^\circ &= 0.3090, & \cos 56^\circ &= 0.5592 \\ \sin 172^\circ &= 0.1392 & \cos 115^\circ &= -0.4226, \\ \sin 241.63^\circ &= -0.8799, & \cos 331.78^\circ &= 0.8811 \\ \tan 29^\circ &= 0.5543, \\ \tan 178^\circ &= -0.0349 \\ \tan 296.42^\circ &= -2.0127 \end{aligned}$$

To evaluate, say, $\sin 42^\circ 23'$ using a calculator means finding $\sin 42 \frac{23}{60}$ since there are 60 minutes in 1 degree.

$$\frac{23}{60} = 0.383\bar{3} \text{ thus } 42^\circ 23' = 42.38\bar{3}^\circ$$

Thus $\sin 42^\circ 23' = \sin 42.38\bar{3}^\circ = 0.6741$, correct to 4 decimal places.

Similarly, $\cos 72^\circ 38' = \cos 72 \frac{38}{60} = 0.2985$, correct to 4 decimal places.

Most calculators contain only sine, cosine and tangent functions. Thus to evaluate secants, cosecants and cotangents, reciprocals need to be used. The following values, correct to 4 decimal places, may be checked:

$$\begin{aligned} \sec 32^\circ &= \frac{1}{\cos 32^\circ} = 1.1792 \\ \csc 75^\circ &= \frac{1}{\sin 75^\circ} = 1.0353 \\ \cot 41^\circ &= \frac{1}{\tan 41^\circ} = 1.1504 \\ \sec 215.12^\circ &= \frac{1}{\cos 215.12^\circ} = -1.2226 \end{aligned}$$

$$\operatorname{cosecant} 321.62^\circ = \frac{1}{\sin 321.62^\circ} = -1.6106$$

$$\operatorname{cotangent} 263.59^\circ = \frac{1}{\tan 263.59^\circ} = 0.1123$$

If we know the value of a trigonometric ratio and need to find the angle we use the **inverse function** on our calculators.

For example, using shift and sin on our calculator gives $\sin^{-1}$ (

If, for example, we know the sine of an angle is 0.5 then the value of the angle is given by:

$$\sin^{-1} 0.5 = 30^\circ \quad (\text{Check that } \sin 30^\circ = 0.5)$$

(Note that $\sin^{-1} x$ does not mean $\frac{1}{\sin x}$; also, $\sin^{-1} x$ may also be written as $\arcsin x$)

Similarly, if $\cos \theta = 0.4371$ then

$$\theta = \cos^{-1} 0.4371 = 64.08^\circ$$

$$\text{and if } \tan A = 3.5984 \text{ then } A = \tan^{-1} 3.5984 = 74.47^\circ$$

each correct to 2 decimal places.

Use your calculator to check the following worked examples.

Problem 6. Determine, correct to 4 decimal places, $\sin 43^\circ 39'$

$$\begin{aligned} \sin 43^\circ 39' &= \sin 43 \frac{39}{60} = \sin 43.65^\circ \\ &= \mathbf{0.6903} \end{aligned}$$

This answer can be obtained using the **calculator** as follows:

1. Press sin 2. Enter 43 3. Press $^\circ$ '''
4. Enter 39 5. Press $^\circ$ ''' 6. Press)
7. Press = Answer = **0.6902512....**

Problem 7. Determine, correct to 3 decimal places, $6 \cos 62^\circ 12'$

$$\begin{aligned} 6 \cos 62^\circ 12' &= 6 \cos 62 \frac{12}{60} = 6 \cos 62.20^\circ \\ &= \mathbf{2.798} \end{aligned}$$

This answer can be obtained using the **calculator** as follows:

1. Enter 6 2. Press cos 3. Enter 62
4. Press $^\circ$ ''' 5. Enter 12 6. Press $^\circ$ '''
7. Press) 8. Press = Answer = **2.798319....**

Problem 8. Evaluate, correct to 4 decimal places:

- (a) $\sin 168^\circ 14'$ (b) $\cos 271.41^\circ$
(c) $\tan 98^\circ 4'$

$$(a) \quad \sin 168^\circ 14' = \sin 168 \frac{14}{60} = \mathbf{0.2039}$$

$$(b) \quad \cos 271.41^\circ = \mathbf{0.0246}$$

$$(c) \quad \tan 98^\circ 4' = \tan 98 \frac{4}{60} = \mathbf{-7.0558}$$

Problem 9. Evaluate, correct to 4 decimal places:

- (a) $\sec 161^\circ$ (b) $\sec 302^\circ 29'$

$$(a) \quad \sec 161^\circ = \frac{1}{\cos 161^\circ} = \mathbf{-1.0576}$$

$$\begin{aligned} (b) \quad \sec 302^\circ 29' &= \frac{1}{\cos 302^\circ 29'} = \frac{1}{\cos 302 \frac{29}{60}} \\ &= \mathbf{1.8620} \end{aligned}$$

Problem 10. Evaluate, correct to 4 significant figures:

- (a) $\operatorname{cosecant} 279.16^\circ$ (b) $\operatorname{cosecant} 49^\circ 7'$

$$(a) \quad \operatorname{cosec} 279.16^\circ = \frac{1}{\sin 279.16^\circ} = \mathbf{-1.013}$$

$$\begin{aligned} (b) \quad \operatorname{cosec} 49^\circ 7' &= \frac{1}{\sin 49^\circ 7'} = \frac{1}{\sin 49 \frac{7}{60}} \\ &= \mathbf{1.323} \end{aligned}$$

Problem 11. Evaluate, correct to 4 decimal places:

- (a) $\cot 17.49^\circ$ (b) $\cot 163^\circ 52'$

$$(a) \quad \cot 17.49^\circ = \frac{1}{\tan 17.49^\circ} = \mathbf{3.1735}$$

$$\begin{aligned} (b) \quad \cot 163^\circ 52' &= \frac{1}{\tan 163^\circ 52'} = \frac{1}{\tan 163 \frac{52}{60}} \\ &= \mathbf{-3.4570} \end{aligned}$$

Problem 12. Evaluate, correct to 4 significant figures:

- (a) $\sin 1.481$ (b) $\cos(3\pi/5)$ (c) $\tan 2.93$

- (a) $\sin 1.481$ means the sine of 1.481 radians. Hence a calculator needs to be on the radian function.
Hence $\sin 1.481 = \mathbf{0.9960}$
- (b) $\cos(3\pi/5) = \cos 1.884955\dots = \mathbf{-0.3090}$
- (c) $\tan 2.93 = \mathbf{-0.2148}$

Problem 13. Evaluate, correct to 4 decimal places:

- (a) $\secant 5.37$ (b) $\operatorname{cosecant} \pi/4$
(c) $\cotangent \pi/24$

- (a) Again, with no degrees sign, it is assumed that 5.37 means 5.37 radians.
Hence $\sec 5.37 = \frac{1}{\cos 5.37} = \mathbf{1.6361}$
- (b) $\operatorname{cosec}(\pi/4) = \frac{1}{\sin(\pi/4)} = \frac{1}{\sin 0.785398\dots} = \mathbf{1.4142}$
- (c) $\cot(5\pi/24) = \frac{1}{\tan(5\pi/24)} = \frac{1}{\tan 0.654498\dots} = \mathbf{1.3032}$

Problem 14. Find, in degrees, the acute angle $\sin^{-1} 0.4128$ correct to 2 decimal places.

$\sin^{-1} 0.4128$ means 'the angle whose sine is 0.4128'

Using a calculator:

1. Press shift 2. Press sin 3. Enter 0.4128
4. Press) 5. Press = The answer 24.380848... is displayed

Hence, $\sin^{-1} 0.4128 = \mathbf{24.38^\circ}$

Problem 15. Find the acute angle $\cos^{-1} 0.2437$ in degrees and minutes.

$\cos^{-1} 0.2437$ means 'the angle whose cosine is 0.2437'

Using a calculator:

1. Press shift 2. Press cos 3. Enter 0.2437
4. Press) 5. Press = The answer 75.894979... is displayed
6. Press $^\circ \text{ ' '}$ and $75^\circ 53' 41.93''$ is displayed

Hence, $\cos^{-1} 0.2437 = \mathbf{75.89^\circ = 77^\circ 54'}$
correct to the nearest minute.

Problem 16. Find the acute angle $\tan^{-1} 7.4523$ in degrees and minutes.

$\tan^{-1} 7.4523$ means 'the angle whose tangent is 7.4523'

Using a calculator:

1. Press shift 2. Press tan 3. Enter 7.4523
4. Press) 5. Press = The answer 82.357318... is displayed
6. Press $^\circ \text{ ' '}$ and $82^\circ 21' 26.35''$ is displayed

Hence, $\tan^{-1} 7.4523 = \mathbf{82.36^\circ = 82^\circ 21'}$
correct to the nearest minute.

Problem 17. Determine the acute angles:

- (a) $\sec^{-1} 2.3164$ (b) $\operatorname{cosec}^{-1} 1.1784$
(c) $\cot^{-1} 2.1273$

$$\begin{aligned} \text{(a)} \quad \sec^{-1} 2.3164 &= \cos^{-1} \left(\frac{1}{2.3164} \right) \\ &= \cos^{-1} 0.4317\dots \\ &= \mathbf{64.42^\circ \text{ or } 64^\circ 25'} \\ &\quad \text{or } \mathbf{1.124 \text{ radians}} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \operatorname{cosec}^{-1} 1.1784 &= \sin^{-1} \left(\frac{1}{1.1784} \right) \\ &= \sin^{-1} 0.8486\dots \\ &= \mathbf{58.06^\circ \text{ or } 58^\circ 4'} \\ &\quad \text{or } \mathbf{1.013 \text{ radians}} \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \cot^{-1} 2.1273 &= \tan^{-1} \left(\frac{1}{2.1273} \right) \\ &= \tan^{-1} 0.4700\dots \\ &= \mathbf{25.18^\circ \text{ or } 25^\circ 11'} \\ &\quad \text{or } \mathbf{0.439 \text{ radians}} \end{aligned}$$

Problem 18. Evaluate the following expression, correct to 4 significant figures:

$$\frac{4 \sec 32^\circ 10' - 2 \cot 15^\circ 19'}{3 \operatorname{cosec} 63^\circ 8' \tan 14^\circ 57'}$$

By calculator:

$$\sec 32^\circ 10' = 1.1813, \cot 15^\circ 19' = 3.6512$$

$$\operatorname{cosec} 63^\circ 8' = 1.1210, \tan 14^\circ 57' = 0.2670$$

$$\begin{aligned} \text{Hence } & \frac{4\sec 32^\circ 10' - 2\cot 15^\circ 19'}{3\operatorname{cosec} 63^\circ 8' \tan 14^\circ 57'} \\ &= \frac{4(1.1813) - 2(3.6512)}{3(1.1210)(0.2670)} \\ &= \frac{4.7252 - 7.3024}{0.8979} \\ &= \frac{-2.5772}{0.8979} = -2.870 \end{aligned}$$

correct to 4 significant figures.

Problem 19. Evaluate correct to 4 decimal places:

(a) $\sec(-115^\circ)$ (b) $\operatorname{cosec}(-95^\circ 47')$

- (a) Positive angles are considered by convention to be anticlockwise and negative angles as clockwise. Hence -115° is actually the same as 245° (i.e. $360^\circ - 115^\circ$)

$$\begin{aligned} \text{Hence } \sec(-115^\circ) &= \sec 245^\circ = \frac{1}{\cos 245^\circ} \\ &= -2.3662 \end{aligned}$$

$$(b) \operatorname{cosec}(-95^\circ 47') = \frac{1}{\sin\left(-95^\circ \frac{47'}{60}\right)} = -1.0051$$

Problem 20. In triangle EFG in Fig. 12.11, calculate angle G .

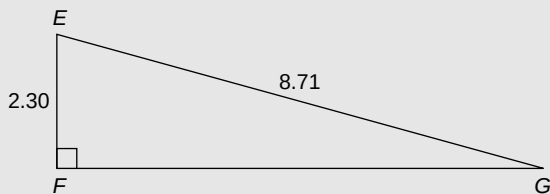


Figure 12.11

With reference to $\angle G$, the two sides of the triangle given are the opposite side EF and the hypotenuse EG ; hence, sine is used,

$$\text{i.e.} \quad \sin G = \frac{2.30}{8.71} = 0.26406429 \dots$$

$$\text{from which,} \quad G = \sin^{-1} 0.26406429 \dots$$

$$\text{i.e.} \quad G = 15.311360 \dots$$

$$\text{Hence,} \quad \angle G = 15.31^\circ \text{ or } 15^\circ 19'$$

Now try the following Practice Exercise

Practice Exercise 53 Evaluating trigonometric ratios (Answers on page 837)

In Problems 1 to 8, evaluate correct to 4 decimal places:

- (a) $\sin 27^\circ$ (b) $\sin 172.41^\circ$
(c) $\sin 302^\circ 52'$
- (a) $\cos 124^\circ$ (b) $\cos 21.46^\circ$
(c) $\cos 284^\circ 10'$
- (a) $\tan 145^\circ$ (b) $\tan 310.59^\circ$
(c) $\tan 49^\circ 16'$
- (a) $\sec 73^\circ$ (b) $\sec 286.45^\circ$
(c) $\sec 155^\circ 41'$
- (a) $\operatorname{cosec} 213^\circ$ (b) $\operatorname{cosec} 15.62^\circ$
(c) $\operatorname{cosec} 311^\circ 50'$
- (a) $\cot 71^\circ$ (b) $\cot 151.62^\circ$
(c) $\cot 321^\circ 23'$
- (a) $\sin \frac{2\pi}{3}$ (b) $\cos 1.681$ (c) $\tan 3.672$
- (a) $\sec \frac{\pi}{8}$ (b) $\operatorname{cosec} 2.961$ (c) $\cot 2.612$

In Problems 9 to 14, determine the acute angle in degrees (correct to 2 decimal places), degrees and minutes, and in radians (correct to 3 decimal places).

- $\sin^{-1} 0.2341$
- $\cos^{-1} 0.8271$
- $\tan^{-1} 0.8106$
- $\sec^{-1} 1.6214$

13. $\operatorname{cosec}^{-1} 2.4891$
 14. $\cot^{-1} 1.9614$
 15. In the triangle shown in Fig. 12.12, determine angle θ , correct to 2 decimal places.

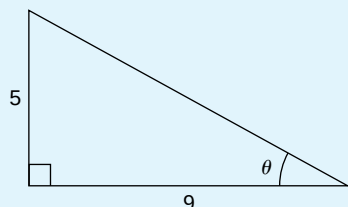


Figure 12.12

16. In the triangle shown in Fig. 12.13, determine angle θ in degrees and minutes.

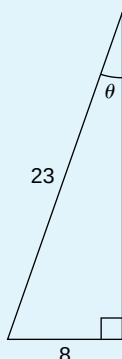


Figure 12.13

In Problems 17 to 20, evaluate correct to 4 significant figures:

17. $4 \cos 56^\circ 19' - 3 \sin 21^\circ 57'$
 18. $\frac{11.5 \tan 49^\circ 11' - \sin 90^\circ}{3 \cos 45^\circ}$
 19. $\frac{5 \sin 86^\circ 3'}{3 \tan 14^\circ 29' - 2 \cos 31^\circ 9'}$
 20. $\frac{6.4 \operatorname{cosec} 29^\circ 5' - \sec 81^\circ}{2 \cot 12^\circ}$
 21. Determine the acute angle, in degrees and minutes, correct to the nearest minute, given by $\sin^{-1} \left(\frac{4.32 \sin 42^\circ 16'}{7.86} \right)$
 22. If $\tan x = 1.5276$, determine $\sec x$, $\operatorname{cosec} x$, and $\cot x$. (Assume x is an acute angle.)

In Problems 23 to 25 evaluate correct to 4 significant figures:

23. $\frac{(\sin 34^\circ 27')(\cos 69^\circ 2')}{(2 \tan 53^\circ 39')}$
 24. $3 \cot 14^\circ 15' \sec 23^\circ 9'$
 25. $\frac{\operatorname{cosec} 27^\circ 19' + \sec 45^\circ 29'}{1 - \operatorname{cosec} 27^\circ 19' \sec 45^\circ 29'}$
 26. Evaluate correct to 4 decimal places:
 (a) $\sin(-125^\circ)$ (b) $\tan(-241^\circ)$
 (c) $\cos(-49^\circ 15')$
 27. Evaluate correct to 5 significant figures:
 (a) $\operatorname{cosec}(-143^\circ)$ (b) $\cot(-252^\circ)$
 (c) $\sec(-67^\circ 22')$

12.5 Solution of right-angled triangles

To 'solve a right-angled triangle' means 'to find the unknown sides and angles'. This is achieved by using (i) the theorem of Pythagoras, and/or (ii) trigonometric ratios. This is demonstrated in the following problems.

Problem 21. In triangle PQR shown in Fig. 12.14, find the lengths of PQ and PR .

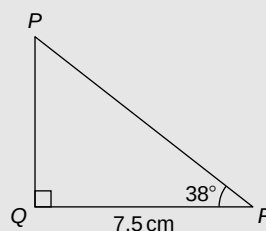


Figure 12.14

$$\tan 38^\circ = \frac{PQ}{QR} = \frac{PQ}{7.5}$$

hence $PQ = 7.5 \tan 38^\circ = 7.5(0.7813)$
 $= 5.860 \text{ cm}$

$$\cos 38^\circ = \frac{QR}{PR} = \frac{7.5}{PR}$$

hence $PR = \frac{7.5}{\cos 38^\circ} = \frac{7.5}{0.7880} = 9.518 \text{ cm}$

[Check: Using Pythagoras' theorem

$$(7.5)^2 + (5.860)^2 = 90.59 = (9.518)^2]$$

Problem 22. Solve the triangle ABC shown in Fig. 12.15.

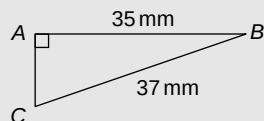


Figure 12.15

To 'solve triangle ABC ' means 'to find the length AC and angles B and C '

$$\sin C = \frac{35}{37} = 0.94595$$

hence $\angle C = \sin^{-1} 0.94595 = 71.08^\circ = 71^\circ 5'$

$\angle B = 180^\circ - 90^\circ - 71^\circ 5' = 18^\circ 55'$ (since angles in a triangle add up to 180°)

$$\sin B = \frac{AC}{37}$$

hence $AC = 37 \sin 18^\circ 55' = 37(0.3242)$

$$= 12.0 \text{ mm}$$

or, using Pythagoras' theorem, $37^2 = 35^2 + AC^2$, from which, $AC = \sqrt{(37^2 - 35^2)} = 12.0 \text{ mm}$.

Problem 23. Solve triangle XYZ given $\angle X = 90^\circ$, $\angle Y = 23^\circ 17'$ and $YZ = 20.0 \text{ mm}$. Determine also its area.

It is always advisable to make a reasonably accurate sketch so as to visualise the expected magnitudes of unknown sides and angles. Such a sketch is shown in Fig. 12.16.

$$\angle Z = 180^\circ - 90^\circ - 23^\circ 17' = 66^\circ 43'$$

$$\sin 23^\circ 17' = \frac{XZ}{20.0}$$

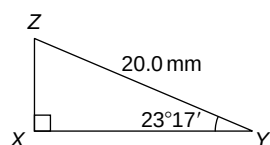


Figure 12.16

hence $XZ = 20.0 \sin 23^\circ 17'$

$$= 20.0(0.3953) = 7.906 \text{ mm}$$

$$\cos 23^\circ 17' = \frac{XY}{20.0}$$

hence $XY = 20.0 \cos 23^\circ 17'$

$$= 20.0(0.9186) = 18.37 \text{ mm}$$

[Check: Using Pythagoras' theorem

$$(18.37)^2 + (7.906)^2 = 400.0 = (20.0)^2]$$

Area of triangle XYZ

$$= \frac{1}{2} (\text{base}) (\text{perpendicular height})$$

$$= \frac{1}{2} (XY)(XZ) = \frac{1}{2} (18.37)(7.906)$$

$$= 72.62 \text{ mm}^2$$

Now try the following Practice Exercise

Practice Exercise 54 The solution of right-angled triangles (Answers on page 837)

1. Solve triangle ABC in Fig. 12.17(i).

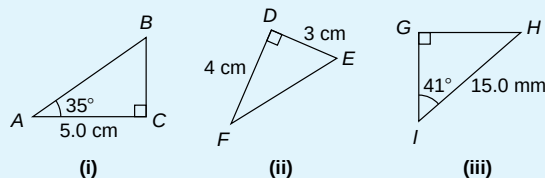


Figure 12.17

2. Solve triangle DEF in Fig. 12.17(ii).
3. Solve triangle GHI in Fig. 12.17(iii).
4. Solve the triangle JKL in Fig. 12.18(i) and find its area.
5. Solve the triangle MNO in Fig. 12.18(ii) and find its area.

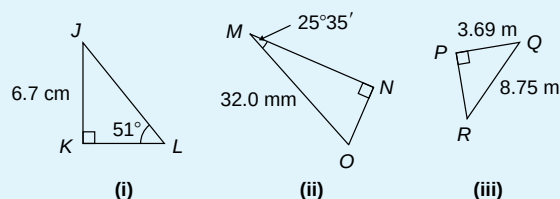


Figure 12.18

6. Solve the triangle PQR in Fig. 12.18(iii) and find its area.

7. A ladder rests against the top of the perpendicular wall of a building and makes an angle of 73° with the ground. If the foot of the ladder is 2 m from the wall, calculate the height of the building.

12.6 Angles of elevation and depression

- (a) If, in Fig. 12.19, BC represents horizontal ground and AB a vertical flagpole, then the **angle of elevation** of the top of the flagpole, A , from the point C is the angle that the imaginary straight line AC must be raised (or elevated) from the horizontal CB , i.e. angle θ .

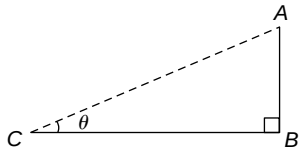


Figure 12.19

- (b) If, in Fig. 12.20, PQ represents a vertical cliff and R a ship at sea, then the **angle of depression** of the ship from point P is the angle through which the imaginary straight line PR must be lowered (or depressed) from the horizontal to the ship, i.e. angle ϕ .

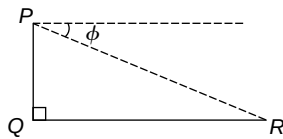


Figure 12.20

(Note, $\angle PRQ$ is also ϕ – alternate angles between parallel lines.)

Problem 24. An electricity pylon stands on horizontal ground. At a point 80 m from the base of the pylon, the angle of elevation of the top of the pylon is 23° . Calculate the height of the pylon to the nearest metre.

Figure 12.21 shows the pylon AB and the angle of elevation of A from point C is 23°

$$\tan 23^\circ = \frac{AB}{BC} = \frac{AB}{80}$$

Hence height of pylon AB

$$\begin{aligned} &= 80 \tan 23^\circ = 80(0.4245) = 33.96 \text{ m} \\ &= \mathbf{34 \text{ m to the nearest metre.}} \end{aligned}$$

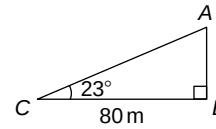


Figure 12.21

Problem 25. A surveyor measures the angle of elevation of the top of a perpendicular building as 19° . He moves 120 m nearer the building and finds the angle of elevation is now 47° . Determine the height of the building.

The building PQ and the angles of elevation are shown in Fig. 12.22.

In triangle PQS ,

$$\tan 19^\circ = \frac{h}{x + 120}$$

hence $h = \tan 19^\circ(x + 120),$

i.e. $h = 0.3443(x + 120)$ (1)

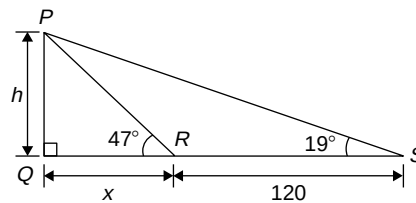


Figure 12.22

In triangle PQR , $\tan 47^\circ = \frac{h}{x}$

hence $h = \tan 47^\circ(x),$ i.e. $h = 1.0724x$ (2)

Equating equations (1) and (2) gives:

$$0.3443(x + 120) = 1.0724x$$

$$0.3443x + (0.3443)(120) = 1.0724x$$

$$(0.3443)(120) = (1.0724 - 0.3443)x$$

$$41.316 = 0.7281x$$

$$x = \frac{41.316}{0.7281} = 56.74 \text{ m}$$

From equation (2), **height of building,**

$$h = 1.0724x = 1.0724(56.74) = \mathbf{60.85 \text{ m}}$$

Problem 26. The angle of depression of a ship viewed at a particular instant from the top of a 75 m vertical cliff is 30° . Find the distance of the ship from the base of the cliff at this instant. The ship is sailing away from the cliff at constant speed and one minute later its angle of depression from the top of the cliff is 20° . Determine the speed of the ship in km/h.

Fig. 12.23 shows the cliff AB , the initial position of the ship at C and the final position at D . Since the angle of depression is initially 30° then $\angle ACB = 30^\circ$ (alternate angles between parallel lines).

$$\tan 30^\circ = \frac{AB}{BC} = \frac{75}{BC}$$

hence $BC = \frac{75}{\tan 30^\circ} = \frac{75}{0.5774} = 129.9 \text{ m}$
= initial position of ship from base of cliff

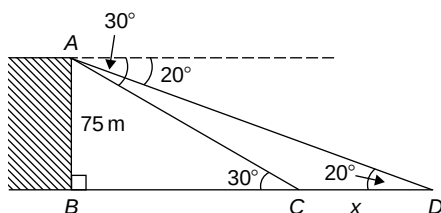


Figure 12.23

In triangle ABD ,

$$\tan 20^\circ = \frac{AB}{BD} = \frac{75}{BC + CD}$$

$$= \frac{75}{129.9 + x}$$

Hence $129.9 + x = \frac{75}{\tan 20^\circ} = \frac{75}{0.3640}$
 $= 206.0 \text{ m}$

from which $x = 206.0 - 129.9 = 76.1 \text{ m}$

Thus the ship sails 76.1 m in one minute, i.e. 60 s, hence speed of ship

$$= \frac{\text{distance}}{\text{time}} = \frac{76.1}{60} \text{ m/s}$$

$$= \frac{76.1 \times 60 \times 60}{60 \times 1000} \text{ km/h} = 4.57 \text{ km/h}$$

Now try the following Practice Exercise

Practice Exercise 55 Angles of elevation and depression (Answers on page 838)

1. If the angle of elevation of the top of a vertical 30 m high aerial is 32° , how far is it to the aerial?
2. From the top of a vertical cliff 80.0 m high the angles of depression of two buoys lying due west of the cliff are 23° and 15° , respectively. How far are the buoys apart?
3. From a point on horizontal ground a surveyor measures the angle of elevation of the top of a flagpole as $18^\circ 40'$. He moves 50 m nearer to the flagpole and measures the angle of elevation as $26^\circ 22'$. Determine the height of the flagpole.
4. A flagpole stands on the edge of the top of a building. At a point 200 m from the building the angles of elevation of the top and bottom of the pole are 32° and 30° respectively. Calculate the height of the flagpole.
5. From a ship at sea, the angles of elevation of the top and bottom of a vertical lighthouse standing on the edge of a vertical cliff are 31° and 26° , respectively. If the lighthouse is 25.0 m high, calculate the height of the cliff.
6. From a window 4.2 m above horizontal ground the angle of depression of the foot of a building across the road is 24° and the angle of elevation of the top of the building is 34° . Determine, correct to the nearest centimetre, the width of the road and the height of the building.
7. The elevation of a tower from two points, one due east of the tower and the other due west of it are 20° and 24° , respectively, and the two points of observation are 300 m apart. Find the height of the tower to the nearest metre.

12.7 Sine and cosine rules

To 'solve a triangle' means 'to find the values of unknown sides and angles'. If a triangle is **right angled**, trigonometric ratios and the theorem of Pythagoras may be used for its solution, as shown in Section 12.5.

However, for a **non-right-angled triangle**, trigonometric ratios and Pythagoras' theorem **cannot** be used. Instead, two rules, called the **sine rule** and the **cosine rule**, are used.

Sine rule

With reference to triangle ABC of Fig. 12.24, the **sine rule** states:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

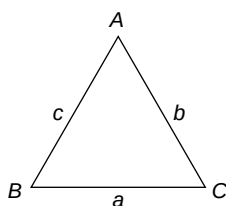


Figure 12.24

The rule may be used only when:

- (i) one side and any two angles are initially given, or
- (ii) two sides and an angle (not the included angle) are initially given.

Cosine rule

With reference to triangle ABC of Fig. 12.24, the **cosine rule** states:

$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos A \\ \text{or } b^2 &= a^2 + c^2 - 2ac \cos B \\ \text{or } c^2 &= a^2 + b^2 - 2ab \cos C \end{aligned}$$

The rule may be used only when:

- (i) two sides and the included angle are initially given, or
- (ii) three sides are initially given.

12.8 Area of any triangle

The **area of any triangle** such as ABC of Fig. 12.24 is given by:

- (i) $\frac{1}{2} \times \text{base} \times \text{perpendicular height}$, or
- (ii) $\frac{1}{2}ab \sin C$ or $\frac{1}{2}ac \sin B$ or $\frac{1}{2}bc \sin A$, or

(iii) $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$

12.9 Worked problems on the solution of triangles and finding their areas

Problem 27. In a triangle XYZ , $\angle X = 51^\circ$, $\angle Y = 67^\circ$ and $YZ = 15.2$ cm. Solve the triangle and find its area.

The triangle XYZ is shown in Fig. 12.25. Since the angles in a triangle add up to 180° , then $Z = 180^\circ - 51^\circ - 67^\circ = 62^\circ$. Applying the sine rule:

$$\frac{15.2}{\sin 51^\circ} = \frac{y}{\sin 67^\circ} = \frac{z}{\sin 62^\circ}$$

Using $\frac{15.2}{\sin 51^\circ} = \frac{y}{\sin 67^\circ}$ and transposing gives:

$$y = \frac{15.2 \sin 67^\circ}{\sin 51^\circ} = 18.00 \text{ cm} = XZ$$

Using $\frac{15.2}{\sin 51^\circ} = \frac{z}{\sin 62^\circ}$ and transposing gives:

$$z = \frac{15.2 \sin 62^\circ}{\sin 51^\circ} = 17.27 \text{ cm} = XY$$

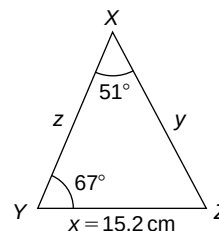


Figure 12.25

$$\begin{aligned} \text{Area of triangle } XYZ &= \frac{1}{2}xy \sin Z \\ &= \frac{1}{2}(15.2)(18.00) \sin 62^\circ = 120.8 \text{ cm}^2 \text{ (or area} \\ &= \frac{1}{2}xz \sin Y = \frac{1}{2}(15.2)(17.27) \sin 67^\circ = 120.8 \text{ cm}^2). \end{aligned}$$

It is always worth checking with triangle problems that the longest side is opposite the largest angle, and vice-versa. In this problem, Y is the largest angle and XZ is the longest of the three sides.

Problem 28. Solve the triangle PQR and find its area given that $QR = 36.5$ mm, $PR = 29.6$ mm and $\angle Q = 36^\circ$.

Triangle PQR is shown in Fig. 12.26.

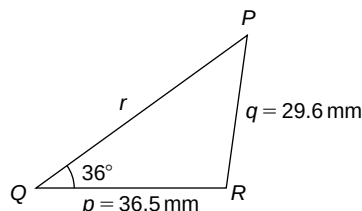


Figure 12.26

Applying the sine rule:

$$\frac{29.6}{\sin 36^\circ} = \frac{36.5}{\sin P}$$

from which,

$$\sin P = \frac{36.5 \sin 36^\circ}{29.6} = 0.7248$$

Hence $P = \sin^{-1} 0.7248 = 46^\circ 27'$ or $133^\circ 33'$

When $P = 46^\circ 27'$ and $Q = 36^\circ$ then

$$R = 180^\circ - 46^\circ 27' - 36^\circ = 97^\circ 33'$$

When $P = 133^\circ 33'$ and $Q = 36^\circ$ then

$$R = 180^\circ - 133^\circ 33' - 36^\circ = 10^\circ 27'$$

Thus, in this problem there are **two** separate sets of results and both are feasible solutions. Such a situation is called the **ambiguous case**.

Case 1. $P = 46^\circ 27'$, $Q = 36^\circ$, $R = 97^\circ 33'$,
 $p = 36.5$ mm and $q = 29.6$ mm.

From the sine rule:

$$\frac{r}{\sin 97^\circ 33'} = \frac{29.6}{\sin 36^\circ}$$

from which,

$$r = \frac{29.6 \sin 97^\circ 33'}{\sin 36^\circ} = 49.92 \text{ mm}$$

$$\begin{aligned} \text{Area} &= \frac{1}{2} pq \sin R = \frac{1}{2} (36.5)(29.6) \sin 97^\circ 33' \\ &= 535.5 \text{ mm}^2 \end{aligned}$$

Case 2. $P = 133^\circ 33'$, $Q = 36^\circ$, $R = 10^\circ 27'$,
 $p = 36.5$ mm and $q = 29.6$ mm.

From the sine rule:

$$\frac{r}{\sin 10^\circ 27'} = \frac{29.6}{\sin 36^\circ}$$

from which,

$$r = \frac{29.6 \sin 10^\circ 27'}{\sin 36^\circ} = 9.134 \text{ mm}$$

$$\begin{aligned} \text{Area} &= \frac{1}{2} pq \sin R = \frac{1}{2} (36.5)(29.6) \sin 10^\circ 27' \\ &= 97.98 \text{ mm}^2 \end{aligned}$$

Triangle PQR for case 2 is shown in Fig. 12.27.

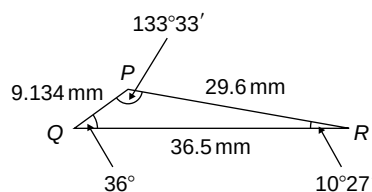


Figure 12.27

Now try the following Practice Exercise

Practice Exercise 56 Solving triangles and finding their areas (Answers on page 838)

In Problems 1 and 2, use the sine rule to solve the triangles ABC and find their areas.

1. $A = 29^\circ$, $B = 68^\circ$, $b = 27$ mm.
2. $B = 71^\circ 26'$, $C = 56^\circ 32'$, $b = 8.60$ cm.

In Problems 3 and 4, use the sine rule to solve the triangles DEF and find their areas.

3. $d = 17$ cm, $f = 22$ cm, $F = 26^\circ$
4. $d = 32.6$ mm, $e = 25.4$ mm, $D = 104^\circ 22'$

In Problems 5 and 6, use the sine rule to solve the triangles JKL and find their areas.

5. $j = 3.85$ cm, $k = 3.23$ cm, $K = 36^\circ$
6. $k = 46$ mm, $l = 36$ mm, $L = 35^\circ$

12.10 Further worked problems on solving triangles and finding their areas

Problem 29. Solve triangle DEF and find its area given that $EF = 35.0$ mm, $DE = 25.0$ mm and $\angle E = 64^\circ$

Triangle DEF is shown in Fig. 12.28.

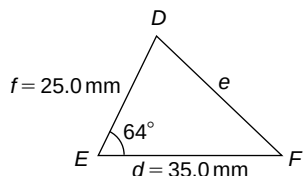


Figure 12.28

Applying the cosine rule:

$$e^2 = d^2 + f^2 - 2df \cos E$$

$$\text{i.e. } e^2 = (35.0)^2 + (25.0)^2 - [2(35.0)(25.0) \cos 64^\circ]$$

$$= 1225 + 625 - 767.1 = 1083$$

from which, $e = \sqrt{1083} = 32.91 \text{ mm}$

Applying the sine rule:

$$\frac{32.91}{\sin 64^\circ} = \frac{25.0}{\sin F}$$

$$\text{from which, } \sin F = \frac{25.0 \sin 64^\circ}{32.91} = 0.6828$$

$$\text{Thus } \angle F = \sin^{-1} 0.6828 = 43^\circ 4' \text{ or } 136^\circ 56'$$

$F = 136^\circ 56'$ is not possible in this case since $136^\circ 56' + 64^\circ$ is greater than 180° . Thus only $F = 43^\circ 4'$ is valid

$$\angle D = 180^\circ - 64^\circ - 43^\circ 4' = 72^\circ 56'$$

$$\begin{aligned} \text{Area of triangle } DEF &= \frac{1}{2} df \sin E \\ &= \frac{1}{2} (35.0)(25.0) \sin 64^\circ = 393.2 \text{ mm}^2 \end{aligned}$$

Problem 30. A triangle ABC has sides $a = 9.0 \text{ cm}$, $b = 7.5 \text{ cm}$ and $c = 6.5 \text{ cm}$. Determine its three angles and its area.

Triangle ABC is shown in Fig. 12.29. It is usual first to calculate the largest angle to determine whether the triangle is acute or obtuse. In this case the largest angle is A (i.e. opposite the longest side).

Applying the cosine rule:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{from which, } 2bc \cos A = b^2 + c^2 - a^2$$

$$\begin{aligned} \text{and } \cos A &= \frac{b^2 + c^2 - a^2}{2bc} = \frac{7.5^2 + 6.5^2 - 9.0^2}{2(7.5)(6.5)} \\ &= 0.1795 \end{aligned}$$

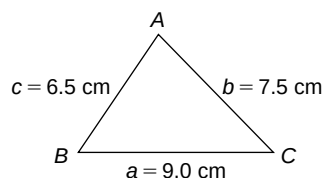


Figure 12.29

Hence $A = \cos^{-1} 0.1795 = 79^\circ 40'$ (or $280^\circ 20'$, which is obviously impossible). The triangle is thus acute angled since $\cos A$ is positive. (If $\cos A$ had been negative, angle A would be obtuse, i.e. lie between 90° and 180° .)

Applying the sine rule:

$$\frac{9.0}{\sin 79^\circ 40'} = \frac{7.5}{\sin B}$$

from which,

$$\sin B = \frac{7.5 \sin 79^\circ 40'}{9.0} = 0.8198$$

$$\text{Hence } B = \sin^{-1} 0.8198 = 55^\circ 4'$$

$$\text{and } C = 180^\circ - 79^\circ 40' - 55^\circ 4' = 45^\circ 16'$$

$$\text{Area} = \sqrt{[s(s-a)(s-b)(s-c)]}$$

$$\begin{aligned} \text{where } s &= \frac{a+b+c}{2} = \frac{9.0+7.5+6.5}{2} \\ &= 11.5 \text{ cm} \end{aligned}$$

Hence **area**

$$\begin{aligned} &= \sqrt{[11.5(11.5-9.0)(11.5-7.5)(11.5-6.5)]} \\ &= \sqrt{[11.5(2.5)(4.0)(5.0)]} = 23.98 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Alternatively, area} &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} (9.0)(7.5) \sin 45^\circ 16' = 23.98 \text{ cm}^2 \end{aligned}$$

Now try the following Practice Exercise

Practice Exercise 57 Solving triangles and finding their areas (Answers on page 838)

In Problems 1 and 2, use the cosine and sine rules to solve the triangles PQR and find their areas.

$$1. \quad q = 12 \text{ cm}, r = 16 \text{ cm}, P = 54^\circ$$

$$2. \quad q = 3.25 \text{ m}, r = 4.42 \text{ m}, P = 105^\circ$$

In problems 3 and 4, use the cosine and sine rules to solve the triangles XYZ and find their areas.

3. $x = 10.0 \text{ cm}$, $y = 8.0 \text{ cm}$, $z = 7.0 \text{ cm}$.
4. $x = 21 \text{ mm}$, $y = 34 \text{ mm}$, $z = 42 \text{ mm}$.

12.11 Practical situations involving trigonometry

There are a number of **practical situations** where the use of trigonometry is needed to find unknown sides and angles of triangles. This is demonstrated in the following problems.

Problem 31. A room 8.0 m wide has a span roof which slopes at 33° on one side and 40° on the other. Find the length of the roof slopes, correct to the nearest centimetre.

A section of the roof is shown in Fig. 12.30.

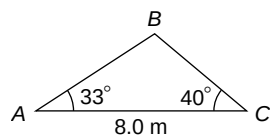


Figure 12.30

Angle at ridge, $B = 180^\circ - 33^\circ - 40^\circ = 107^\circ$
From the sine rule:

$$\frac{8.0}{\sin 107^\circ} = \frac{a}{\sin 33^\circ}$$

from which,

$$a = \frac{8.0 \sin 33^\circ}{\sin 107^\circ} = 4.556 \text{ m}$$

Also from the sine rule:

$$\frac{8.0}{\sin 107^\circ} = \frac{c}{\sin 40^\circ}$$

from which,

$$c = \frac{8.0 \sin 40^\circ}{\sin 107^\circ} = 5.377 \text{ m}$$

Hence the roof slopes are 4.56 m and 5.38 m, correct to the nearest centimetre.

Problem 32. Two voltage phasors are shown in Fig. 12.31. If $V_1 = 40 \text{ V}$ and $V_2 = 100 \text{ V}$ determine

the value of their resultant (i.e. length OA) and the angle the resultant makes with V_1

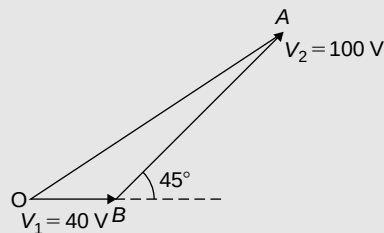


Figure 12.31

Angle $OBA = 180^\circ - 45^\circ = 135^\circ$

Applying the cosine rule:

$$\begin{aligned} OA^2 &= V_1^2 + V_2^2 - 2V_1V_2 \cos OBA \\ &= 40^2 + 100^2 - \{2(40)(100) \cos 135^\circ\} \\ &= 1600 + 10\,000 - \{-5657\} \\ &= 1600 + 10\,000 + 5657 = 17\,257 \end{aligned}$$

The resultant

$$OA = \sqrt{(17\,257)} = 131.4 \text{ V}$$

Applying the sine rule:

$$\frac{131.4}{\sin 135^\circ} = \frac{100}{\sin AOB}$$

$$\begin{aligned} \text{from which, } \sin AOB &= \frac{100 \sin 135^\circ}{131.4} \\ &= 0.5381 \end{aligned}$$

Hence angle $AOB = \sin^{-1} 0.5381 = 32^\circ 33'$ (or $147^\circ 27'$, which is impossible in this case).

Hence the resultant voltage is 131.4 volts at $32^\circ 33'$ to V_1

Problem 33. In Fig. 12.32, PR represents the inclined jib of a crane and is 10.0 long. PQ is 4.0 m long. Determine the inclination of the jib to the vertical and the length of tie QR .

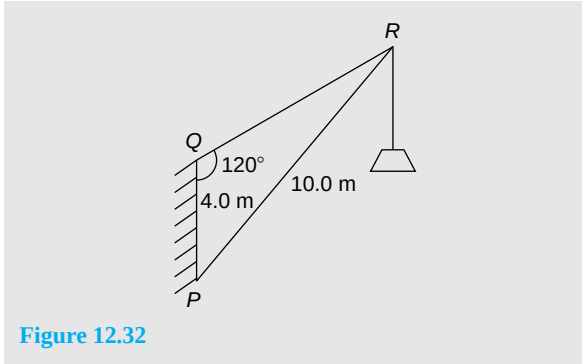


Figure 12.32

Applying the sine rule:

$$\frac{PR}{\sin 120^\circ} = \frac{PQ}{\sin R}$$

from which,

$$\sin R = \frac{PQ \sin 120^\circ}{PR} = \frac{(4.0) \sin 120^\circ}{10.0} = 0.3464$$

Hence $\angle R = \sin^{-1} 0.3464 = 20^\circ 16'$ (or $159^\circ 44'$, which is impossible in this case).

$\angle P = 180^\circ - 120^\circ - 20^\circ 16' = 39^\circ 44'$, which is the inclination of the jib to the vertical.

Applying the sine rule:

$$\frac{10.0}{\sin 120^\circ} = \frac{QR}{\sin 39^\circ 44'}$$

from which, **length of tie,**

$$QR = \frac{10.0 \sin 39^\circ 44'}{\sin 120^\circ} = 7.38 \text{ m}$$

Now try the following Practice Exercise

Practice Exercise 58 Practical situations involving trigonometry (Answers on page 838)

1. A ship P sails at a steady speed of 45 km/h in a direction of $W 32^\circ N$ (i.e. a bearing of 302°) from a port. At the same time another ship Q leaves the port at a steady speed of 35 km/h in a direction $N 15^\circ E$ (i.e. a bearing of 015°). Determine their distance apart after four hours.
2. Two sides of a triangular plot of land are 52.0 m and 34.0 m, respectively. If the area

of the plot is 620 m^2 find (a) the length of fencing required to enclose the plot and (b) the angles of the triangular plot.

3. A jib crane is shown in Fig. 12.33. If the tie rod PR is 8.0 long and PQ is 4.5 m long, determine (a) the length of jib RQ and (b) the angle between the jib and the tie rod.

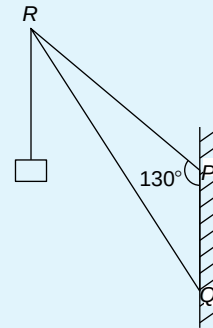


Figure 12.33

4. A building site is in the form of a quadrilateral, as shown in Fig. 12.34, and its area is 1510 m^2 . Determine the length of the perimeter of the site.

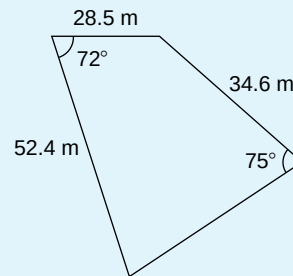


Figure 12.34

5. Determine the length of members BF and EB in the roof truss shown in Fig. 12.35.

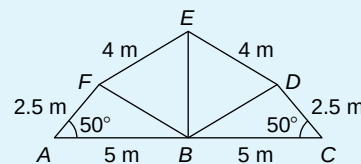


Figure 12.35

6. A laboratory 9.0 m wide has a span roof which slopes at 36° on one side and 44° on the other. Determine the lengths of the roof slopes.

12.12 Further practical situations involving trigonometry

Problem 34. A vertical aerial stands on horizontal ground. A surveyor positioned due east of the aerial measures the elevation of the top as 48° . He moves due south 30.0 m and measures the elevation as 44° . Determine the height of the aerial.

In Fig. 12.36, DC represents the aerial, A is the initial position of the surveyor and B his final position.

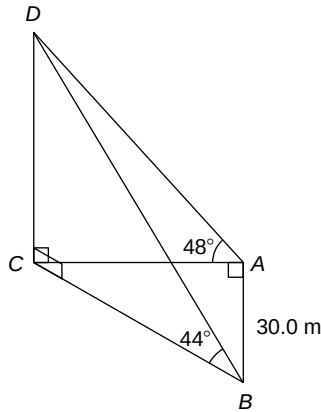


Figure 12.36

From triangle ACD , $\tan 48^\circ = \frac{DC}{AC}$

from which $AC = \frac{DC}{\tan 48^\circ}$

Similarly, from triangle BCD ,

$$BC = \frac{DC}{\tan 44^\circ}$$

For triangle ABC , using Pythagoras' theorem:

$$BC^2 = AB^2 + AC^2$$

$$\left(\frac{DC}{\tan 44^\circ}\right)^2 = (30.0)^2 + \left(\frac{DC}{\tan 48^\circ}\right)^2$$

$$DC^2 \left(\frac{1}{\tan^2 44^\circ} - \frac{1}{\tan^2 48^\circ}\right) = 30.0^2$$

$$DC^2(1.072323 - 0.810727) = 30.0^2$$

$$DC^2 = \frac{30.0^2}{0.261596} = 3440.4$$

Hence, height of aerial,

$$DC = \sqrt{3440.4} = 58.65 \text{ m}$$

Problem 35. A crank mechanism of a petrol engine is shown in Fig. 12.37. Arm OA is 10.0 cm long and rotates clockwise about O . The connecting rod AB is 30.0 cm long and end B is constrained to move horizontally.

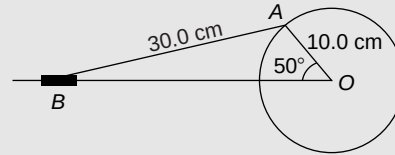


Figure 12.37

- For the position shown in Fig. 12.37 determine the angle between the connecting rod AB and the horizontal and the length of OB .
- How far does B move when angle AOB changes from 50° to 120° ?

- Applying the sine rule:

$$\frac{AB}{\sin 50^\circ} = \frac{AO}{\sin B}$$

from which,

$$\sin B = \frac{AO \sin 50^\circ}{AB} = \frac{10.0 \sin 50^\circ}{30.0}$$

$$= 0.2553$$

Hence $B = \sin^{-1} 0.2553 = 14^\circ 47'$ (or $165^\circ 13'$, which is impossible in this case).

Hence the connecting rod AB makes an angle of $14^\circ 47'$ with the horizontal.

$$\text{Angle } OAB = 180^\circ - 50^\circ - 14^\circ 47' = 115^\circ 13'$$

Applying the sine rule:

$$\frac{30.0}{\sin 50^\circ} = \frac{OB}{\sin 115^\circ 13'}$$

from which,

$$OB = \frac{30.0 \sin 115^\circ 13'}{\sin 50^\circ} = 35.43 \text{ cm}$$

- Fig. 12.38 shows the initial and final positions of the crank mechanism. In triangle $OA'B'$, applying

the sine rule:

$$\frac{30.0}{\sin 120^\circ} = \frac{10.0}{\sin A'B'O}$$

from which,

$$\sin A'B'O = \frac{10.0 \sin 120^\circ}{30.0} = 0.2887$$

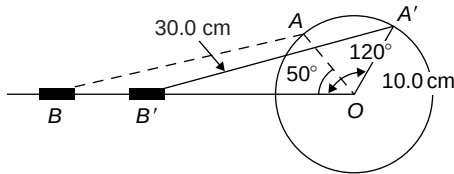


Figure 12.38

Hence $A'B'O = \sin^{-1} 0.2887 = 16^\circ 47'$ (or $163^\circ 13'$ which is impossible in this case).

Angle $OA'B' = 180^\circ - 120^\circ - 16^\circ 47' = 43^\circ 13'$

Applying the sine rule:

$$\frac{30.0}{\sin 120^\circ} = \frac{OB'}{\sin 43^\circ 13'}$$

from which,

$$OB' = \frac{30.0 \sin 43^\circ 13'}{\sin 120^\circ} = 23.72 \text{ cm}$$

Since $OB = 35.43 \text{ cm}$ and $OB' = 23.72 \text{ cm}$ then $BB' = 35.43 - 23.72 = 11.71 \text{ cm}$.

Hence B moves 11.71 cm when angle AOB changes from 50° to 120°

Problem 36. The area of a field is in the form of a quadrilateral $ABCD$ as shown in Fig. 12.39. Determine its area.

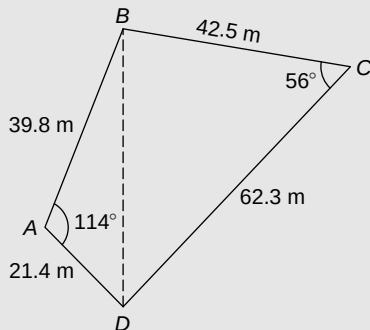


Figure 12.39

A diagonal drawn from B to D divides the quadrilateral into two triangles.

Area of quadrilateral $ABCD$

$$\begin{aligned} &= \text{area of triangle } ABD + \text{area of triangle } BCD \\ &= \frac{1}{2}(39.8)(21.4) \sin 114^\circ + \frac{1}{2}(42.5)(62.3) \sin 56^\circ \\ &= 389.04 + 1097.5 = \mathbf{1487 \text{ m}^2} \end{aligned}$$

Now try the following Practice Exercise

Practice Exercise 59 Practical situations involving trigonometry (Answers on page 838)

1. PQ and QR are the phasors representing the alternating currents in two branches of a circuit. Phasor PQ is 20.0 A and is horizontal. Phasor QR (which is joined to the end of PQ to form triangle PQR) is 14.0 A and is at an angle of 35° to the horizontal. Determine the resultant phasor PR and the angle it makes with phasor PQ .
2. Three forces acting on a fixed point are represented by the sides of a triangle of dimensions 7.2 cm , 9.6 cm and 11.0 cm . Determine the angles between the lines of action and the three forces.
3. Calculate, correct to 3 significant figures, the co-ordinates x and y to locate the hole centre at P shown in Fig. 12.40.

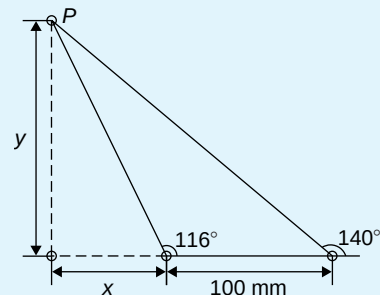


Figure 12.40

4. An idler gear, 30 mm in diameter, has to be fitted between a 70 mm diameter driving gear and a 90 mm diameter driven gear as shown in Fig. 12.41. Determine the value of angle θ between the centre lines.

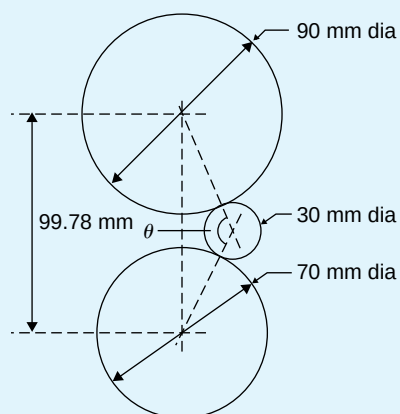


Figure 12.41

5. A reciprocating engine mechanism is shown in Fig. 12.42. The crank AB is 12.0 cm long and the connecting rod BC is 32.0 cm long. For the position shown determine the length of AC and the angle between the crank and the connecting rod.

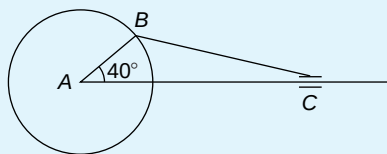


Figure 12.42

6. From Fig. 12.42, determine how far C moves, correct to the nearest millimetre, when angle CAB changes from 40° to 160° , B moving in an anticlockwise direction.
7. A surveyor, standing $W\ 25^\circ\ S$ of a tower, measures the angle of elevation of the top of the tower as $46^\circ 30'$. From a position $E\ 23^\circ\ S$ from the tower the elevation of the top is $37^\circ 15'$. Determine the height of the tower if the distance between the two observations is 75 m.
8. An aeroplane is sighted due east from a radar station at an elevation of 40° and a height of 8000 m and later at an elevation of 35° and height 5500 m in a direction $E\ 70^\circ\ S$. If it is descending uniformly, find the angle of descent. Determine also the speed of the aeroplane in km/h if the time between the two observations is 45 s.
9. Sixteen holes are equally spaced on a pitch circle of 70 mm diameter. Determine the length of the chord joining the centres of two adjacent holes.

For fully worked solutions to each of the problems in Practice Exercises 51 to 59 in this chapter, go to the website:

www.routledge.com/cw/bird

Chapter 13

Cartesian and polar co-ordinates

Why it is important to understand: Cartesian and polar co-ordinates

Applications where polar co-ordinates would be used include terrestrial navigation with sonar-like devices, and those in engineering and science involving energy radiation patterns. Applications where Cartesian co-ordinates would be used include any navigation on a grid and anything involving raster graphics (e.g. bitmap – a dot matrix data structure representing a generally rectangular grid of pixels). The ability to change from Cartesian to polar co-ordinates is vitally important when using complex numbers and their use in a.c. electrical circuit theory and with vector geometry.

At the end of this chapter, you should be able to:

- change from Cartesian to polar co-ordinates
- change from polar to Cartesian co-ordinates
- use a scientific notation calculator to change from Cartesian to polar co-ordinates and vice-versa

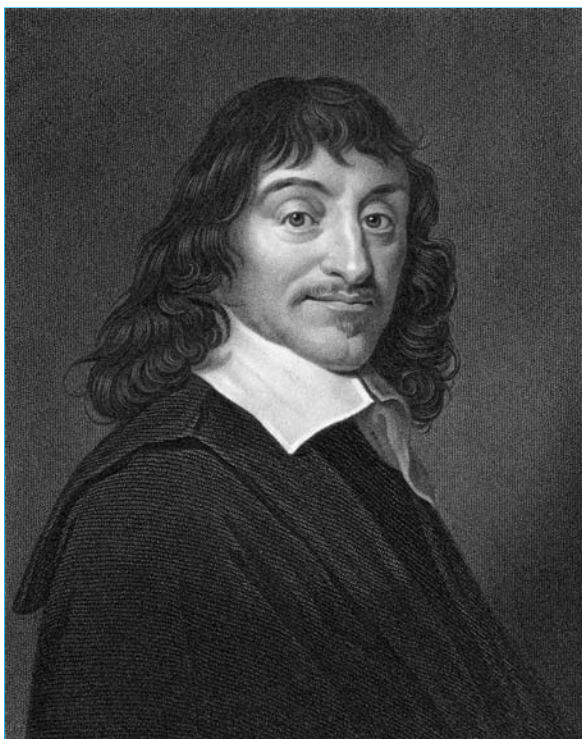
13.1 Introduction

There are two ways in which the position of a point in a plane can be represented. These are

- (a) by **Cartesian co-ordinates**, (named after **Descartes***), i.e. (x, y) , and
- (b) by **polar co-ordinates**, i.e. (r, θ) , where r is a 'radius' from a fixed point and θ is an angle from a fixed point.

13.2 Changing from Cartesian into polar co-ordinates

In Fig. 13.1, if lengths x and y are known, then the length of r can be obtained from Pythagoras' theorem (see Chapter 12) since OPQ is a right-angled triangle. Hence $r^2 = (x^2 + y^2)$



* **Who was Descartes?** René Descartes (31 March 1596–11 February 1650) was a French philosopher, mathematician, and writer. He wrote many influential texts including *Meditations on First Philosophy*. Descartes is best known for the philosophical statement '*Cogito ergo sum*' (I think, therefore I am), found in part IV of *Discourse on the Method*. To find out more go to www.routledge.com/cw/bird

from which, $r = \sqrt{x^2 + y^2}$

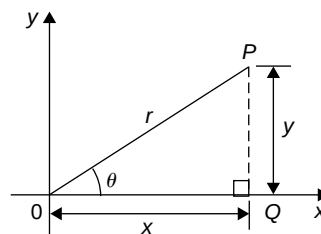


Figure 13.1

From trigonometric ratios (see Chapter 12),

$$\tan \theta = \frac{y}{x}$$

from which $\theta = \tan^{-1} \frac{y}{x}$

$r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \frac{y}{x}$ are the two formulae we need to change from Cartesian to polar co-ordinates. The angle θ , which may be expressed in degrees or radians, must **always** be measured from the positive x -axis, i.e., measured from the line OQ in Fig. 13.1. It is suggested that when changing from Cartesian to polar co-ordinates a diagram should always be sketched.

Problem 1. Change the Cartesian co-ordinates (3, 4) into polar co-ordinates.

A diagram representing the point (3, 4) is shown in Fig. 13.2.

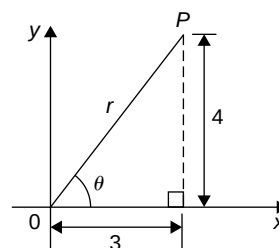


Figure 13.2

From Pythagoras' theorem, $r = \sqrt{3^2 + 4^2} = 5$ (note that -5 has no meaning in this context). By trigonometric ratios, $\theta = \tan^{-1} \frac{4}{3} = 53.13^\circ$ or 0.927 rad.

[Note that $53.13^\circ = 53.13 \times (\pi / 180)$ rad $= 0.927$ rad.]

Hence (3, 4) in Cartesian co-ordinates corresponds to (5, 53.13°) or (5, 0.927 rad) in polar co-ordinates.

Problem 2. Express in polar co-ordinates the position (−4, 3)

A diagram representing the point using the Cartesian co-ordinates (−4, 3) is shown in Fig. 13.3.

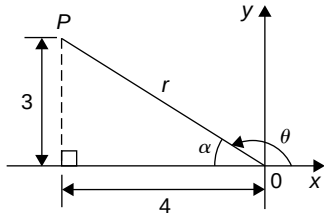


Figure 13.3

From Pythagoras' theorem, $r = \sqrt{4^2 + 3^2} = 5$

By trigonometric ratios, $\alpha = \tan^{-1} \frac{3}{4} = 36.87^\circ$ or 0.644 rad.

Hence $\theta = 180^\circ - 36.87^\circ = 143.13^\circ$ or $\theta = \pi - 0.644 = 2.498$ rad.

Hence the position of point P in polar co-ordinate form is (5, 143.13°) or (5, 2.498 rad).

Problem 3. Express (−5, −12) in polar co-ordinates.

A sketch showing the position (−5, −12) is shown in Fig. 13.4.

$$r = \sqrt{5^2 + 12^2} = 13$$

$$\begin{aligned} \text{and } \alpha &= \tan^{-1} \frac{12}{5} \\ &= 67.38^\circ \text{ or } 1.176 \text{ rad} \end{aligned}$$

Hence $\theta = 180^\circ + 67.38^\circ = 247.38^\circ$ or

$$\theta = \pi + 1.176 = 4.318 \text{ rad}$$

Thus (−5, −12) in Cartesian co-ordinates corresponds to (13, 247.38°) or (13, 4.318 rad) in polar co-ordinates.

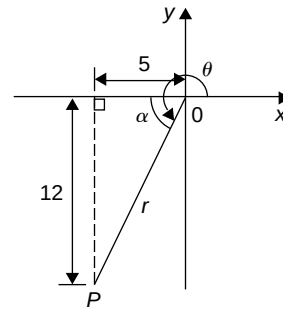


Figure 13.4

Problem 4. Express (2, −5) in polar co-ordinates.

A sketch showing the position (2, −5) is shown in Fig. 13.5.

$$r = \sqrt{2^2 + 5^2} = \sqrt{29} = 5.385 \text{ correct to 3 decimal places.}$$

$$\alpha = \tan^{-1} \frac{5}{2} = 68.20^\circ \text{ or } 1.190 \text{ rad}$$

Hence $\theta = 360^\circ - 68.20^\circ = 291.80^\circ$ or

$$\theta = 2\pi - 1.190 = 5.093 \text{ rad}$$

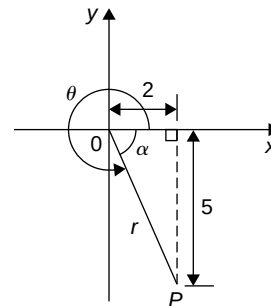


Figure 13.5

Thus (2, −5) in Cartesian co-ordinates corresponds to (5.385, 291.80°) or (5.385, 5.093 rad) in polar co-ordinates.

Now try the following Practice Exercise

Practice Exercise 60 Changing Cartesian into polar co-ordinates (Answers on page 838)

In Problems 1 to 8, express the given Cartesian co-ordinates as polar co-ordinates, correct to 2 decimal places, in both degrees and in radians.

1. (3, 5)
2. (6.18, 2.35)
3. (-2, 4)
4. (-5.4, 3.7)
5. (-7, -3)
6. (-2.4, -3.6)
7. (5, -3)
8. (9.6, -12.4)

13.3 Changing from polar into Cartesian co-ordinates

From the right-angled triangle OPQ in Fig. 13.6,

$$\cos \theta = \frac{x}{r} \text{ and } \sin \theta = \frac{y}{r}, \text{ from}$$

trigonometric ratios

Hence $x = r \cos \theta$ and $y = r \sin \theta$

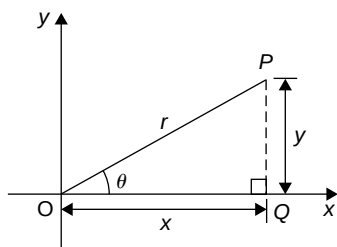


Figure 13.6

If lengths r and angle θ are known then $x = r \cos \theta$ and $y = r \sin \theta$ are the two formulae we need to change from polar to Cartesian co-ordinates.

Problem 5. Change $(4, 32^\circ)$ into Cartesian co-ordinates.

A sketch showing the position $(4, 32^\circ)$ is shown in Fig. 13.7.

$$\text{Now } x = r \cos \theta = 4 \cos 32^\circ = 3.39$$

$$\text{and } y = r \sin \theta = 4 \sin 32^\circ = 2.12$$

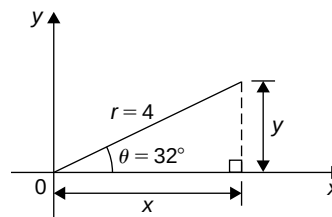


Figure 13.7

Hence $(4, 32^\circ)$ in polar co-ordinates corresponds to $(3.39, 2.12)$ in Cartesian co-ordinates.

Problem 6. Express $(6, 137^\circ)$ in Cartesian co-ordinates.

A sketch showing the position $(6, 137^\circ)$ is shown in Fig. 13.8.

$$x = r \cos \theta = 6 \cos 137^\circ = -4.388$$

which corresponds to length OA in Fig. 13.8.

$$y = r \sin \theta = 6 \sin 137^\circ = 4.092$$

which corresponds to length AB in Fig. 13.8.

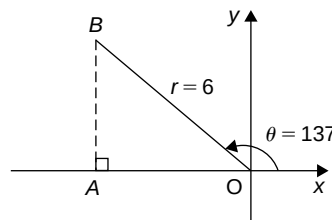


Figure 13.8

Thus $(6, 137^\circ)$ in polar co-ordinates corresponds to $(-4.388, 4.092)$ in Cartesian co-ordinates.

(Note that when changing from polar to Cartesian co-ordinates it is not quite so essential to draw a sketch. Use of $x = r \cos \theta$ and $y = r \sin \theta$ automatically produces the correct signs.)

Problem 7. Express $(4.5, 5.16 \text{ rad})$ in Cartesian co-ordinates.

A sketch showing the position $(4.5, 5.16 \text{ rad})$ is shown in Fig. 13.9.

$$x = r \cos \theta = 4.5 \cos 5.16 = 1.948$$

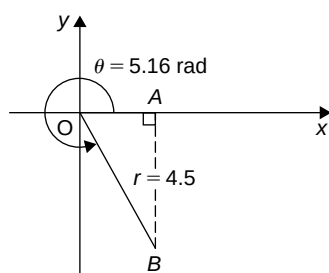


Figure 13.9

which corresponds to length OA in Fig. 13.9.

$$y = r \sin \theta = 4.5 \sin 5.16 = -4.057$$

which corresponds to length AB in Fig. 13.9.

Thus $(1.948, -4.057)$ in Cartesian co-ordinates corresponds to $(4.5, 5.16 \text{ rad})$ in polar co-ordinates.

Now try the following Practice Exercise

Practice Exercise 61 Changing polar into Cartesian co-ordinates (Answers on page 838)

In Problems 1 to 8, express the given polar co-ordinates as Cartesian co-ordinates, correct to 3 decimal places.

1. $(5, 75^\circ)$
2. $(4.4, 1.12 \text{ rad})$
3. $(7, 140^\circ)$
4. $(3.6, 2.5 \text{ rad})$
5. $(10.8, 210^\circ)$
6. $(4, 4 \text{ rad})$

7. $(1.5, 300^\circ)$
8. $(6, 5.5 \text{ rad})$
9. Fig. 13.10 shows five equally spaced holes on an 80 mm pitch circle diameter. Calculate their co-ordinates relative to axes Ox and Oy in (a) polar form, (b) Cartesian form.
(c) Calculate also the shortest distance between the centres of two adjacent holes.

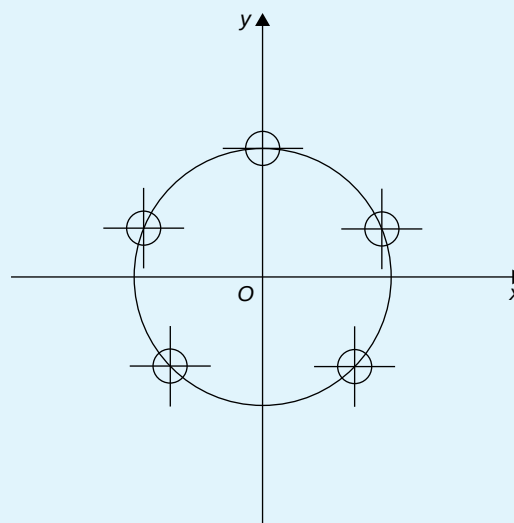


Figure 13.10

13.4 Use of Pol/Rec functions on calculators

Another name for Cartesian co-ordinates is **rectangular** co-ordinates. Many scientific notation calculators possess **Pol** and **Rec** functions. 'Rec' is an abbreviation of 'rectangular' (i.e. Cartesian) and 'Pol' is an abbreviation of 'polar'. Check the operation manual for your particular calculator to determine how to use these two functions. They make changing from Cartesian to polar co-ordinates, and vice-versa, so much quicker and easier.

For example, with the Casio fx-991ES PLUS calculator, or similar, to change the Cartesian number $(3, 4)$ into polar form, the following procedure is adopted:

1. Press 'shift' 2. Press 'Pol' 3. Enter 3
4. Enter 'comma' (obtained by 'shift' then)
5. Enter 4 6. Press)
7. Press = The answer is: $r = 5, \theta = 53.13^\circ$

Hence, (3, 4) in Cartesian form is the same as (5, 53.13°) in polar form.

If the angle is required in **radians**, then before repeating the above procedure press 'shift', 'mode' and then 4 to change your calculator to radian mode.

Similarly, to change the polar form number (7, 126°) into Cartesian or rectangular form, adopt the following procedure:

1. Press 'shift' 2. Press 'Rec' 3. Enter 7
 4. Enter 'comma'
-

5. Enter 126 (assuming your calculator is in degrees mode)

6. Press) 7. Press =

The answer is: **$X = -4.11$, and scrolling across, $Y = 5.66$** , correct to 2 decimal places.

Hence, (7, 126°) in polar form is the same as (-4.11, 5.66) in rectangular or Cartesian form.

Now return to Practice Exercises 60 and 61 in this chapter and use your calculator to determine the answers, and see how much more quickly they may be obtained.

Chapter 14

The circle and its properties

Why it is important to understand: The circle and its properties

A circle is one of the fundamental shapes of geometry; it consists of all the points that are equidistant from a central point. Knowledge of calculations involving circles is needed with crank mechanisms, with determinations of latitude and longitude, with pendulums, and even in the design of paper clips. The floodlit area at a football ground, the area an automatic garden sprayer sprays and the angle of lap of a belt drive all rely on calculations involving the arc of a circle. The ability to handle calculations involving circles and its properties is clearly essential in several branches of engineering design.

At the end of this chapter, you should be able to:

- define a circle
- state some properties of a circle – including radius, circumference, diameter, semicircle, quadrant, tangent, sector, chord, segment and arc
- appreciate the angle in a semicircle is a right angle
- define a radian, change radians to degrees, and vice-versa
- determine arc length, area of a circle and area of a sector of a circle
- state the equation of a circle
- sketch a circle given its equation
- understand linear and angular velocity
- understand centripetal force

14.1 Introduction

A **circle** is a plain figure enclosed by a curved line, every point on which is equidistant from a point within, called the **centre**.

14.2 Properties of circles

- (i) The distance from the centre to the curve is called the **radius**, r , of the circle (see OP in Fig. 14.1).

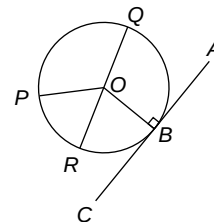


Figure 14.1

- (ii) The boundary of a circle is called the **circumference**, c .

- (iii) Any straight line passing through the centre and touching the circumference at each end is called the **diameter**, d (see QR in Fig. 14.1). Thus $d = 2r$.
- (iv) The ratio $\frac{\text{circumference}}{\text{diameter}} = \text{a constant for any circle}$.
This constant is denoted by the Greek letter π (pronounced 'pie'), where $\pi = 3.14159$, correct to 5 decimal places.
Hence $c/d = \pi$ or $c = \pi d$ or $c = 2\pi r$.
- (v) A **semicircle** is one half of the whole circle.
- (vi) A **quadrant** is one quarter of a whole circle.
- (vii) A **tangent** to a circle is a straight line which meets the circle in one point only and does not cut the circle when produced. AC in Fig. 14.1 is a tangent to the circle since it touches the curve at point B only. If radius OB is drawn, then angle ABO is a right angle.
- (viii) A **sector** of a circle is the part of a circle between radii (for example, the portion OXY of Fig. 14.2 is a sector). If a sector is less than a semicircle it is called a **minor sector**, if greater than a semicircle it is called a **major sector**.

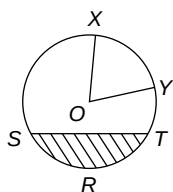


Figure 14.2

- (ix) A **chord** of a circle is any straight line which divides the circle into two parts and is terminated at each end by the circumference. ST , in Fig. 14.2 is a chord.
- (x) A **segment** is the name given to the parts into which a circle is divided by a chord. If the segment is less than a semicircle it is called a **minor segment** (see shaded area in Fig. 14.2). If the segment is greater than a semicircle it is called a **major segment** (see the unshaded area in Fig. 14.2).
- (xi) An **arc** is a portion of the circumference of a circle. The distance SRT in Fig. 14.2 is called a **minor arc** and the distance $SXYT$ is called a **major arc**.
- (xii) The angle at the centre of a circle, subtended by an arc, is double the angle at the circumference

subtended by the same arc. With reference to Fig. 14.3, **Angle $AOC = 2 \times \text{angle } ABC$** .

- (xiii) The angle in a semicircle is a right angle (see angle BQP in Fig. 14.3).

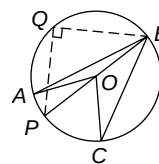


Figure 14.3

Problem 1. If the diameter of a circle is 75 mm, find its circumference.

$$\begin{aligned}\text{Circumference, } c &= \pi \times \text{diameter} = \pi d \\ &= \pi (75) = 235.6 \text{ mm}\end{aligned}$$

Problem 2. In Fig. 14.4, AB is a tangent to the circle at B . If the circle radius is 40 mm and $AB = 150$ mm, calculate the length AO .

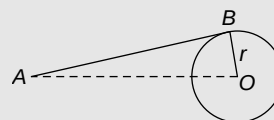


Figure 14.4

A tangent to a circle is at right angles to a radius drawn from the point of contact, i.e. $ABO = 90^\circ$. Hence, using Pythagoras' theorem:

$$\begin{aligned}AO^2 &= AB^2 + OB^2 \\ AO &= \sqrt{(AB^2 + OB^2)} = \sqrt{[(150)^2 + (40)^2]} \\ &= 155.2 \text{ mm}\end{aligned}$$

Now try the following Practice Exercise

Practice Exercise 62 Properties of circles (Answers on page 838)

- If the radius of a circle is 41.3 mm, calculate the circumference of the circle.
- Find the diameter of a circle whose perimeter is 149.8 cm.
- A crank mechanism is shown in Fig. 14.5, where XY is a tangent to the circle at point X . If

the circle radius OX is 10 cm and length OY is 40 cm, determine the length of the connecting rod XY .

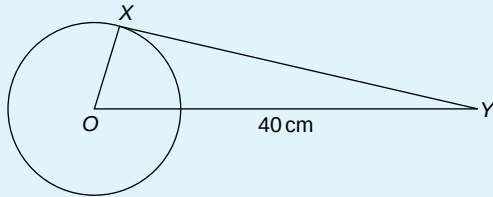


Figure 14.5

4. If the circumference of the Earth is 40 000 km at the equator, calculate its diameter.
5. Calculate the length of wire in the paper clip shown in Fig. 14.6. The dimensions are in millimetres.

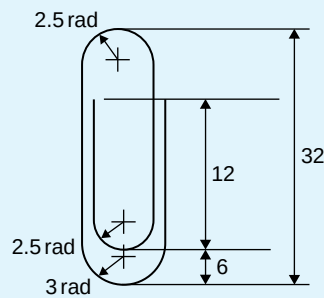


Figure 14.6

14.3 Radians and degrees

One **radian** is defined as the angle subtended at the centre of a circle by an arc equal in length to the radius.

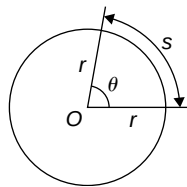


Figure 14.7

With reference to Fig. 14.7, for arc length s ,

$$\theta \text{ radians} = \frac{s}{r}$$

When $s =$ whole circumference $(= 2\pi r)$ then

$$\theta = \frac{s}{r} = \frac{2\pi r}{r} = 2\pi$$

i.e. 2π radians = 360° or π radians = 180°

Thus, $1 \text{ rad} = \frac{180^\circ}{\pi} = 57.30^\circ$, correct to 2 decimal places.

Since $\pi \text{ rad} = 180^\circ$, then $\frac{\pi}{2} = 90^\circ$, $\frac{\pi}{3} = 60^\circ$, $\frac{\pi}{4} = 45^\circ$, and so on.

Problem 3. Convert to radians: (a) 125°
(b) $69^\circ 47'$

- (a) Since $180^\circ = \pi$ rad then $1^\circ = \pi/180$ rad, therefore

$$125^\circ = 125 \left(\frac{\pi}{180} \right)^c = \mathbf{2.182 \text{ rad}}$$

(Note that c means ‘circular measure’ and indicates radian measure.)

- $$(b) \quad 69^{\circ}47' = 69\frac{47^{\circ}}{60} = 69.783^{\circ}$$

$$69.783^\circ = 69.783 \left(\frac{\pi}{180} \right)^c = \mathbf{1.218 \text{ rad}}$$

Problem 4. Convert to degrees and minutes:

- (a) 0.749 rad (b) $3\pi/4$ rad.

- (a) Since $\pi \text{ rad} = 180^\circ$ then $1 \text{ rad} = 180^\circ/\pi$, therefore

$$0.749 = 0.749 \left(\frac{180}{\pi} \right)^\circ = 42.915^\circ$$

$0.915^\circ = (0.915 \times 60)' = 55'$, correct to the nearest minute, hence

$$0.749 \text{ rad} = 42^\circ 55'$$

- (b) Since $1 \text{ rad} = \left(\frac{180}{\pi}\right)^\circ$ then

$$\frac{3\pi}{4} \text{ rad} = \frac{3\pi}{4} \left(\frac{180}{\pi} \right)^\circ = \frac{3}{4} (180)^\circ = \mathbf{135^\circ}$$

Problem 5. Express in radians, in terms of π ,

- (a) 150° (b) 270° (c) 37.5°

Since $180^\circ = \pi$ rad then $1^\circ = \pi/180$, hence

$$(a) \quad 150^\circ = 150 \left(\frac{\pi}{180} \right) \text{ rad} = \frac{5\pi}{6} \text{ rad}$$

$$(b) \quad 270^\circ = 270 \left(\frac{\pi}{180} \right) \text{ rad} = \frac{3\pi}{2} \text{ rad}$$

$$(c) \quad 37.5^\circ = 37.5 \left(\frac{\pi}{180} \right) \text{ rad} = \frac{75\pi}{360} \text{ rad} = \frac{5\pi}{24} \text{ rad}$$

Now try the following Practice Exercise

Practice Exercise 63 Radians and degrees (Answers on page 839)

- Convert to radians in terms of π : (a) 30°
(b) 75° (c) 225°
- Convert to radians: (a) 48° (b) $84^\circ 51'$
(c) $232^\circ 15'$
- Convert to degrees: (a) $\frac{7\pi}{6}$ rad (b) $\frac{4\pi}{9}$ rad
(c) $\frac{7\pi}{12}$ rad.
- Convert to degrees and minutes: (a) 0.0125 rad
(b) 2.69 rad (c) 7.241 rad.
- A car engine speed is 1000 rev/min. Convert this speed into rad/s.

14.4 Arc length and area of circles and sectors

Arc length

From the definition of the radian in the previous section and Fig. 14.7,

$$\text{arc length, } s = r\theta \quad \text{where } \theta \text{ is in radians}$$

Area of circle

For any circle, area = $\pi \times (\text{radius})^2$

$$\text{i.e.} \quad \text{area} = \pi r^2$$

$$\text{Since } r = \frac{d}{2}, \text{ then } \text{area} = \pi r^2 \text{ or } \frac{\pi d^2}{4}$$

Area of sector

$$\text{Area of a sector} = \frac{\theta}{360} (\pi r^2) \quad \text{when } \theta \text{ is in degrees}$$

$$= \frac{\theta}{2\pi} (\pi r^2) = \frac{1}{2} r^2 \theta$$

when θ is in radians

Problem 6. A hockey pitch has a semicircle of radius 14.63 m around each goal net. Find the area enclosed by the semicircle, correct to the nearest square metre.

$$\text{Area of a semicircle} = \frac{1}{2} \pi r^2$$

$$\text{When } r = 14.63 \text{ m, area} = \frac{1}{2} \pi (14.63)^2$$

$$\text{i.e. area of semicircle} = 336 \text{ m}^2$$

Problem 7. Find the area of a circular metal plate, correct to the nearest square millimetre, having a diameter of 35.0 mm.

$$\text{Area of a circle} = \pi r^2 = \frac{\pi d^2}{4}$$

$$\text{When } d = 35.0 \text{ mm, area} = \frac{\pi (35.0)^2}{4}$$

$$\text{i.e. area of circular plate} = 962 \text{ mm}^2$$

Problem 8. Find the area of a circle having a circumference of 60.0 mm.

$$\text{Circumference, } c = 2\pi r$$

$$\text{from which radius } r = \frac{c}{2\pi} = \frac{60.0}{2\pi} = \frac{30.0}{\pi}$$

$$\text{Area of a circle} = \pi r^2$$

$$\text{i.e. area} = \pi \left(\frac{30.0}{\pi} \right)^2 = 286.5 \text{ mm}^2$$

Problem 9. Find the length of arc of a circle of radius 5.5 cm when the angle subtended at the centre is 1.20 rad.

Length of arc, $s = r\theta$, where θ is in radians, hence

$$s = (5.5)(1.20) = 6.60 \text{ cm}$$

Problem 10. Determine the diameter and circumference of a circle if an arc of length 4.75 cm subtends an angle of 0.91 rad.

Since $s = r\theta$ then $r = \frac{s}{\theta} = \frac{4.75}{0.91} = 5.22 \text{ cm}$

Diameter $= 2 \times \text{radius} = 2 \times 5.22 = \mathbf{10.44 \text{ cm}}$

Circumference, $c = \pi d = \pi(10.44) = \mathbf{32.80 \text{ cm}}$

Problem 11. If an angle of 125° is subtended by an arc of a circle of radius 8.4 cm, find the length of (a) the minor arc, and (b) the major arc, correct to 3 significant figures.

(a) Since $180^\circ = \pi \text{ rad}$ then $1^\circ = \left(\frac{\pi}{180}\right) \text{ rad}$ and $125^\circ = 125 \left(\frac{\pi}{180}\right) \text{ rad}$.

Length of minor arc,

$$s = r\theta = (8.4)(125) \left(\frac{\pi}{180}\right) = \mathbf{18.3 \text{ cm}},$$

correct to 3 significant figures.

(b) Length of major arc

$= (\text{circumference} - \text{minor arc})$

$$= 2\pi(8.4) - 18.3 = 34.5 \text{ cm},$$

correct to 3 significant figures.

(Alternatively, major arc $= r\theta$
 $= 8.4(360 - 125) \left(\frac{\pi}{180}\right) = \mathbf{34.5 \text{ cm}}.$

Problem 12. Determine the angle, in degrees and minutes, subtended at the centre of a circle of diameter 42 mm by an arc of length 36 mm. Calculate also the area of the minor sector formed.

Since length of arc, $s = r\theta$ then $\theta = s/r$

$$\text{Radius, } r = \frac{\text{diameter}}{2} = \frac{42}{2} = 21 \text{ mm}$$

$$\text{hence } \theta = \frac{s}{r} = \frac{36}{21} = 1.7143 \text{ rad}$$

$1.7143 \text{ rad} = 1.7143 \times (180/\pi)^\circ = 98.22^\circ = \mathbf{98^\circ 13'}$
 = angle subtended at centre of circle.

Area of sector

$$= \frac{1}{2}r^2\theta = \frac{1}{2}(21)^2(1.7143) = \mathbf{378 \text{ mm}^2}$$

Problem 13. A football stadium floodlight can spread its illumination over an angle of 45° to a distance of 55 m. Determine the maximum area that is floodlit.

Floodlit area = area of sector

$$= \frac{1}{2}r^2\theta = \frac{1}{2}(55)^2 \left(45 \times \frac{\pi}{180}\right) \\ = \mathbf{1188 \text{ m}^2}$$

Problem 14. An automatic garden spray produces a spray to a distance of 1.8 m and revolves through an angle α which may be varied. If the desired spray catchment area is to be 2.5 m^2 , to what should angle α be set, correct to the nearest degree.

Area of sector $= \frac{1}{2}r^2\theta$, hence $2.5 = \frac{1}{2}(1.8)^2\alpha$

from which, $\alpha = \frac{2.5 \times 2}{1.8^2} = 1.5432 \text{ rad}$

$$1.5432 \text{ rad} = \left(1.5432 \times \frac{180^\circ}{\pi}\right) = 88.42^\circ$$

Hence **angle $\alpha = 88^\circ$** , correct to the nearest degree.

Problem 15. The angle of a tapered groove is checked using a 20 mm diameter roller as shown in Fig. 14.8. If the roller lies 2.12 mm below the top of the groove, determine the value of angle θ .

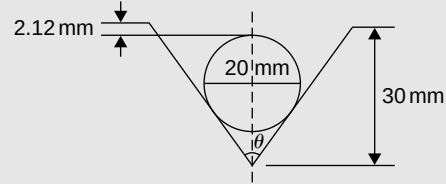


Figure 14.8

In Fig. 14.9, triangle ABC is right-angled at C (see Section 14.2 (vii)).

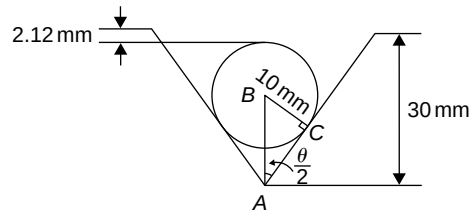


Figure 14.9

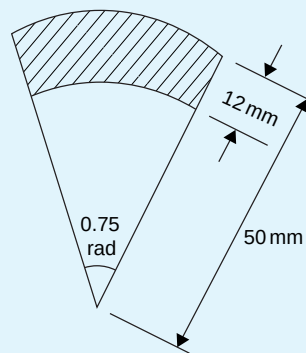
Length $BC = 10 \text{ mm}$ (i.e. the radius of the circle), and $AB = 30 - 10 - 2.12 = 17.88 \text{ mm}$ from Fig. 14.9.

Hence, $\sin \frac{\theta}{2} = \frac{10}{17.88}$ and $\frac{\theta}{2} = \sin^{-1} \left(\frac{10}{17.88} \right) = 34^\circ$
 and **angle $\theta = 68^\circ$**

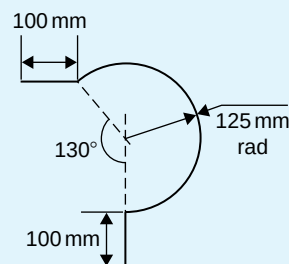
Now try the following Practice Exercise**Practice Exercise 64 Arc length and area of circles and sectors (Answers on page 839)**

1. Calculate the area of a circle of radius 6.0 cm, correct to the nearest square centimetre.
2. The diameter of a circle is 55.0 mm. Determine its area, correct to the nearest square millimetre.
3. The perimeter of a circle is 150 mm. Find its area, correct to the nearest square millimetre.
4. Find the area of the sector, correct to the nearest square millimetre, of a circle having a radius of 35 mm, with angle subtended at centre of 75°
5. An annulus has an outside diameter of 49.0 mm and an inside diameter of 15.0 mm. Find its area correct to 4 significant figures.
6. Find the area, correct to the nearest square metre, of a 2 m wide path surrounding a circular plot of land 200 m in diameter.
7. A rectangular park measures 50 m by 40 m. A 3 m flower bed is made round the two longer sides and one short side. A circular fish pond of diameter 8.0 m is constructed in the centre of the park. It is planned to grass the remaining area. Find, correct to the nearest square metre, the area of grass.
8. Find the length of an arc of a circle of radius 8.32 cm when the angle subtended at the centre is 2.14 rad. Calculate also the area of the minor sector formed.
9. If the angle subtended at the centre of a circle of diameter 82 mm is 1.46 rad, find the lengths of the (a) minor arc (b) major arc.
10. A pendulum of length 1.5 m swings through an angle of 10° in a single swing. Find, in centimetres, the length of the arc traced by the pendulum bob.
11. Determine the length of the radius and circumference of a circle if an arc length of 32.6 cm subtends an angle of 3.76 rad.
12. Determine the angle of lap, in degrees and minutes, if 180 mm of a belt drive are in contact with a pulley of diameter 250 mm.

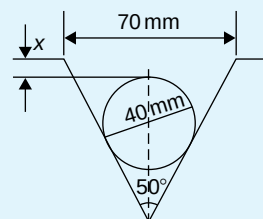
13. Determine the number of complete revolutions a motorcycle wheel will make in travelling 2 km, if the wheel's diameter is 85.1 cm.
14. The floodlight at a sports ground spreads its illumination over an angle of 40° to a distance of 48 m. Determine (a) the angle in radians, and (b) the maximum area that is floodlit.
15. Determine (a) the shaded area in Fig. 14.10 (b) the percentage of the whole sector that the area of the shaded portion represents.

**Figure 14.10**

16. Determine the length of steel strip required to make the clip shown in Fig. 14.11.

**Figure 14.11**

17. A 50° tapered hole is checked with a 40 mm diameter ball as shown in Fig. 14.12. Determine the length shown as x.

**Figure 14.12**

14.5 The equation of a circle

The simplest equation of a circle, centre at the origin, radius r , is given by:

$$x^2 + y^2 = r^2$$

For example, Fig. 14.13 shows a circle $x^2 + y^2 = 9$. More generally, the equation of a circle, centre (a, b) , radius r , is given by:

$$(x - a)^2 + (y - b)^2 = r^2 \quad (1)$$

Figure 14.14 shows a circle $(x - 2)^2 + (y - 3)^2 = 4$. The general equation of a circle is:

$$x^2 + y^2 + 2ex + 2fy + c = 0 \quad (2)$$

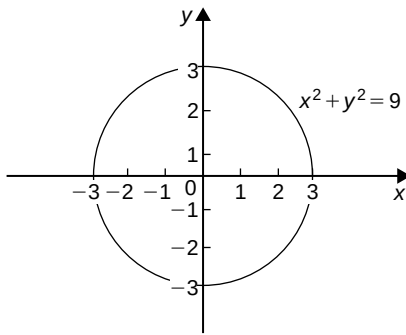


Figure 14.13

Multiplying out the bracketed terms in equation (1) gives:

$$x^2 - 2ax + a^2 + y^2 - 2by + b^2 = r^2$$

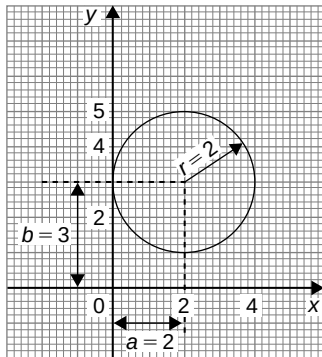


Figure 14.14

Comparing this with equation (2) gives:

$$2e = -2a, \text{ i.e. } a = -\frac{2e}{2}$$

$$\text{and } 2f = -2b, \text{ i.e. } b = -\frac{2f}{2}$$

$$\text{and } c = a^2 + b^2 - r^2,$$

$$\text{i.e., } r = \sqrt{(a^2 + b^2 - c)}$$

Thus, for example, the equation

$$x^2 + y^2 - 4x - 6y + 9 = 0$$

represents a circle with centre $a = -\left(\frac{-4}{2}\right)$,

$$b = -\left(\frac{-6}{2}\right), \text{ i.e. at } (2, 3) \text{ and}$$

$$\text{radius } r = \sqrt{(2^2 + 3^2 - 9)} = 2$$

Hence $x^2 + y^2 - 4x - 6y + 9 = 0$ is the circle shown in Fig. 14.14 (which may be checked by multiplying out the brackets in the equation

$$(x - 2)^2 + (y - 3)^2 = 4$$

Problem 16. Determine (a) the radius and (b) the co-ordinates of the centre of the circle given by the equation: $x^2 + y^2 + 8x - 2y + 8 = 0$

$x^2 + y^2 + 8x - 2y + 8 = 0$ is of the form shown in equation (2),

$$\text{where } a = -\left(\frac{8}{2}\right) = -4, b = -\left(\frac{-2}{2}\right) = 1$$

$$\text{and } r = \sqrt{[(-4)^2 + (1)^2 - 8]} = \sqrt{9} = 3$$

Hence $x^2 + y^2 + 8x - 2y + 8 = 0$ represents a circle **centre $(-4, 1)$ and radius 3**, as shown in Fig. 14.15.

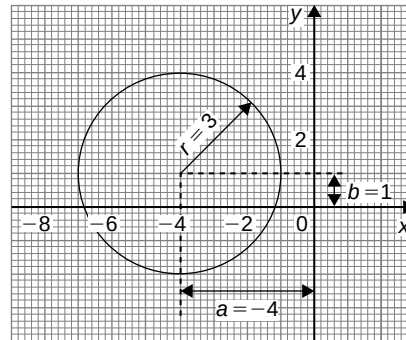


Figure 14.15

Alternatively, $x^2 + y^2 + 8x - 2y + 8 = 0$ may be re-arranged as:

$$(x + 4)^2 + (y - 1)^2 - 9 = 0$$

i.e. $(x + 4)^2 + (y - 1)^2 = 3^2$

which represents a circle, **centre $(-4, 1)$ and radius 3**, as stated above.

Problem 17. Sketch the circle given by the equation: $x^2 + y^2 - 4x + 6y - 3 = 0$

The equation of a circle, centre (a, b) , radius r is given by:

$$(x - a)^2 + (y - b)^2 = r^2$$

The general equation of a circle is

$$x^2 + y^2 + 2ex + 2fy + c = 0$$

From above, $a = -\frac{2e}{2}$, $b = -\frac{2f}{2}$ and

$$r = \sqrt{(a^2 + b^2 - c)}$$

Hence if $x^2 + y^2 - 4x + 6y - 3 = 0$

then $a = -\left(\frac{-4}{2}\right) = 2$, $b = -\left(\frac{6}{2}\right) = -3$

and $r = \sqrt{[(2)^2 + (-3)^2 - (-3)]}$
 $= \sqrt{16} = 4$

Thus **the circle has centre $(2, -3)$ and radius 4**, as shown in Fig. 14.16.

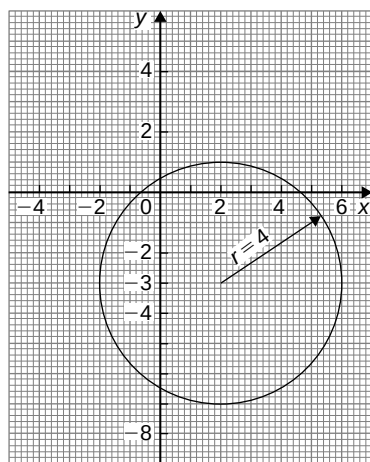


Figure 14.16

Alternatively, $x^2 + y^2 - 4x + 6y - 3 = 0$ may be re-arranged as:

$$(x - 2)^2 + (y + 3)^2 - 3 - 13 = 0$$

i.e. $(x - 2)^2 + (y + 3)^2 = 4^2$

which represents a circle, **centre $(2, -3)$ and radius 4**, as stated above.

Now try the following Practice Exercise

Practice Exercise 65 The equation of a circle (Answers on page 839)

1. Determine (a) the radius, and (b) the co-ordinates of the centre of the circle given by the equation $x^2 + y^2 + 6x - 2y - 26 = 0$
2. Sketch the circle given by the equation $x^2 + y^2 - 6x + 4y - 3 = 0$
3. Sketch the curve $x^2 + (y - 1)^2 - 25 = 0$
4. Sketch the curve $x = 6\sqrt{[1 - (y/6)^2]}$

14.6 Linear and angular velocity

Linear velocity

Linear velocity v is defined as the rate of change of linear displacement s with respect to time t . For motion in a straight line:

$$\text{linear velocity} = \frac{\text{change of displacement}}{\text{change of time}}$$

i.e. $v = \frac{s}{t}$ (1)

The unit of linear velocity is metres per second (m/s).

Angular velocity

The speed of revolution of a wheel or a shaft is usually measured in revolutions per minute or revolutions per second but these units do not form part of a coherent system of units. The basis in SI units is the angle turned through in one second.

Angular velocity is defined as the rate of change of angular displacement θ , with respect to time t . For an object rotating about a fixed axis at a constant speed:

$$\text{angular velocity} = \frac{\text{angle turned through}}{\text{time taken}}$$

$$\text{i.e.} \quad \omega = \frac{\theta}{t} \quad (2)$$

The unit of angular velocity is radians per second (rad/s). An object rotating at a constant speed of n revolutions per second subtends an angle of $2\pi n$ radians in one second, i.e. its angular velocity ω is given by:

$$\omega = 2\pi n \text{ rad/s} \quad (3)$$

From page 150, $s = r\theta$ and from equation (2) above, $\theta = \omega t$

$$\text{hence} \quad s = r(\omega t)$$

$$\text{from which} \quad \frac{s}{t} = \omega r$$

$$\text{However, from equation (1) } v = \frac{s}{t}$$

$$\text{hence} \quad v = \omega r \quad (4)$$

Equation (4) gives the relationship between linear velocity v and angular velocity ω .

Problem 18. A wheel of diameter 540 mm is rotating at $\frac{1500}{\pi}$ rev/min. Calculate the angular velocity of the wheel and the linear velocity of a point on the rim of the wheel.

From equation (3), angular velocity $\omega = 2\pi n$ where n is the speed of revolution in rev/s. Since in this case $n = \frac{1500}{\pi}$ rev/min = $\frac{1500}{60\pi}$ rev/s, then

$$\text{angular velocity } \omega = 2\pi \left(\frac{1500}{60\pi} \right) = 50 \text{ rad/s}$$

The linear velocity of a point on the rim, $v = \omega r$, where r is the radius of the wheel, i.e.

$$\frac{540}{2} \text{ mm} = \frac{0.54}{2} \text{ m} = 0.27 \text{ m}$$

$$\text{Thus linear velocity } v = \omega r = (50)(0.27) = 13.5 \text{ m/s}$$

Problem 19. A car is travelling at 64.8 km/h and has wheels of diameter 600 mm.

- Find the angular velocity of the wheels in both rad/s and rev/min.
- If the speed remains constant for 1.44 km, determine the number of revolutions made by the wheel, assuming no slipping occurs.

$$\begin{aligned} \text{(a) Linear velocity } v &= 64.8 \text{ km/h} \\ &= 64.8 \frac{\text{km}}{\text{h}} \times 1000 \frac{\text{m}}{\text{km}} \times \frac{1}{3600} \frac{\text{h}}{\text{s}} = 18 \text{ m/s} \end{aligned}$$

$$\begin{aligned} \text{The radius of a wheel} &= \frac{600}{2} = 300 \text{ mm} \\ &= 0.3 \text{ m} \end{aligned}$$

From equation (5), $v = \omega r$, from which,

$$\begin{aligned} \text{angular velocity } \omega &= \frac{v}{r} = \frac{18}{0.3} \\ &= 60 \text{ rad/s} \end{aligned}$$

From equation (4), angular velocity, $\omega = 2\pi n$, where n is in rev/s.

$$\begin{aligned} \text{Hence angular speed } n &= \frac{\omega}{2\pi} = \frac{60}{2\pi} \text{ rev/s} \\ &= 60 \times \frac{60}{2\pi} \text{ rev/min} \\ &= 573 \text{ rev/min} \end{aligned}$$

- From equation (1), since $v = s/t$ then the time taken to travel 1.44 km, i.e. 1440 m at a constant speed of 18 m/s is given by:

$$\text{time } t = \frac{s}{v} = \frac{1440 \text{ m}}{18 \text{ m/s}} = 80 \text{ s}$$

Since a wheel is rotating at 573 rev/min, then in 80/60 minutes it makes

$$573 \text{ rev/min} \times \frac{80}{60} \text{ min} = 764 \text{ revolutions}$$

Now try the following Practice Exercise

Practice Exercise 66 Linear and angular velocity (Answers on page 839)

- A pulley driving a belt has a diameter of 300 mm and is turning at $2700/\pi$ revolutions per minute. Find the angular velocity of the pulley and the linear velocity of the belt assuming that no slip occurs.
- A bicycle is travelling at 36 km/h and the diameter of the wheels of the bicycle is 500 mm. Determine the linear velocity of a point on the rim of one of the wheels of the bicycle, and the angular velocity of the wheels.

3. A train is travelling at 108 km/h and has wheels of diameter 800 mm.
- Determine the angular velocity of the wheels in both rad/s and rev/min.
 - If the speed remains constant for 2.70 km, determine the number of revolutions made by a wheel, assuming no slipping occurs.

14.7 Centripetal force

When an object moves in a circular path at constant speed, its direction of motion is continually changing and hence its velocity (which depends on both magnitude and direction) is also continually changing. Since acceleration is the (change in velocity)/(time taken), the object has an acceleration. Let the object be moving with a constant angular velocity of ω and a tangential velocity of magnitude v and let the change of velocity for a small change of angle of θ ($=\omega t$) be V in Fig. 14.17. Then $v_2 - v_1 = V$. The vector diagram is shown in Fig. 14.17(b) and since the magnitudes of v_1 and v_2 are the same, i.e. v , the vector diagram is an isosceles triangle.

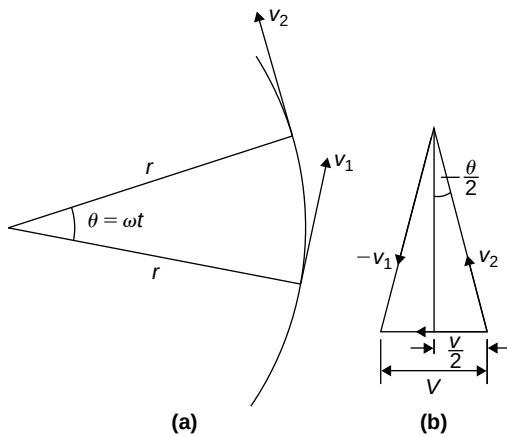


Figure 14.17

Bisecting the angle between v_2 and v_1 gives:

$$\sin \frac{\theta}{2} = \frac{V/2}{v} = \frac{V}{2v}$$

$$\text{i.e. } V = 2v \sin \frac{\theta}{2} \quad (1)$$

Since $\theta = \omega t$ then

$$t = \frac{\theta}{\omega} \quad (2)$$

Dividing equation (1) by equation (2) gives:

$$\frac{V}{t} = \frac{2v \sin(\theta/2)}{(\theta/\omega)} = \frac{v\omega \sin(\theta/2)}{(\theta/2)}$$

For small angles $\frac{\sin(\theta/2)}{(\theta/2)} \approx 1$,

$$\begin{aligned} \text{hence } \frac{V}{t} &= \frac{\text{change of velocity}}{\text{change of time}} \\ &= \text{acceleration } a = v\omega \end{aligned}$$

However, $\omega = \frac{v}{r}$ (from Section 14.6)

$$\text{thus } v\omega = v \cdot \frac{v}{r} = \frac{v^2}{r}$$

i.e. the acceleration a is $\frac{v^2}{r}$ and is towards the centre of the circle of motion (along V). It is called the **centripetal acceleration**. If the mass of the rotating object is m , then by Newton's second law, the **centripetal force** is $\frac{mv^2}{r}$ and its direction is towards the centre of the circle of motion.

Problem 20. A vehicle of mass 750 kg travels around a bend of radius 150 m, at 50.4 km/h. Determine the centripetal force acting on the vehicle.

The centripetal force is given by $\frac{mv^2}{r}$ and its direction is towards the centre of the circle.

$$\begin{aligned} \text{Mass } m &= 750 \text{ kg, } v = 50.4 \text{ km/h} \\ &= \frac{50.4 \times 1000}{60 \times 60} \text{ m/s} \\ &= 14 \text{ m/s} \end{aligned}$$

and radius $r = 150 \text{ m}$,

$$\text{thus centripetal force} = \frac{750(14)^2}{150} = 980 \text{ N}$$

Problem 21. An object is suspended by a thread 250 mm long and both object and thread move in a horizontal circle with a constant angular velocity of 2.0 rad/s. If the tension in the thread is 12.5 N, determine the mass of the object.

Centripetal force (i.e. tension in thread),

$$F = \frac{mv^2}{r} = 12.5 \text{ N}$$

Angular velocity $\omega = 2.0 \text{ rad/s}$ and
radius $r = 250 \text{ mm} = 0.25 \text{ m}$.

Since linear velocity $v = \omega r$, $v = (2.0)(0.25)$
 $= 0.5 \text{ m/s}$.

Since $F = \frac{mv^2}{r}$, then mass $m = \frac{Fr}{v^2}$,

i.e. mass of object, $m = \frac{(12.5)(0.25)}{0.5^2} = 12.5 \text{ kg}$

Problem 22. An aircraft is turning at constant altitude, the turn following the arc of a circle of radius 1.5 km. If the maximum allowable acceleration of the aircraft is $2.5g$, determine the maximum speed of the turn in km/h. Take g as 9.8 m/s^2 .

The acceleration of an object turning in a circle is $\frac{v^2}{r}$. Thus, to determine the maximum speed of turn, $\frac{v^2}{r} = 2.5g$, from which,

$$\begin{aligned}\text{velocity, } v &= \sqrt{(2.5gr)} = \sqrt{(2.5)(9.8)(1500)} \\ &= \sqrt{36\,750} = 191.7 \text{ m/s}\end{aligned}$$

$$\text{and } 191.7 \text{ m/s} = 191.7 \times \frac{60 \times 60}{1000} \text{ km/h} = 690 \text{ km/h}$$

Now try the following Practice Exercise

Practice Exercise 67 Centripetal force
(Answers on page 839)

1. Calculate the tension in a string when it is used to whirl a stone of mass 200 g round in a horizontal circle of radius 90 cm with a constant speed of 3 m/s.
2. Calculate the centripetal force acting on a vehicle of mass 1 tonne when travelling around a bend of radius 125 m at 40 km/h. If this force should not exceed 750 N, determine the reduction in speed of the vehicle to meet this requirement.
3. A speed-boat negotiates an S-bend consisting of two circular arcs of radii 100 m and 150 m. If the speed of the boat is constant at 34 km/h, determine the change in acceleration when leaving one arc and entering the other.

For fully worked solutions to each of the problems in Practice Exercises 62 to 67 in this chapter,
go to the website:

www.routledge.com/cw/bird



Revision Test 4 Trigonometry, Cartesian and polar co-ordinates and properties of circles

This Revision Test covers the material contained in Chapters 12 to 14. The marks for each question are shown in brackets at the end of each question.

1. A 2.0m long ladder is placed against a perpendicular pylon with its foot 52 cm from the pylon.
(a) Find how far up the pylon (correct to the nearest mm) the ladder reaches. (b) If the foot of the ladder is moved 10 cm towards the pylon, how far does the top of the ladder rise? (7)

2. Evaluate correct to 4 significant figures:

(a) $\cos 124^\circ 13'$ (b) $\cot 72.68^\circ$ (4)

3. From a point on horizontal ground a surveyor measures the angle of elevation of a church spire as 15° . He moves 30m nearer to the church and measures the angle of elevation as 20° . Calculate the height of the spire. (9)

4. If $\sec \theta = 2.4613$, determine the acute angle θ (4)

5. Evaluate, correct to 3 significant figures:

$$\frac{3.5 \operatorname{cosec} 31^\circ 17' - \cot(-12^\circ)}{3 \sec 79^\circ 41'} \quad (5)$$

6. A man leaves a point walking at 6.5km/h in a direction E 20° N (i.e. a bearing of 70°). A cyclist leaves the same point at the same time in a direction E 40° S (i.e. a bearing of 130°) travelling at a constant speed. Find the average speed of the cyclist if the walker and cyclist are 80km apart after five hours. (8)

7. A crank mechanism shown in Fig. RT4.1 comprises arm OP , of length 0.90m, which rotates anticlockwise about the fixed point O , and connecting rod PQ of length 4.20m. End Q moves horizontally in a straight line OR .

- (a) If $\angle POR$ is initially zero, how far does end Q travel in $\frac{1}{4}$ revolution.
(b) If $\angle POR$ is initially 40° find the angle between the connecting rod and the horizontal and the length OQ .

- (c) Find the distance Q moves (correct to the nearest cm) when $\angle POR$ changes from 40° to 140° . (16)

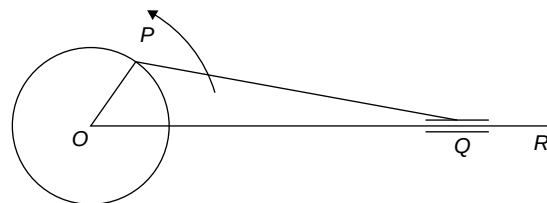


Figure RT4.1

8. Change the following Cartesian co-ordinates into polar co-ordinates, correct to 2 decimal places, in both degrees and in radians:

(a) $(-2.3, 5.4)$ (b) $(7.6, -9.2)$ (10)

9. Change the following polar co-ordinates into Cartesian co-ordinates, correct to 3 decimal places: (a) $(6.5, 132^\circ)$ (b) $(3, 3 \text{ rad})$ (6)

10. (a) Convert 2.154 radians into degrees and minutes.

(b) Change $71^\circ 17'$ into radians. (4)

11. 140mm of a belt drive is in contact with a pulley of diameter 180mm which is turning at 300 revolutions per minute. Determine (a) the angle of lap, (b) the angular velocity of the pulley, and (c) the linear velocity of the belt, assuming that no slipping occurs. (9)

12. Figure RT4.2 shows a cross-section through a circular water container where the shaded area represents the water in the container. Determine: (a) the depth, h , (b) the area of the shaded portion, and (c) the area of the unshaded area. (11)

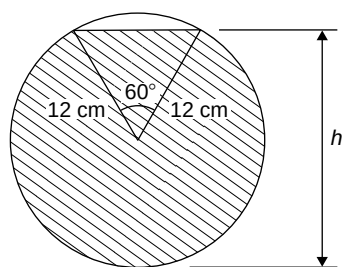


Figure RT4.2

13. Determine (a) the co-ordinates of the centre of the circle and (b) the radius, given the equation

$$x^2 + y^2 - 2x + 6y + 6 = 0 \quad (7)$$

For lecturers/instructors/teachers, fully worked solutions to each of the problems in Revision Test 4, together with a full marking scheme, are available at the website:

www.routledge.com/cw/bird

Chapter 15

Trigonometric waveforms

Why it is important to understand: Trigonometric waveforms

Trigonometric graphs are commonly used in all areas of science and engineering for modelling many different natural and mechanical phenomena such as waves, engines, acoustics, electronics, populations, UV intensity, growth of plants and animals, and so on. Periodic trigonometric graphs mean that the shape repeats itself exactly after a certain amount of time. Anything that has a regular cycle, like the tides, temperatures, rotation of the Earth, and so on, can be modelled using a sine or cosine curve. The most common periodic signal waveform that is used in electrical and electronic engineering is the sinusoidal waveform. However, an alternating a.c. waveform may not always take a smooth shape based around the sine and cosine function; a.c. waveforms can also take the shape of square or triangular waves, i.e. complex waves. In engineering, it is therefore important to have a clear understanding of sine and cosine waveforms.

At the end of this chapter, you should be able to:

- sketch sine, cosine and tangent waveforms
- determine angles of any magnitude
- understand cycle, amplitude, period, periodic time, frequency, lagging/leading angles with reference to sine and cosine waves
- perform calculations involving sinusoidal form $A\sin(\omega t \pm \alpha)$
- define a complex wave and harmonic analysis
- use harmonic synthesis to construct a complex waveform

15.1 Graphs of trigonometric functions

By drawing up tables of values from 0° to 360° , graphs of $y = \sin A$, $y = \cos A$ and $y = \tan A$ may be plotted. Values obtained with a calculator (correct to 3 decimal places – which is more than sufficient for plotting graphs), using 30° intervals, are shown below, with the respective graphs shown in Fig. 15.1.

(a) $y = \sin A$

A	0	30°	60°	90°	120°	150°	180°
$\sin A$	0	0.500	0.866	1.000	0.866	0.500	0

A	210°	240°	270°	300°	330°	360°
$\sin A$	-0.500	-0.866	-1.000	-0.866	-0.500	0

(b) $y = \cos A$

A	0	30°	60°	90°	120°	150°	180°
cos A	1.000	0.866	0.500	0	-0.500	-0.866	-1.000

A	210°	240°	270°	300°	330°	360°
cos A	-0.866	-0.500	0	0.500	0.866	1.000

(c) $y = \tan A$

A	0	30°	60°	90°	120°	150°	180°
tan A	0	0.577	1.732	∞	-1.732	-0.577	0

A	210°	240°	270°	300°	330°	360°
tan A	0.577	1.732	∞	-1.732	-0.577	0

From Fig. 15.1 it is seen that:

- Sine and cosine graphs oscillate between peak values of ± 1 .
- The cosine curve is the same shape as the sine curve but displaced by 90° .

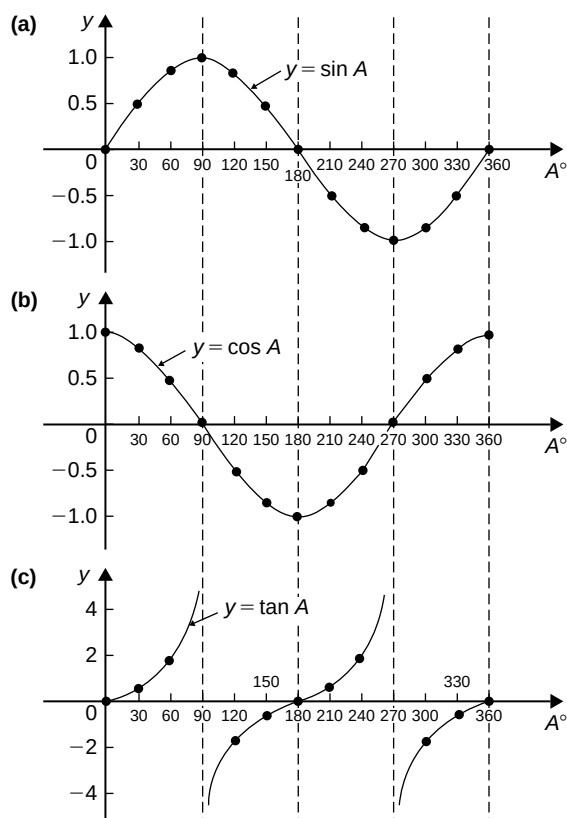


Figure 15.1

- The sine and cosine curves are continuous and they repeat at intervals of 360° ; the tangent curve appears to be discontinuous and repeats at intervals of 180° .

15.2 Angles of any magnitude

- Fig. 15.2 shows rectangular axes XX' and YY' intersecting at origin O . As with graphical work, measurements made to the right and above O are positive while those to the left and downwards are negative. Let OA be free to rotate about O . By convention, when OA moves anticlockwise angular measurement is considered positive, and vice-versa.

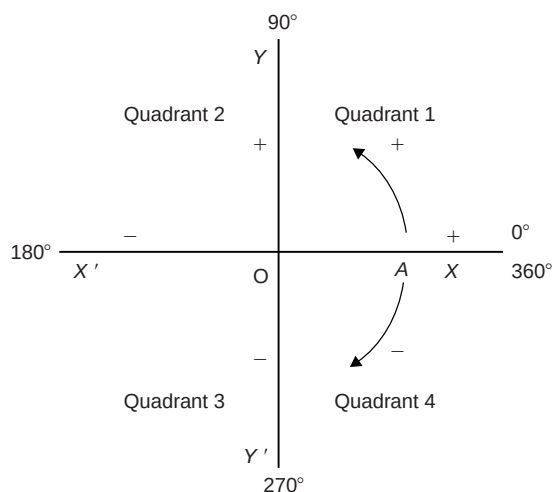


Figure 15.2

- Let OA be rotated anticlockwise so that θ_1 is any angle in the first quadrant and let perpendicular AB be constructed to form the right-angled triangle OAB (see Fig. 15.3). Since all three sides of the triangle are positive, all six trigonometric ratios are positive in the first quadrant. (Note: OA is always positive since it is the radius of a circle.)
- Let OA be further rotated so that θ_2 is any angle in the second quadrant and let AC be constructed to form the right-angled triangle OAC . Then:

$$\begin{aligned} \sin \theta_2 &= \frac{+}{+} = + & \cos \theta_2 &= \frac{-}{+} = - \\ \tan \theta_2 &= \frac{+}{-} = - & \operatorname{cosec} \theta_2 &= \frac{+}{+} = + \\ \sec \theta_2 &= \frac{+}{-} = - & \cot \theta_2 &= \frac{-}{+} = - \end{aligned}$$

- Let OA be further rotated so that θ_3 is any angle in the third quadrant and let AD be constructed to form the right-angled triangle OAD . Then:

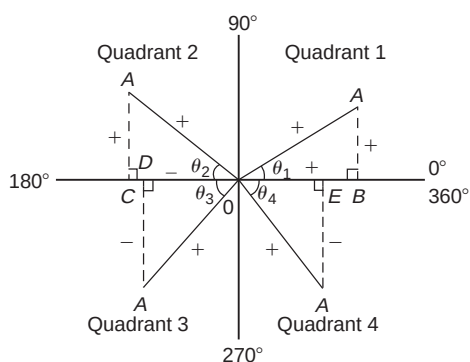


Figure 15.3

$$\sin \theta_3 = \frac{-}{+} = - \text{ (and hence cosec } \theta_3 \text{ is -)}$$

$$\cos \theta_3 = \frac{-}{+} = - \text{ (and hence sec } \theta_3 \text{ is +)}$$

$$\tan \theta_3 = \frac{-}{-} = + \text{ (and hence cot } \theta_3 \text{ is -)}$$

- (v) Let OA be further rotated so that θ_4 is any angle in the fourth quadrant and let AE be constructed to form the right-angled triangle OAE . Then:

$$\sin \theta_4 = \frac{-}{+} = - \text{ (and hence cosec } \theta_4 \text{ is -)}$$

$$\cos \theta_4 = \frac{+}{+} = + \text{ (and hence sec } \theta_4 \text{ is +)}$$

$$\tan \theta_4 = \frac{-}{+} = - \text{ (and hence cot } \theta_4 \text{ is -)}$$

- (vi) The results obtained in (ii) to (v) are summarised in Fig. 15.4. The letters underlined spell the word CAST when starting in the fourth quadrant and moving in an anticlockwise direction.
- (vii) In the first quadrant of Fig. 15.1 all the curves have positive values; in the second only sine is positive; in the third only tangent is positive; in the fourth only cosine is positive (exactly as summarised in Fig. 15.4).

A knowledge of angles of any magnitude is needed when finding, for example, all the angles between 0° and 360° whose sine is, say, 0.3261. If 0.3261 is entered into a calculator and then the inverse sine key pressed (or $\sin^{-1}$ key) the answer 19.03° appears. However, there is a second angle between 0° and 360° which the calculator does not give. Sine is also positive in the second quadrant (either from CAST or from Fig. 15.1(a)). The other angle is shown in Fig. 15.5 as angle θ where $\theta = 180^\circ - 19.03^\circ = 160.97^\circ$. Thus 19.03° and 160.97° are the angles between 0° and 360° whose sine is 0.3261 (check that $\sin 160.97^\circ = 0.3261$ on your calculator).

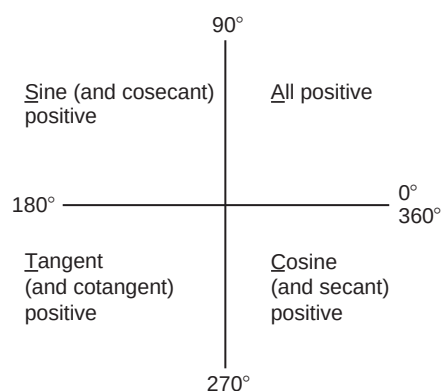


Figure 15.4

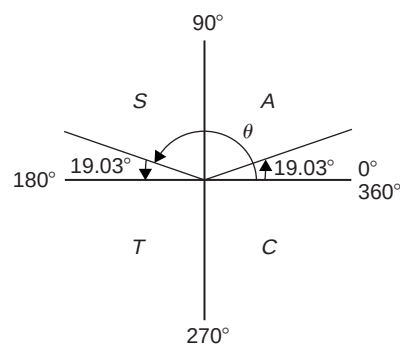


Figure 15.5

Be careful! Your calculator only gives you one of these answers. The second answer needs to be deduced from a knowledge of angles of any magnitude, as shown in the following problems.

Problem 1. Determine all the angles between 0° and 360° whose sine is -0.4638

The angles whose sine is -0.4638 occurs in the third and fourth quadrants since sine is negative in these quadrants (see Fig. 15.6(a)). From Fig. 15.6(b), $\theta = \sin^{-1} 0.4638 = 27^\circ 38'$

Measured from 0° , the two angles between 0° and 360° whose sine is -0.4638 are $180^\circ + 27^\circ 38'$, i.e. $207^\circ 38'$ and $360^\circ - 27^\circ 38'$, i.e. $332^\circ 22'$. (Note that a calculator generally only gives one answer, i.e. -27.632588°)

Problem 2. Determine all the angles between 0° and 360° whose tangent is 1.7629

A tangent is positive in the first and third quadrants (see Fig. 15.7(a)). From Fig. 15.7(b), $\theta = \tan^{-1} 1.7629 = 60^\circ 26'$. Measured from 0° , the two

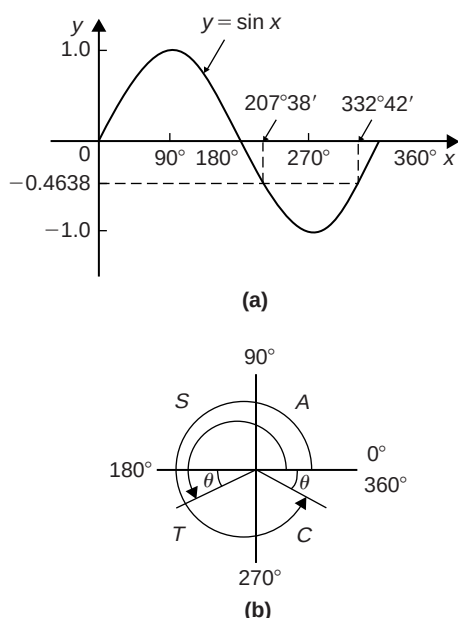


Figure 15.6

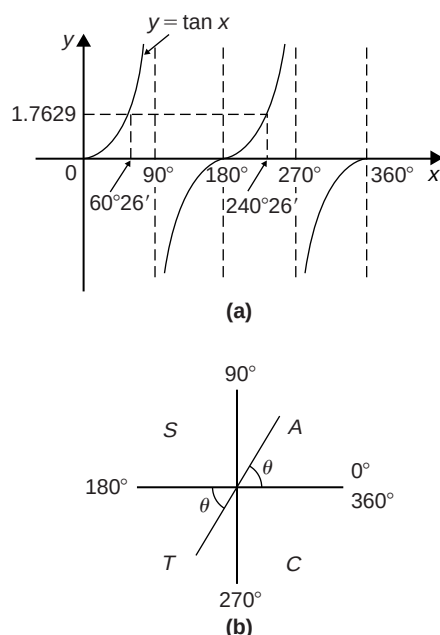


Figure 15.7

angles between 0° and 360° whose tangent is 1.7629 are $60^\circ 26'$ and $180^\circ + 60^\circ 26'$, i.e. $240^\circ 26'$

Problem 3. Solve $\sec^{-1}(-2.1499) = \alpha$ for angles of α between 0° and 360° .

Secant is negative in the second and third quadrants (i.e. the same as for cosine). From Fig. 15.8,

$$\theta = \sec^{-1} 2.1499 = \cos^{-1} \left(\frac{1}{2.1499} \right) = 62^\circ 17'$$

Measured from 0° , the two angles between 0° and 360° whose secant is -2.1499 are

$$\alpha = 180^\circ - 62^\circ 17' = 117^\circ 43' \quad \text{and}$$

$$\alpha = 180^\circ + 62^\circ 17' = 242^\circ 17'$$

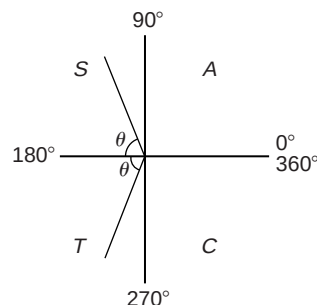


Figure 15.8

Problem 4. Solve $\cot^{-1} 1.3111 = \alpha$ for angles of α between 0° and 360° .

Cotangent is positive in the first and third quadrants (i.e. same as for tangent). From Fig. 15.9,

$$\theta = \cot^{-1} 1.3111 = \tan^{-1} \left(\frac{1}{1.3111} \right) = 37^\circ 20'$$

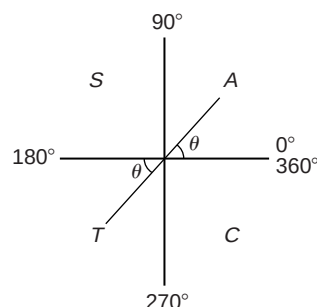


Figure 15.9

Hence $\alpha = 37^\circ 20'$

and $\alpha = 180^\circ + 37^\circ 20' = 217^\circ 20'$

Now try the following Practice Exercise

Practice Exercise 68 Evaluating trigonometric ratios of any magnitude (Answers on page 839)

- Find all the angles between 0° and 360° whose sine is -0.7321

2. Determine the angles between 0° and 360° whose cosecant is 2.5317
3. If cotangent $x = -0.6312$, determine the values of x in the range $0^\circ \leq x \leq 360^\circ$

In Problems 4 to 6 solve the given equations.

4. $\cos^{-1}(-0.5316) = t$
5. $\sec^{-1} 2.3162 = x$
6. $\tan^{-1} 0.8314 = \theta$

15.3 The production of a sine and cosine wave

In Fig. 15.10, let OR be a vector 1 unit long and free to rotate anticlockwise about O . In one revolution a circle is produced and is shown with 15° sectors. Each

radius arm has a vertical and a horizontal component. For example, at 30° , the vertical component is TS and the horizontal component is OS .

From trigonometric ratios,

$$\sin 30^\circ = \frac{TS}{TO} = \frac{TS}{1}, \text{ i.e. } TS = \sin 30^\circ$$

$$\text{and } \cos 30^\circ = \frac{OS}{TO} = \frac{OS}{1}, \text{ i.e. } OS = \cos 30^\circ$$

The vertical component TS may be projected across to $T'S'$, which is the corresponding value of 30° on the graph of y against angle x° . If all such vertical components as TS are projected onto the graph, then a **sine wave** is produced as shown in Fig. 15.10.

If all horizontal components such as OS are projected onto a graph of y against angle x° , then a **cosine wave** is produced. It is easier to visualise these projections by redrawing the circle with the radius arm OR initially in a vertical position, as shown in Fig. 15.11.

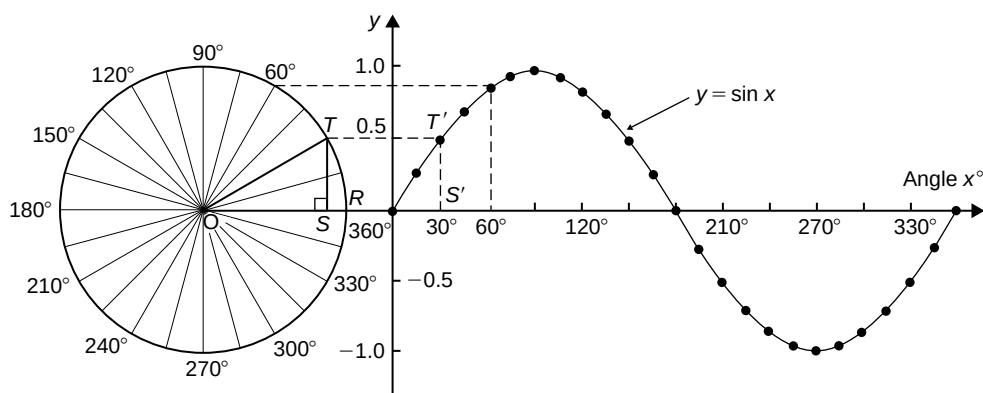


Figure 15.10

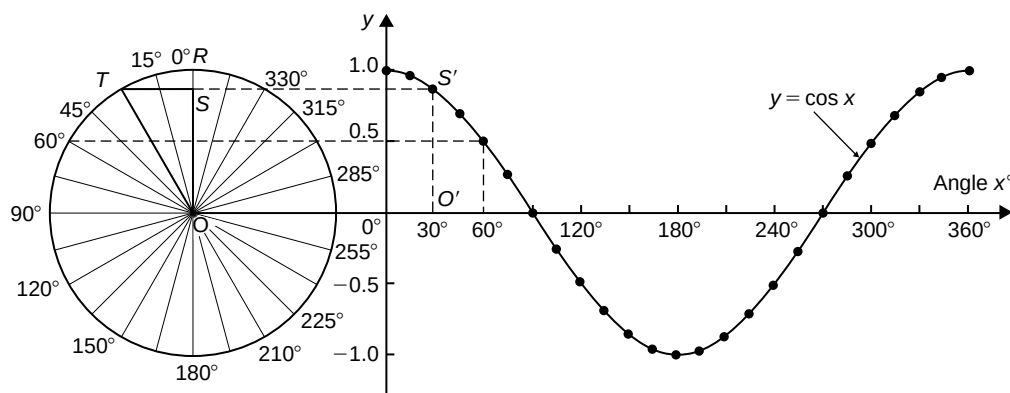


Figure 15.11

From Figs. 15.10 and 15.11 it is seen that a cosine curve is of the same form as the sine curve but is displaced by 90° (or $\pi/2$ radians).

15.4 Sine and cosine curves

Graphs of sine and cosine waveforms

(i) A graph of $y = \sin A$ is shown by the broken line in Fig. 15.12 and is obtained by drawing up a table of values as in Section 15.1. A similar table may be produced for $y = \sin 2A$.

A°	$2A$	$\sin 2A$	A°	$2A$	$\sin 2A$
0	0	0	225	450	1.0
30	60	0.866	240	480	0.866
45	90	1.0	270	540	0
60	120	0.866	300	600	-0.866
90	180	0	315	630	-1.0
120	240	-0.866	330	660	-0.866
135	270	-1.0	360	720	0
150	300	-0.866			
180	360	0			
210	420	0.866			

A graph of $y = \sin 2A$ is shown in Fig. 15.12.

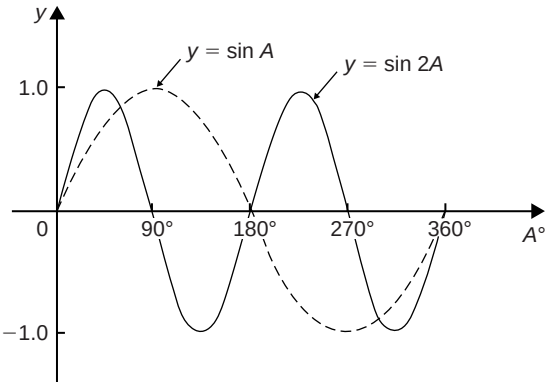


Figure 15.12

(ii) A graph of $y = \sin \frac{1}{2}A$ is shown in Fig. 15.13 using the following table of values.

A°	$\frac{1}{2}A$	$\sin \frac{1}{2}A$
0	0	0
30	15	0.259
60	30	0.500
90	45	0.707
120	60	0.866
150	75	0.966
180	90	1.00
210	105	0.966
240	120	0.866
270	135	0.707
300	150	0.500
330	165	0.259
360	180	0

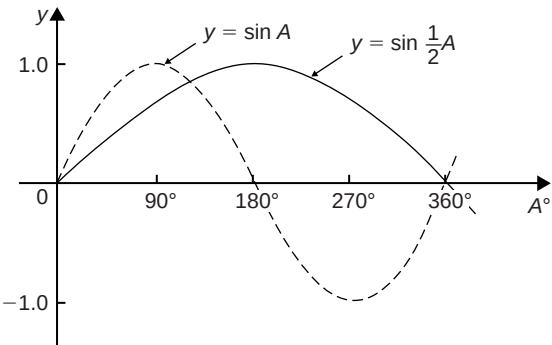


Figure 15.13

(iii) A graph of $y = \cos A$ is shown by the broken line in Fig. 15.14 and is obtained by drawing up a

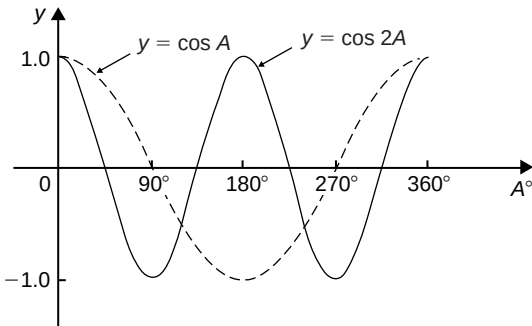


Figure 15.14

table of values. A similar table may be produced for $y = \cos 2A$ with the result as shown.

- (iv) A graph of $y = \cos \frac{1}{2}A$ is shown in Fig. 15.15, which may be produced by drawing up a table of values, similar to above.

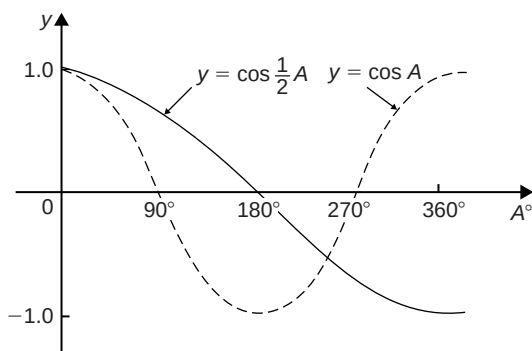


Figure 15.15

Periodic functions and period

- (i) Each of the graphs shown in Figs. 15.12 to 15.15 will repeat themselves as angle A increases and are thus called **periodic functions**.
- (ii) $y = \sin A$ and $y = \cos A$ repeat themselves every 360° (or 2π radians); thus 360° is called the **period** of these waveforms. $y = \sin 2A$ and $y = \cos 2A$ repeat themselves every 180° (or π radians); thus 180° is the period of these waveforms.
- (iii) In general, if $y = \sin pA$ or $y = \cos pA$ (where p is a constant) then the period of the waveform is $360^\circ/p$ (or $2\pi/p$ rad). Hence if $y = \sin 3A$ then the period is $360/3$, i.e. 120° , and if $y = \cos 4A$ then the period is $360/4$, i.e. 90° .

Amplitude

Amplitude is the name given to the maximum or peak value of a sine wave. Each of the graphs shown in Figs. 15.12 to 15.15 has an amplitude of +1 (i.e. they oscillate between +1 and -1). However, if $y = 4 \sin A$, each of the values in the table is multiplied by 4 and the maximum value, and thus amplitude, is 4. Similarly, if $y = 5 \cos 2A$, the amplitude is 5 and the period is $360^\circ/2$, i.e. 180° .

Problem 5. Sketch $y = \sin 3A$ between $A = 0^\circ$ and $A = 360^\circ$.

Amplitude = 1; period = $360^\circ/3 = 120^\circ$.

A sketch of $y = \sin 3A$ is shown in Fig. 15.16.

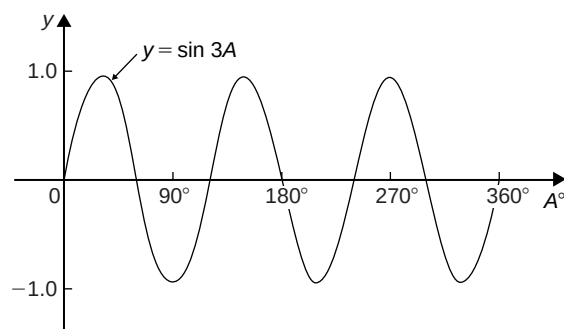


Figure 15.16

Problem 6. Sketch $y = 3 \sin 2A$ from $A = 0$ to $A = 2\pi$ radians.

Amplitude = 3; period = $2\pi/2 = \pi$ rads (or 180°)

A sketch of $y = 3 \sin 2A$ is shown in Fig. 15.17.

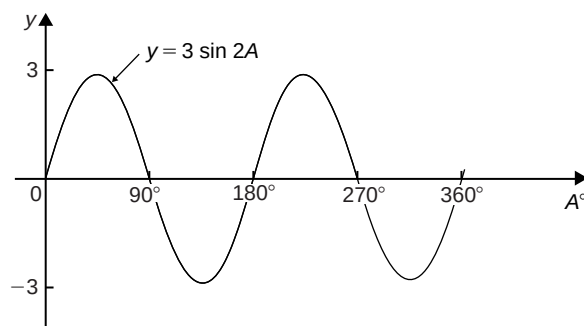


Figure 15.17

Problem 7. Sketch $y = 4 \cos 2x$ from $x = 0^\circ$ to $x = 360^\circ$.

Amplitude = 4; period = $360^\circ/2 = 180^\circ$

A sketch of $y = 4 \cos 2x$ is shown in Fig. 15.18.

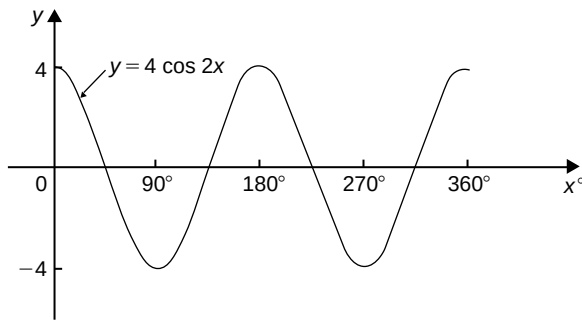


Figure 15.18

Problem 8. Sketch $y = 2 \sin \frac{3}{5} A$ over one cycle.

$$\text{Amplitude} = 2; \text{period} = \frac{360^\circ}{\frac{3}{5}} = \frac{360^\circ \times 5}{3} = 600^\circ$$

A sketch of $y = 2 \sin \frac{3}{5} A$ is shown in Fig. 15.19.

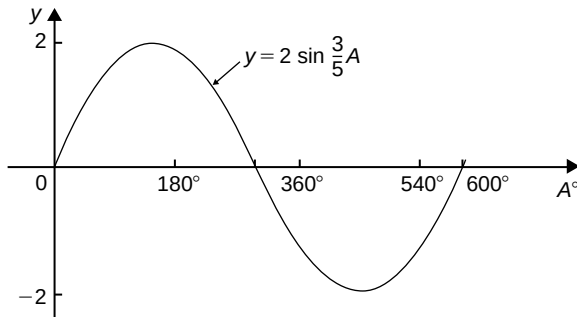


Figure 15.19

Lagging and leading angles

- (i) A sine or cosine curve may not always start at 0° . To show this a periodic function is represented by $y = \sin(A \pm \alpha)$ or $y = \cos(A \pm \alpha)$ where α is a phase displacement compared with $y = \sin A$ or $y = \cos A$.
- (ii) By drawing up a table of values, a graph of $y = \sin(A - 60^\circ)$ may be plotted as shown in Fig. 15.20. If $y = \sin A$ is assumed to start at 0° then $y = \sin(A - 60^\circ)$ starts 60° later (i.e. has a zero value 60° later). Thus $y = \sin(A - 60^\circ)$ is said to **lag** $y = \sin A$ by 60° .

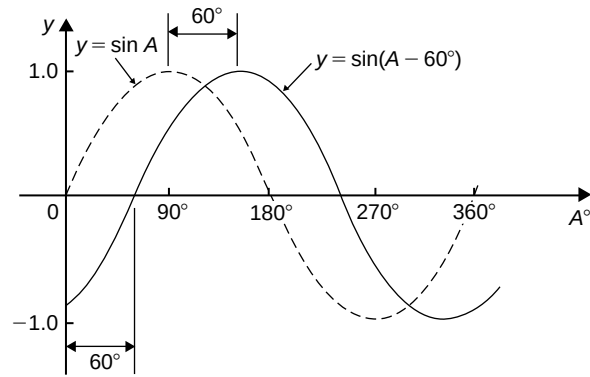


Figure 15.20

- (iii) By drawing up a table of values, a graph of $y = \cos(A + 45^\circ)$ may be plotted as shown in Fig. 15.21. If $y = \cos A$ is assumed to start at 0° then $y = \cos(A + 45^\circ)$ starts 45° earlier (i.e. has a zero value 45° earlier). Thus $y = \cos(A + 45^\circ)$ is said to **lead** $y = \cos A$ by 45° .

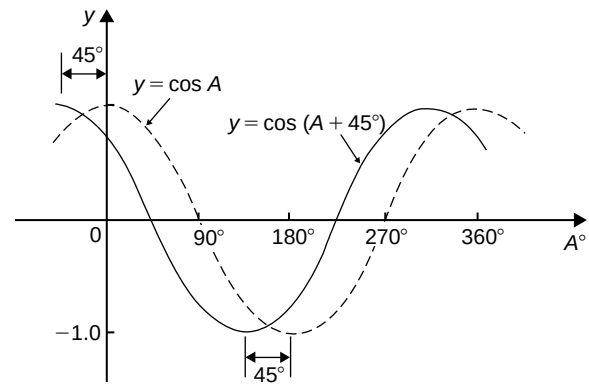


Figure 15.21

- (iv) Generally, a graph of $y = \sin(A - \alpha)$ lags $y = \sin A$ by angle α , and a graph of $y = \sin(A + \alpha)$ leads $y = \sin A$ by angle α .
- (v) A cosine curve is the same shape as a sine curve but starts 90° earlier, i.e. leads by 90° . Hence $\cos A = \sin(A + 90^\circ)$.

Problem 9. Sketch $y = 5 \sin(A + 30^\circ)$ from $A = 0^\circ$ to $A = 360^\circ$.

$$\text{Amplitude} = 5; \text{period} = 360^\circ / 1 = 360^\circ.$$

$5 \sin(A + 30^\circ)$ leads $5 \sin A$ by 30° (i.e. starts 30° earlier).

A sketch of $y = 5 \sin(A + 30^\circ)$ is shown in Fig. 15.22.

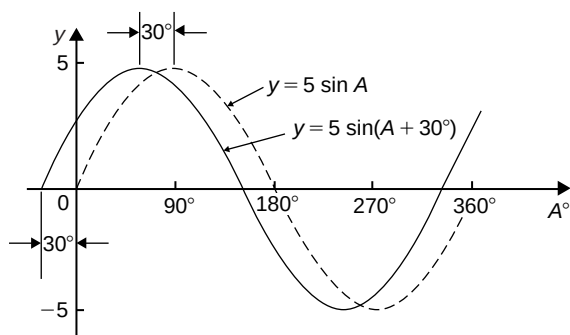


Figure 15.22

Problem 10. Sketch $y = 7 \sin(2A - \pi/3)$ in the range $0 \leq A \leq 2\pi$

Amplitude = 7; period = $2\pi/2 = \pi$ radians.

In general, $y = \sin(pt - \alpha)$ lags $y = \sin pt$ by α/p , hence $7 \sin(2A - \pi/3)$ lags $7 \sin 2A$ by $(\pi/3)/2$, i.e. $\pi/6$ rad or 30° .

A sketch of $y = 7 \sin(2A - \pi/3)$ is shown in Fig. 15.23.

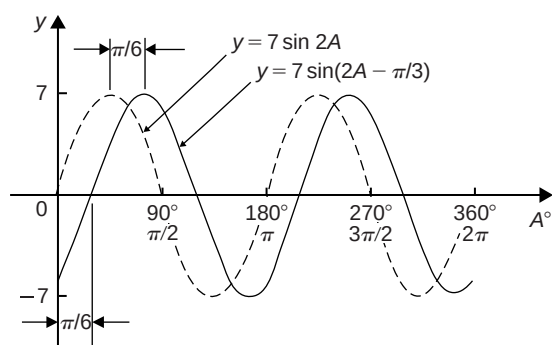


Figure 15.23

Problem 11. Sketch $y = 2 \cos(\omega t - 3\pi/10)$ over one cycle.

Amplitude = 2; period = $2\pi/\omega$ rad.

$2 \cos(\omega t - 3\pi/10)$ lags $2 \cos \omega t$ by $3\pi/10\omega$ seconds.

A sketch of $y = 2 \cos(\omega t - 3\pi/10)$ is shown in Fig. 15.24.

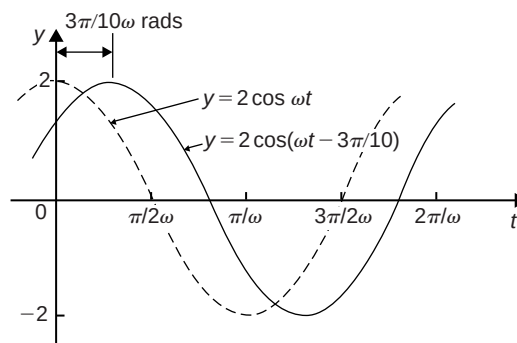


Figure 15.24

Graphs of $\sin^2 A$ and $\cos^2 A$

- (i) A graph of $y = \sin^2 A$ is shown in Fig. 15.25 using the following table of values.

A°	$\sin A$	$(\sin A)^2 = \sin^2 A$
0	0	0
30	0.50	0.25
60	0.866	0.75
90	1.0	1.0
120	0.866	0.75
150	0.50	0.25
180	0	0
210	-0.50	0.25
240	-0.866	0.75
270	-1.0	1.0
300	-0.866	0.75
330	-0.50	0.25
360	0	0

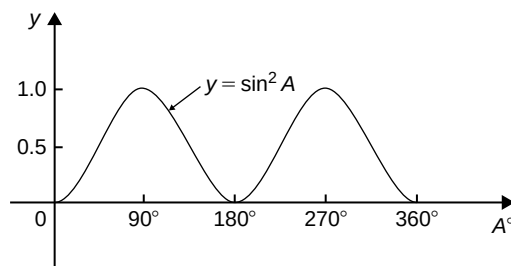


Figure 15.25

- (ii) A graph of $y = \cos^2 A$ is shown in Fig. 15.26, obtained by drawing up a table of values, similar to above.

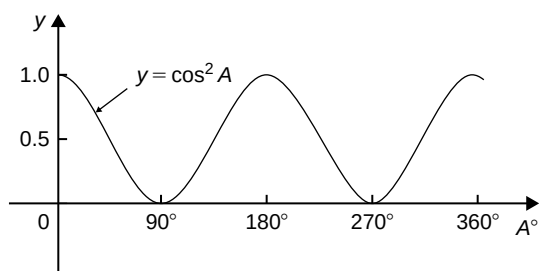


Figure 15.26

- (iii) $y = \sin^2 A$ and $y = \cos^2 A$ are both periodic functions of period 180° (or π rad) and both contain only positive values. Thus a graph of $y = \sin^2 2A$ has a period $180^\circ/2$, i.e. 90° . Similarly, a graph of $y = 4\cos^2 3A$ has a maximum value of 4 and a period of $180^\circ/3$, i.e. 60° .

Problem 12. Sketch $y = 3\sin^2 \frac{1}{2}A$ in the range $0 < A < 360^\circ$.

Maximum value = 3; period = $180^\circ / (1/2) = 360^\circ$
A sketch of $3\sin^2 \frac{1}{2}A$ is shown in Fig. 15.27.

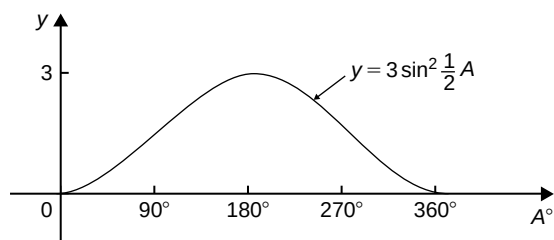


Figure 15.27

Problem 13. Sketch $y = 7\cos^2 2A$ between $A = 0^\circ$ and $A = 360^\circ$

Maximum value = 7; period = $180^\circ / 2 = 90^\circ$
A sketch of $y = 7\cos^2 2A$ is shown in Fig. 15.28.

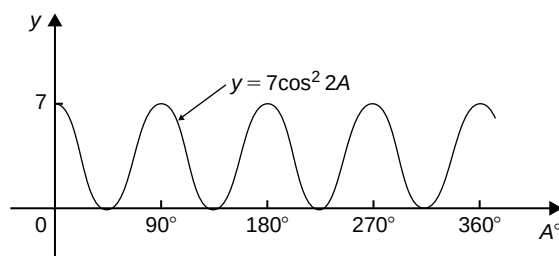


Figure 15.28

Now try the following Practice Exercise

Practice Exercise 69 Sine and cosine curves (Answers on page 839)

In Problems 1 to 9 state the amplitude and period of the waveform and sketch the curve between 0° and 360° .

1. $y = \cos 3A$
2. $y = 2\sin \frac{5x}{2}$
3. $y = 3\sin 4t$
4. $y = 3\cos \frac{\theta}{2}$
5. $y = \frac{7}{2}\sin \frac{3x}{8}$
6. $y = 6\sin(t - 45^\circ)$
7. $y = 4\cos(2\theta + 30^\circ)$
8. $y = 2\sin^2 2t$
9. $y = 5\cos^2 \frac{3}{2}\theta$

15.5 Sinusoidal form $A \sin(\omega t \pm \alpha)$

In Fig. 15.29, let OR represent a vector that is free to rotate anticlockwise about O at a velocity of ω rad/s. A rotating vector is called a **phasor**. After a time t seconds OR will have turned through an angle ωt radians (shown as angle TOR in Fig. 15.29). If ST is constructed perpendicular to OR , then $\sin \omega t = ST/TO$, i.e. $ST = TO \sin \omega t$.

If all such vertical components are projected onto a graph of y against ωt , a sine wave results of amplitude OR (as shown in Section 15.3).

If phasor OR makes one revolution (i.e. 2π radians) in T seconds, then the angular velocity,

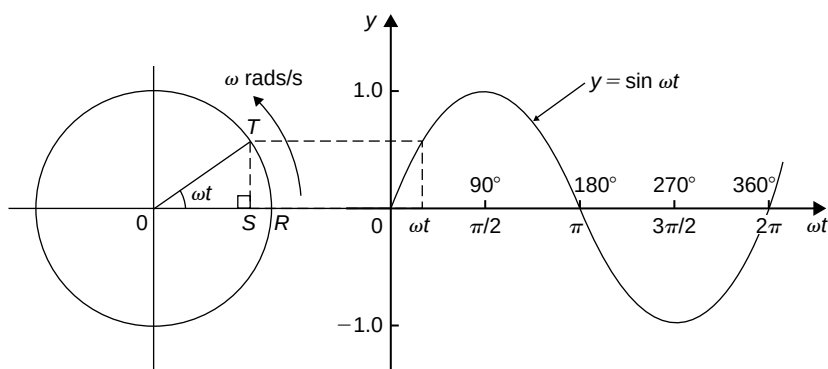


Figure 15.29

$\omega = 2\pi / T$ rad/s, from which, **$T = 2\pi / \omega$ seconds.**

T is known as the **periodic time**.

The number of complete cycles occurring per second is called the **frequency, f**

$$\begin{aligned} \text{Frequency} &= \frac{\text{number of cycles}}{\text{second}} = \frac{1}{T} \\ &= \frac{\omega}{2\pi} \text{ i.e. } \mathbf{f = \frac{\omega}{2\pi} \text{ Hz}} \end{aligned}$$

Hence **angular velocity, $\omega = 2\pi f$ rad/s**

Amplitude is the name given to the maximum or peak value of a sine wave, as explained in Section 15.3. The amplitude of the sine wave shown in Fig. 15.29 is 1.

A sine or cosine wave may not always start at 0° . To show this a periodic function is represented by $y = \sin(\omega t \pm \alpha)$ or $y = \cos(\omega t \pm \alpha)$, where α is a phase displacement compared with $y = \sin A$ or $y = \cos A$. A graph of $y = \sin(\omega t - \alpha)$ **lags** $y = \sin \omega t$ by angle α , and a graph of $y = \sin(\omega t + \alpha)$ **leads** $y = \sin \omega t$ by angle α .

The angle ωt is measured in **radians** (i.e. $\left(\omega \frac{\text{rad}}{\text{s}}\right)(\text{ts}) = \omega t$ radians) hence angle α should also be in radians.

The relationship between degrees and radians is:

$$360^\circ = 2\pi \text{ radians or } \mathbf{180^\circ = \pi \text{ radians}}$$

Hence $1 \text{ rad} = \frac{180}{\pi} = 57.30^\circ$ and, for example,

$$71^\circ = 71 \times \frac{\pi}{180} = 1.239 \text{ rad.}$$

Given a general sinusoidal function

$y = A \sin(\omega t \pm \alpha)$, then

(i) A = amplitude

(ii) ω = angular velocity = $2\pi f$ rad/s

(iii) $\frac{2\pi}{\omega}$ = periodic time T seconds

(iv) $\frac{\omega}{2\pi}$ = frequency, f hertz

(v) α = angle of lead or lag (compared with $y = A \sin \omega t$)

Problem 14. An alternating current is given by $i = 30 \sin(100\pi t + 0.27)$ amperes. Find the amplitude, periodic time, frequency and phase angle (in degrees and minutes).

$i = 30 \sin(100\pi t + 0.27)$ A, hence **amplitude = 30 A**
Angular velocity $\omega = 100\pi$, hence

$$\begin{aligned} \text{periodic time, } T &= \frac{2\pi}{\omega} = \frac{2\pi}{100\pi} = \frac{1}{50} \\ &= \mathbf{0.02 \text{ s or } 20 \text{ ms}} \end{aligned}$$

$$\text{Frequency, } f = \frac{1}{T} = \frac{1}{0.02} = \mathbf{50 \text{ Hz}}$$

$$\begin{aligned} \text{Phase angle, } \alpha &= 0.27 \text{ rad} = \left(0.27 \times \frac{180}{\pi}\right)^\circ \\ &= \mathbf{15.47^\circ \text{ or } 15^\circ 28' \text{ leading}} \end{aligned}$$

$$\mathbf{i = 30 \sin(100\pi t)}$$

Problem 15. An oscillating mechanism has a maximum displacement of 2.5 m and a frequency of 60 Hz. At time $t = 0$ the displacement is 90 cm. Express the displacement in the general form $A \sin(\omega t \pm \alpha)$.

Amplitude = maximum displacement = 2.5 m.
 Angular velocity, $\omega = 2\pi f = 2\pi(60) = 120\pi$ rad/s.
 Hence displacement = $2.5 \sin(120\pi t + \alpha)$ m.
 When $t = 0$, displacement = 90 cm = 0.90 m.

Hence $0.90 = 2.5 \sin(0 + \alpha)$

i.e. $\sin \alpha = \frac{0.90}{2.5} = 0.36$

Hence $\alpha = \arcsin 0.36 = 21.10^\circ = 21^\circ 6'$
 $= 0.368$ rad

Thus **displacement = $2.5 \sin(120\pi t + 0.368)$ m**

Problem 16. The instantaneous value of voltage in an a.c. circuit at any time t seconds is given by $v = 340 \sin(50\pi t - 0.541)$ volts. Determine:

- the amplitude, periodic time, frequency and phase angle (in degrees)
 - the value of the voltage when $t = 0$
 - the value of the voltage when $t = 10$ ms
 - the time when the voltage first reaches 200 V
 - the time when the voltage is a maximum.
- Sketch one cycle of the waveform.

- (a) **Amplitude = 340 V**
 Angular velocity, $\omega = 50\pi$

Hence **periodic time, $T = \frac{2\pi}{\omega} = \frac{2\pi}{50\pi} = \frac{1}{25}$**
= 0.04 s or 40 ms

Frequency, $f = \frac{1}{T} = \frac{1}{0.04} = 25$ Hz

Phase angle = 0.541 rad = $\left(0.541 \times \frac{180}{\pi}\right)$
= 31° lagging $v = 340 \sin(50\pi t)$

- (b) **When $t = 0$,**

$v = 340 \sin(0 - 0.541) = 340 \sin(-31^\circ)$
= -175.1 V

- (c) **When $t = 10$ ms**

then $v = 340 \sin\left(50\pi \frac{10}{10^3} - 0.541\right)$
 $= 340 \sin(1.0298) = 340 \sin 59^\circ$
= 291.4 V

- (d) **When $v = 200$ volts**

then $200 = 340 \sin(50\pi t - 0.541)$

$\frac{200}{340} = \sin(50\pi t - 0.541)$

Hence $(50\pi t - 0.541) = \sin^{-1} \frac{200}{340}$
 $= 36.03^\circ$ or 0.6288 rad
 $50\pi t = 0.6288 + 0.541$
 $= 1.1698$

Hence when $v = 200$ V,

time, $t = \frac{1.1698}{50\pi} = 7.447$ ms

- (e) **When the voltage is a maximum, $v = 340$ V.**

Hence $340 = 340 \sin(50\pi t - 0.541)$
 $1 = \sin(50\pi t - 0.541)$
 $50\pi t - 0.541 = \sin^{-1} 1$
 $= 90^\circ$ or 1.5708 rad
 $50\pi t = 1.5708 + 0.541 = 2.1118$
 Hence time, $t = \frac{2.1118}{50\pi} = 13.44$ ms

A sketch of $v = 340 \sin(50\pi t - 0.541)$ volts is shown in Fig. 15.30.

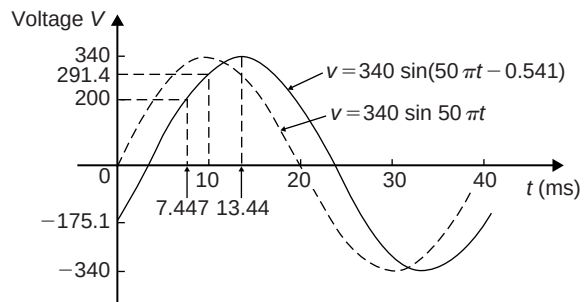


Figure 15.30

Now try the following Practice Exercise**Practice Exercise 70 The sinusoidal form $A \sin(\omega t \pm \alpha)$ (Answers on page 839)**

In Problems 1 to 3, find (a) the amplitude, (b) the frequency, (c) the periodic time, and (d) the phase angle (stating whether it is leading or lagging $A \sin \omega t$) of the alternating quantities given.

1. $i = 40 \sin(50\pi t + 0.29)$ mA
2. $y = 75 \sin(40t - 0.54)$ cm
3. $v = 300 \sin(200\pi t - 0.412)$ V
4. A sinusoidal voltage has a maximum value of 120 V and a frequency of 50 Hz. At time $t = 0$, the voltage is (a) zero and (b) 50 V. Express the instantaneous voltage v in the form $v = A \sin(\omega t \pm \alpha)$
5. An alternating current has a periodic time of 25 ms and a maximum value of 20 A. When time $t = 0$, current $i = -10$ amperes. Express the current i in the form $i = A \sin(\omega t \pm \alpha)$
6. An oscillating mechanism has a maximum displacement of 3.2 m and a frequency of 50 Hz. At time $t = 0$ the displacement is 150 cm. Express the displacement in the general form $A \sin(\omega t \pm \alpha)$
7. The current in an a.c. circuit at any time t seconds is given by:

$$i = 5 \sin(100\pi t - 0.432) \text{ amperes}$$

Determine (a) the amplitude, frequency, periodic time and phase angle (in degrees), (b) the value of current at $t = 0$, (c) the value of current at $t = 8$ ms, (d) the time when the current is first a maximum, (e) the time when the current first reaches 3 A. Sketch one cycle of the waveform showing relevant points.

15.6 Harmonic synthesis with complex waveforms

A waveform that is not sinusoidal is called a **complex wave**. **Harmonic analysis** is the process of resolving a complex periodic waveform into a series of sinusoidal

components of ascending order of frequency. Many of the waveforms met in practice can be represented by the following mathematical expression.

$$v = V_{1m} \sin(\omega t + \alpha_1) + V_{2m} \sin(2\omega t + \alpha_2) + \dots + V_{nm} \sin(n\omega t + \alpha_n)$$

and the magnitude of their harmonic components together with their phase may be calculated using **Fourier series** (see Chapters 73 to 76). **Numerical methods** are used to analyse waveforms for which simple mathematical expressions cannot be obtained. A numerical method of harmonic analysis is explained in Chapter 77 on page 791. In a laboratory, waveform analysis may be performed using a **waveform analyser** which produces a direct readout of the component waves present in a complex wave.

By adding the instantaneous values of the fundamental and progressive harmonics of a complex wave for given instants in time, the shape of a complex waveform can be gradually built up. This graphical procedure is known as **harmonic synthesis** (synthesis meaning ‘the putting together of parts or elements so as to make up a complex whole’). For more on the addition of alternating waveforms, see Chapter 27.

Some examples of harmonic synthesis are considered in the following worked problems.

Problem 17. Use harmonic synthesis to construct the complex voltage given by:

$$v_1 = 100 \sin \omega t + 30 \sin 3\omega t \text{ volts.}$$

The waveform is made up of a fundamental wave of maximum value 100 V and frequency, $f = \omega/2\pi$ hertz and a third harmonic component of maximum value 30 V and frequency $= 3\omega/2\pi (= 3f)$, the fundamental and third harmonics being initially in phase with each other.

In Fig. 15.31, the fundamental waveform is shown by the broken line plotted over one cycle, the periodic time T being $2\pi/\omega$ seconds. On the same axis is plotted $30 \sin 3\omega t$, shown by the dotted line, having a maximum value of 30 V and for which three cycles are completed in time T seconds. At zero time, $30 \sin 3\omega t$ is in phase with $100 \sin \omega t$.

The fundamental and third harmonic are combined by adding ordinates at intervals to produce the waveform for v_1 , as shown. For example, at time $T/12$ seconds, the fundamental has a value of 50 V and the third harmonic a value of 30 V. Adding gives a value of 80 V for waveform v_1 at time $T/12$ seconds. Similarly, at time

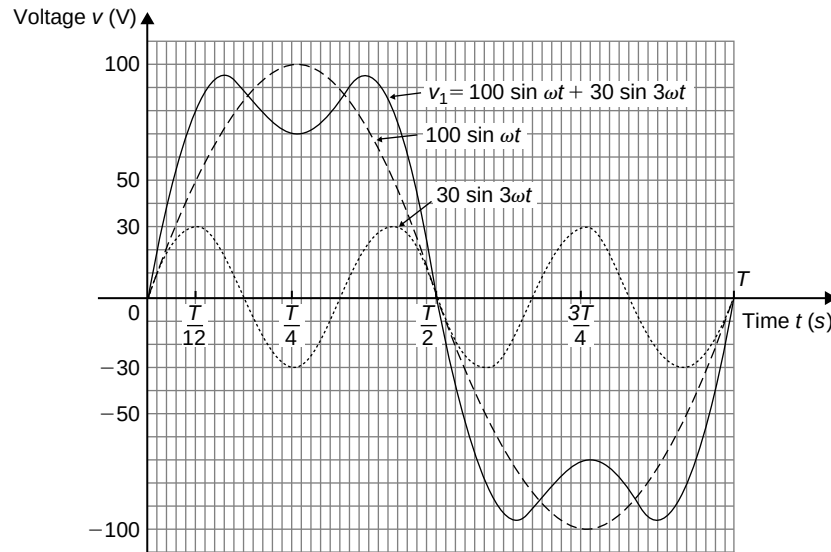


Figure 15.31

$T/4$ seconds, the fundamental has a value of 100 V and the third harmonic a value of -30 V. After addition, the resultant waveform v_1 is 70 V at $T/4$. The procedure is continued between $t=0$ and $t=T$ to produce the complex waveform for v_1 . The negative half-cycle of waveform v_1 is seen to be identical in shape to the positive half-cycle.

If further odd harmonics of the appropriate amplitude and phase were added to v_1 a good approximation to a **square wave** would result.

Problem 18. Construct the complex voltage given by:

$$v_2 = 100 \sin \omega t + 30 \sin \left(3\omega t + \frac{\pi}{2} \right) \text{ volts.}$$

The peak value of the fundamental is 100 volts and the peak value of the third harmonic is 30 V. However, the third harmonic has a phase displacement of $\frac{\pi}{2}$ radian leading (i.e. leading $30 \sin 3\omega t$ by $\frac{\pi}{2}$ radian). Note that, since the periodic time of the fundamental is T seconds, the periodic time of the third harmonic is $T/3$ seconds, and a phase displacement of $\frac{\pi}{2}$ radian or $\frac{1}{4}$ cycle of the third harmonic represents a time interval of $(T/3) \div 4$, i.e. $T/12$ seconds.

Fig. 15.32 shows graphs of $100 \sin \omega t$ and $30 \sin \left(3\omega t + \frac{\pi}{2} \right)$ over the time for one cycle of the fundamental. When ordinates of the two graphs are added at intervals, the resultant waveform v_2 is

as shown. If the negative half-cycle in Fig. 15.32 is reversed it can be seen that the shape of the positive and negative half-cycles are identical.

Problems 17 and 18 demonstrate that whenever odd harmonics are added to a fundamental waveform, whether initially in phase with each other or not, the positive and negative half-cycles of the resultant complex wave are identical in shape. This is a feature of waveforms containing the fundamental and odd harmonics.

Problem 19. Use harmonic synthesis to construct the complex current given by:

$$i_1 = 10 \sin \omega t + 4 \sin 2\omega t \text{ amperes.}$$

Current i_1 consists of a fundamental component, $10 \sin \omega t$, and a second harmonic component, $4 \sin 2\omega t$, the components being initially in phase with each other. The fundamental and second harmonic are shown plotted separately in Fig. 15.33. By adding ordinates at intervals, the complex waveform representing i_1 is produced as shown. It is noted that if all the values in the negative half-cycle were reversed then this half-cycle would appear as a mirror image of the positive half-cycle about a vertical line drawn through time, $t = T/2$.

Problem 20. Construct the complex current given by:

$$i_2 = 10 \sin \omega t + 4 \sin \left(2\omega t + \frac{\pi}{2} \right) \text{ amperes.}$$

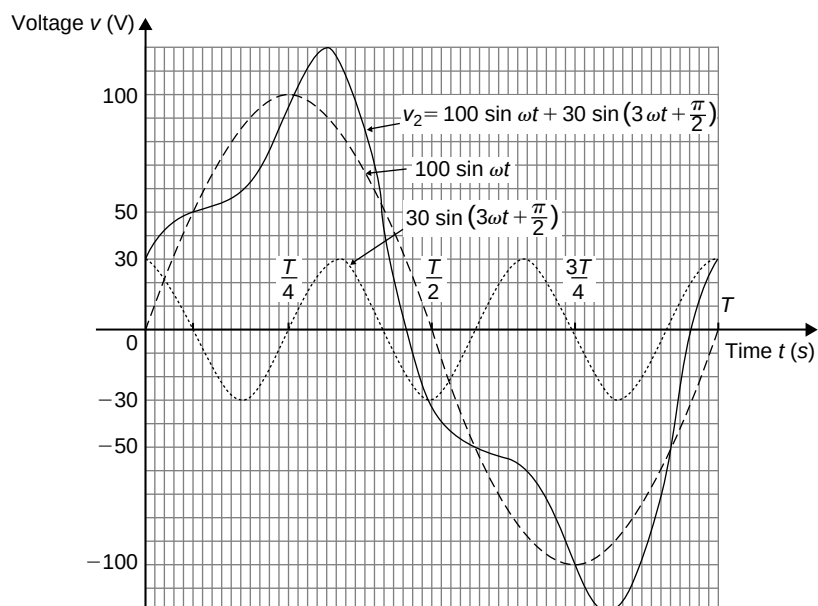


Figure 15.32

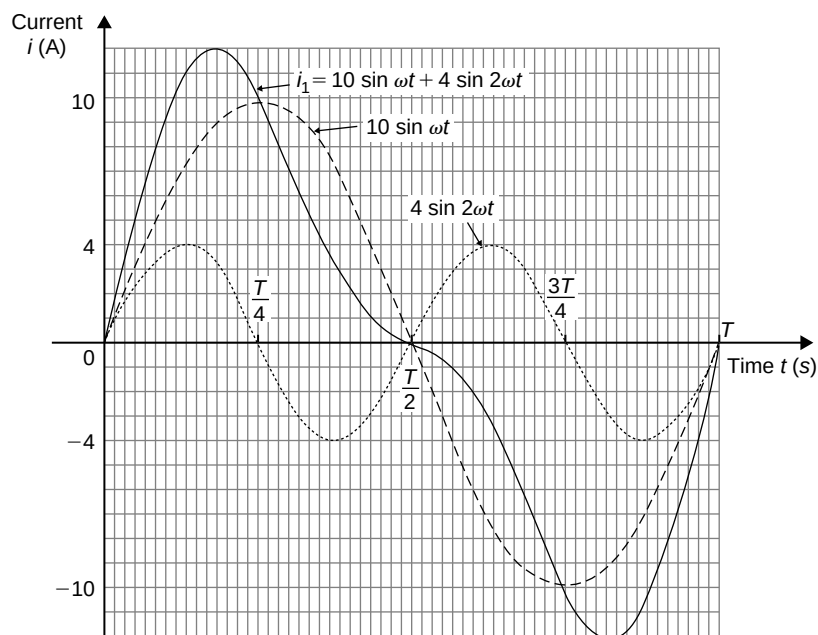


Figure 15.33

The fundamental component, $10 \sin \omega t$, and the second harmonic component, having an amplitude of 4 A and a phase displacement of $\frac{\pi}{2}$ radian leading (i.e. leading $4 \sin 2\omega t$ by $\frac{\pi}{2}$ radian or $T/8$ seconds), are shown plotted separately in Fig. 15.34. By adding ordinates

at intervals, the complex waveform for i_2 is produced as shown. The positive and negative half-cycles of the resultant waveform are seen to be quite dissimilar.

From Problems 18 and 19 it is seen that whenever even harmonics are added to a fundamental component:

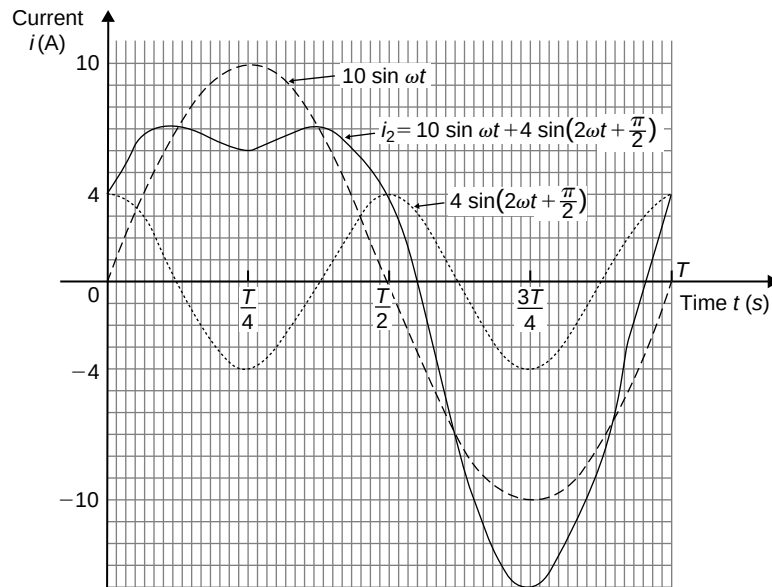


Figure 15.34

- if the harmonics are initially in phase, the negative half-cycle, when reversed, is a mirror image of the positive half-cycle about a vertical line drawn through time, $t = T/2$.
- if the harmonics are initially out of phase with each other, the positive and negative half-cycles are dissimilar.

These are features of waveforms containing the fundamental and even harmonics.

Problem 21. Use harmonic synthesis to construct the complex current expression given by:

$$i = 32 + 50 \sin \omega t + 20 \sin \left(2\omega t - \frac{\pi}{2} \right) \text{ mA}$$

The current i comprises three components – a 32 mA d.c. component, a fundamental of amplitude 50 mA and a second harmonic of amplitude 20 mA, lagging by $\frac{\pi}{2}$ radian. The fundamental and second harmonic are shown separately in Fig. 15.35. Adding ordinates at intervals gives the complex waveform $50 \sin \omega t + 20 \sin \left(2\omega t - \frac{\pi}{2} \right)$

This waveform is then added to the 32 mA d.c. component to produce the waveform i as shown.

The effect of the d.c. component is to shift the whole wave 32 mA upward. The waveform approaches that expected from a **half-wave rectifier**.

Problem 22. A complex waveform v comprises a fundamental voltage of 240 V rms and frequency 50 Hz, together with a 20% third harmonic which has a phase angle lagging by $3\pi/4$ rad at time $t = 0$. (a) Write down an expression to represent voltage v . (b) Use harmonic synthesis to sketch the complex waveform representing voltage v over one cycle of the fundamental component.

- A fundamental voltage having an rms value of 240 V has a maximum value, or amplitude of $\sqrt{2}$ (240), i.e. 339.4 V.

If the fundamental frequency is 50 Hz then angular velocity, $\omega = 2\pi f = 2\pi(50) = 100\pi$ rad/s. Hence the fundamental voltage is represented by $339.4 \sin 100\pi t$ volts. Since the fundamental frequency is 50 Hz, the time for one cycle of the fundamental is given by $T = 1/f = 1/50$ s or 20 ms.

The third harmonic has an amplitude equal to 20% of 339.4 V, i.e. 67.9 V. The frequency of the third harmonic component is $3 \times 50 = 150$ Hz, thus the angular velocity is $2\pi(150)$, i.e. 300π rad/s. Hence

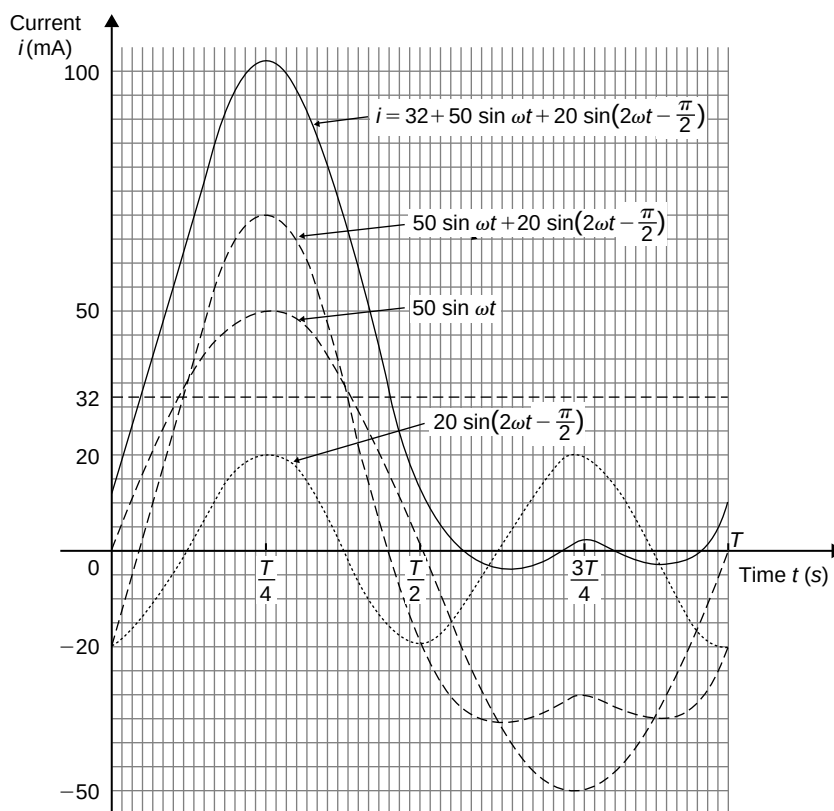


Figure 15.35

the third harmonic voltage is represented by $67.9 \sin(300\pi t - 3\pi/4)$ volts. Thus

voltage, $v = 339.4 \sin 100\pi t$
+ $67.9 \sin(300\pi t - 3\pi/4)$ volts

(b) One cycle of the fundamental, $339.4 \sin 100\pi t$, is shown sketched in Fig. 15.36, together with three cycles of the third harmonic component, $67.9 \sin(300\pi t - 3\pi/4)$ initially lagging by $3\pi/4$ rad. By adding ordinates at intervals,

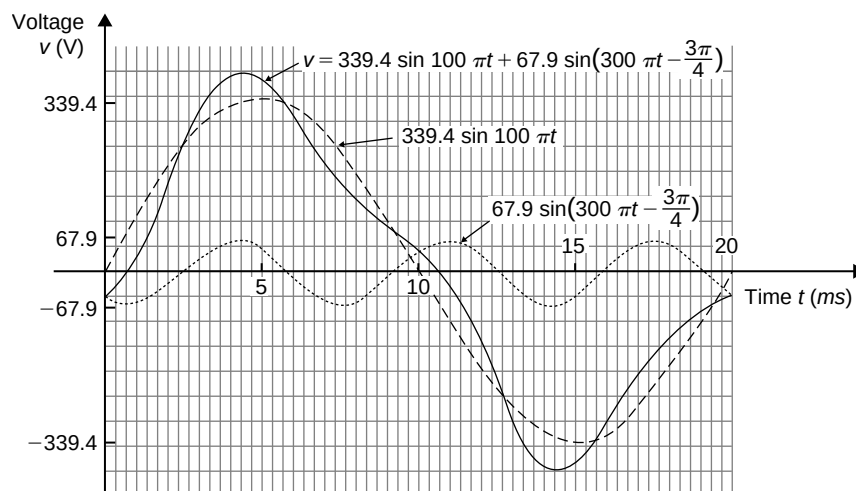


Figure 15.36

the complex waveform representing voltage is produced as shown.

Now try the following Practice Exercise

Practice Exercise 71 Harmonic synthesis with complex waveforms (Answers on page 839)

1. A complex current waveform i comprises a fundamental current of 50 A rms and frequency 100 Hz, together with a 24% third harmonic, both being in phase with each other at zero time. (a) Write down an expression to represent current i . (b) Sketch the complex waveform of current using harmonic synthesis over one cycle of the fundamental.
2. A complex voltage waveform v comprises of a 212.1 V rms fundamental voltage at a frequency of 50 Hz, a 30% second harmonic component lagging by $\pi/2$ rad, and a 10% fourth harmonic component leading by $\pi/3$ rad. (a) Write down an expression to represent voltage v . (b) Sketch the complex voltage waveform using harmonic synthesis over one cycle of the fundamental waveform.

3. A voltage waveform is represented by:

$$v = 20 + 50 \sin \omega t + 20 \sin(2\omega t - \pi/2) \text{ volts.}$$

Draw the complex waveform over one cycle of the fundamental by using harmonic synthesis.

4. Write down an expression representing a current i having a fundamental component of amplitude 16 A and frequency 1 kHz, together with its third and fifth harmonics being respectively one-fifth and one-tenth the amplitude of the fundamental, all components being in phase at zero time. Sketch the complex current waveform for one cycle of the fundamental using harmonic synthesis.
5. A voltage waveform is described by

$$v = 200 \sin 377t + 80 \sin \left(1131t + \frac{\pi}{4} \right) + 20 \sin \left(1885t - \frac{\pi}{3} \right) \text{ volts}$$

Determine (a) the fundamental and harmonic frequencies of the waveform, (b) the percentage third harmonic and (c) the percentage fifth harmonic. Sketch the voltage waveform using harmonic synthesis over one cycle of the fundamental.

For fully worked solutions to each of the problems in Practice Exercises 68 to 71 in this chapter, go to the website:

www.routledge.com/cw/bird

Chapter 16

Hyperbolic functions

Why it is important to understand: Hyperbolic functions

There are two combinations of e^x and e^{-x} which are used so often in engineering that they are given their own name. They are the hyperbolic sine, $\sinh$, and the hyperbolic cosine, $\cosh$. They are of interest because they have many properties analogous to those of trigonometric functions and because they arise in the study of falling bodies, hanging cables, ocean waves, and many other phenomena in science and engineering. The shape of a chain hanging under gravity is well described by $\cosh x$ and the deformation of uniform beams can be expressed in terms of hyperbolic tangents. Other applications of hyperbolic functions are found in fluid dynamics, optics, heat, mechanical engineering, and in astronomy when dealing with the curvature of light in the presence of black holes.

At the end of this chapter, you should be able to:

- define a hyperbolic function
- state practical applications of hyperbolic functions
- define $\sinh x$, $\cosh x$, $\tanh x$, $\operatorname{cosech} x$, $\operatorname{sech} x$ and $\operatorname{coth} x$
- evaluate hyperbolic functions
- sketch graphs of hyperbolic functions
- state Osborne's rule
- prove simple hyperbolic identities
- solve equations involving hyperbolic functions
- derive the series expansions for $\cosh x$ and $\sinh x$

16.1 Introduction to hyperbolic functions

Functions which are associated with the geometry of the conic section called a hyperbola are called **hyperbolic functions**. Applications of hyperbolic functions include transmission line theory and catenary problems. By definition:

- (i) Hyperbolic sine of x ,

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad (1)$$

' $\sinh x$ ' is often abbreviated to ' $\operatorname{sh} x$ ' and is pronounced as 'shine x '

- (ii) Hyperbolic cosine of x ,

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad (2)$$

' $\cosh x$ ' is often abbreviated to ' $\operatorname{ch} x$ ' and is pronounced as 'kosh x '

- (iii) Hyperbolic tangent of x ,

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (3)$$

' $\tanh x$ ' is often abbreviated to ' $\text{th } x$ ' and is pronounced as ' $\text{than } x$ '

(iv) Hyperbolic cosecant of x ,

$$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}} \quad (4)$$

' $\operatorname{cosech} x$ ' is pronounced as ' $\text{coshec } x$ '

(v) Hyperbolic secant of x ,

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} \quad (5)$$

' $\operatorname{sech} x$ ' is pronounced as ' $\text{shec } x$ '

(vi) Hyperbolic cotangent of x ,

$$\operatorname{coth} x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad (6)$$

' $\operatorname{coth} x$ ' is pronounced as ' $\text{koth } x$ '

Some properties of hyperbolic functions

Replacing x by 0 in equation (1) gives:

$$\sinh 0 = \frac{e^0 - e^{-0}}{2} = \frac{1 - 1}{2} = 0$$

Replacing x by 0 in equation (2) gives:

$$\cosh 0 = \frac{e^0 + e^{-0}}{2} = \frac{1 + 1}{2} = 1$$

If a function of x , $f(-x) = -f(x)$, then $f(x)$ is called an **odd function** of x . Replacing x by $-x$ in equation (1) gives:

$$\begin{aligned} \sinh(-x) &= \frac{e^{-x} - e^{-(-x)}}{2} = \frac{e^{-x} - e^x}{2} \\ &= -\left(\frac{e^x - e^{-x}}{2}\right) = -\sinh x \end{aligned}$$

Replacing x by $-x$ in equation (3) gives:

$$\begin{aligned} \tanh(-x) &= \frac{e^{-x} - e^{-(-x)}}{e^{-x} + e^{-(-x)}} = \frac{e^{-x} - e^x}{e^{-x} + e^x} \\ &= -\left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right) = -\tanh x \end{aligned}$$

Hence **$\sinh x$ and $\tanh x$ are both odd functions**

(see Section 16.1), as also are $\operatorname{cosech} x \left(= \frac{1}{\sinh x} \right)$ and

$$\operatorname{coth} x \left(= \frac{1}{\tanh x} \right)$$

If a function of x , $f(-x) = f(x)$, then $f(x)$ is called an **even function** of x . Replacing x by $-x$ in equation (2) gives:

$$\begin{aligned} \cosh(-x) &= \frac{e^{-x} + e^{-(-x)}}{2} = \frac{e^{-x} + e^x}{2} \\ &= \cosh x \end{aligned}$$

Hence **$\cosh x$ is an even function** (see Section 16.2), as

$$\text{also is } \operatorname{sech} x \left(= \frac{1}{\cosh x} \right)$$

Hyperbolic functions may be evaluated most easily using a calculator. Many scientific notation calculators actually possess $\sinh$ and $\cosh$ functions; however, if a calculator does not contain these functions, then the definitions given above may be used.

Problem 1. Evaluate $\sinh 5.4$, correct to 4 significant figures.

Using a calculator,

- (i) press hyp
- (ii) press 1 and $\sinh$ (appears
- (iii) type in 5.4
- (iv) press) to close the brackets
- (v) press = and 110.7009498 appears

Hence, **$\sinh 5.4 = 110.7$** , correct to 4 significant figures.

$$\text{Alternatively, } \sinh 5.4 = \frac{1}{2}(e^{5.4} - e^{-5.4})$$

$$= \frac{1}{2}(221.406416... - 0.00451658...)$$

$$= \frac{1}{2}(221.401899...)$$

$$= \mathbf{110.7}, \text{ correct to 4 significant figures.}$$

Problem 2. Evaluate $\cosh 1.86$, correct to 3 decimal places.

Using a calculator with the procedure similar to that used in Problem 1,

$$\mathbf{\cosh 1.86 = 3.290}, \text{ correct to 3 decimal places.}$$

Problem 3. Evaluate $\text{th} 0.52$, correct to 4 significant figures.

Using a calculator with the procedure similar to that used in Problem 1,

$\text{th} 0.52 = 0.4777$, correct to 4 significant figures.

Problem 4. Evaluate $\text{cosech} 1.4$, correct to 4 significant figures.

$$\text{cosech} 1.4 = \frac{1}{\sinh 1.4}$$

Using a calculator,

- (i) press hyp
- (ii) press 1 and $\sinh$ (appears
- (iii) type in 1.4
- (iv) press) to close the brackets
- (v) press = and 1.904301501 appears
- (vi) press x^{-1}
- (vii) press = and 0.5251269293 appears

Hence, **$\text{cosech} 1.4 = 0.5251$** , correct to 4 significant figures.

Problem 5. Evaluate $\text{sech} 0.86$, correct to 4 significant figures.

$$\text{sech} 0.86 = \frac{1}{\cosh 0.86}$$

Using a calculator with the procedure similar to that used in Problem 4,

$\text{sech} 0.86 = 0.7178$, correct to 4 significant figures.

Problem 6. Evaluate $\text{coth} 0.38$, correct to 3 decimal places.

$$\text{coth} 0.38 = \frac{1}{\tanh 0.38}$$

Using a calculator with the procedure similar to that used in Problem 4,

$\text{coth} 0.38 = 2.757$, correct to 3 decimal places.

Now try the following Practice Exercise

Practice Exercise 72 Evaluating hyperbolic functions (Answers on page 840)

In Problems 1 to 6, evaluate correct to 4 significant figures.

- (a) $\text{sh} 0.64$ (b) $\text{sh} 2.182$
- (a) $\text{ch} 0.72$ (b) $\text{ch} 2.4625$
- (a) $\text{th} 0.65$ (b) $\text{th} 1.81$
- (a) $\text{cosech} 0.543$ (b) $\text{cosech} 3.12$
- (a) $\text{sech} 0.39$ (b) $\text{sech} 2.367$
- (a) $\text{coth} 0.444$ (b) $\text{coth} 1.843$
- A telegraph wire hangs so that its shape is described by $y = 50 \text{ch} \frac{x}{50}$. Evaluate, correct to 4 significant figures, the value of y when $x = 25$
- The length l of a heavy cable hanging under gravity is given by $l = 2c \text{sh}(L/2c)$. Find the value of l when $c = 40$ and $L = 30$
- $V^2 = 0.55L \tanh(6.3d/L)$ is a formula for velocity V of waves over the bottom of shallow water, where d is the depth and L is the wavelength. If $d = 8.0$ and $L = 96$, calculate the value of V .

16.2 Graphs of hyperbolic functions

A graph of $y = \sinh x$ may be plotted using calculator values of hyperbolic functions. The curve is shown in Fig. 16.1. Since the graph is symmetrical about the origin, $\sinh x$ is an **odd function** (as stated in Section 16.1).

A graph of $y = \cosh x$ may be plotted using calculator values of hyperbolic functions. The curve is shown in Fig. 16.2. Since the graph is symmetrical about the y -axis, $\cosh x$ is an **even function** (as stated in Section 16.1). The shape of $y = \cosh x$ is that of a heavy rope or chain hanging freely under gravity and is called a **catenary**. Examples include transmission lines, a telegraph wire or a fisherman's line, and is used in the design of roofs and arches. Graphs of $y = \tanh x$,

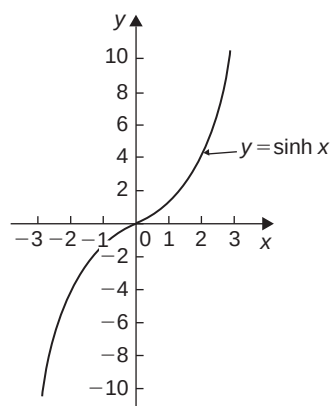


Figure 16.1

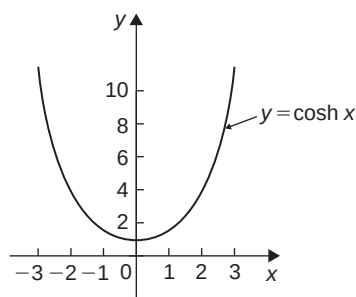


Figure 16.2

$y = \operatorname{cosech} x$, $y = \operatorname{sech} x$ and $y = \operatorname{coth} x$ are deduced in Problems 7 and 8.

Problem 7. Sketch graphs of (a) $y = \tanh x$ and (b) $y = \operatorname{coth} x$ for values of x between -3 and 3

A table of values is drawn up as shown below

x	-3	-2	-1
$\operatorname{sh} x$	-10.02	-3.63	-1.18
$\operatorname{ch} x$	10.07	3.76	1.54
$y = \tanh x = \frac{\operatorname{sh} x}{\operatorname{ch} x}$	-0.995	-0.97	-0.77
$y = \operatorname{coth} x = \frac{\operatorname{ch} x}{\operatorname{sh} x}$	-1.005	-1.04	-1.31

x	0	1	2	3
$\operatorname{sh} x$	0	1.18	3.63	10.02
$\operatorname{ch} x$	1	1.54	3.76	10.07
$y = \tanh x = \frac{\operatorname{sh} x}{\operatorname{ch} x}$	0	0.77	0.97	0.995
$y = \operatorname{coth} x = \frac{\operatorname{ch} x}{\operatorname{sh} x}$	$\pm\infty$	1.31	1.04	1.005

(a) A graph of $y = \tanh x$ is shown in Fig. 16.3(a)

(b) A graph of $y = \operatorname{coth} x$ is shown in Fig. 16.3(b)

Both graphs are symmetrical about the origin, thus $\tanh x$ and $\operatorname{coth} x$ are odd functions.

Problem 8. Sketch graphs of (a) $y = \operatorname{cosech} x$ and (b) $y = \operatorname{sech} x$ from $x = -4$ to $x = 4$, and, from the graphs, determine whether they are odd or even functions.

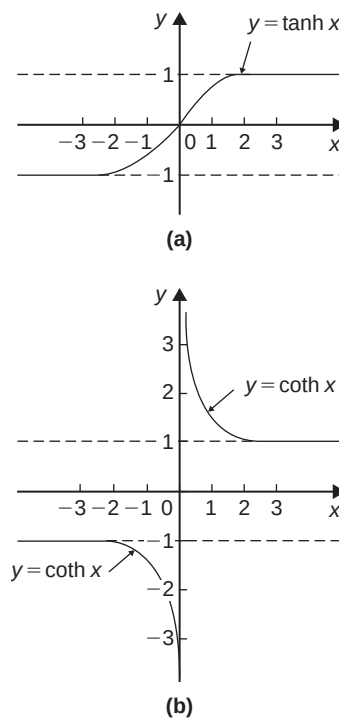


Figure 16.3

A table of values is drawn up as shown below

x	-4	-3	-2	-1
$\text{sh } x$	-22.29	-10.02	-3.63	-1.18
$\text{cosech } x = \frac{1}{\text{sh } x}$	-0.04	-0.10	-0.28	-0.85
$\text{ch } x$	27.31	10.07	3.76	1.54
$\text{sech } x = \frac{1}{\text{ch } x}$	0.04	0.10	0.27	0.65

x	0	1	2	3	4
$\text{sh } x$	0	1.18	3.63	10.02	27.29
$\text{cosech } x = \frac{1}{\text{sh } x}$	$\pm\infty$	0.85	0.28	0.10	0.04
$\text{ch } x$	1	1.54	3.76	10.07	27.31
$\text{sech } x = \frac{1}{\text{ch } x}$	1	0.65	0.27	0.10	0.04

- (a) A graph of $y = \text{cosech } x$ is shown in Fig. 16.4(a). The graph is symmetrical about the origin and is thus an **odd function**.
- (b) A graph of $y = \text{sech } x$ is shown in Fig. 16.4(b). The graph is symmetrical about the y -axis and is thus an **even function**.

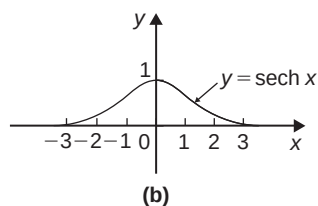
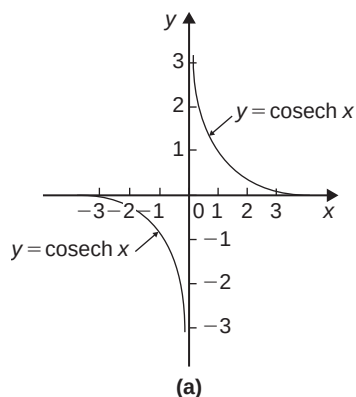


Figure 16.4

16.3 Hyperbolic identities

For every trigonometric identity there is a corresponding hyperbolic identity. **Hyperbolic identities** may be proved by either

- replacing $\text{sh } x$ by $\frac{e^x - e^{-x}}{2}$ and $\text{ch } x$ by $\frac{e^x + e^{-x}}{2}$, or
- by using **Osborne's rule**, which states: 'the six trigonometric ratios used in trigonometrical identities relating general angles may be replaced by their corresponding hyperbolic functions, but the sign of any direct or implied product of two sines must be changed'.

For example, since $\cos^2 x + \sin^2 x = 1$ then, by Osborne's rule, $\text{ch}^2 x - \text{sh}^2 x = 1$, i.e. the trigonometric functions have been changed to their corresponding hyperbolic functions and since $\sin^2 x$ is a product of two sines the sign is changed from $+$ to $-$. Table 16.1 shows some trigonometric identities and their corresponding hyperbolic identities.

Problem 9. Prove the hyperbolic identities

- (a) $\text{ch}^2 x - \text{sh}^2 x = 1$ (b) $1 - \text{th}^2 x = \text{sech}^2 x$
 (c) $\text{coth}^2 x - 1 = \text{cosech}^2 x$

$$(a) \quad \text{ch } x + \text{sh } x = \left(\frac{e^x + e^{-x}}{2} \right) + \left(\frac{e^x - e^{-x}}{2} \right) = e^x$$

$$\begin{aligned} \text{ch } x - \text{sh } x &= \left(\frac{e^x + e^{-x}}{2} \right) - \left(\frac{e^x - e^{-x}}{2} \right) \\ &= e^{-x} \end{aligned}$$

$$(\text{ch } x + \text{sh } x)(\text{ch } x - \text{sh } x) = (e^x)(e^{-x}) = e^0 = 1$$

$$\text{i.e. } \text{ch}^2 x - \text{sh}^2 x = 1 \quad (1)$$

- (b) Dividing each term in equation (1) by $\text{ch}^2 x$ gives:

$$\frac{\text{ch}^2 x}{\text{ch}^2 x} - \frac{\text{sh}^2 x}{\text{ch}^2 x} = \frac{1}{\text{ch}^2 x}$$

$$\text{i.e. } 1 - \text{th}^2 x = \text{sech}^2 x$$

Table 16.1

Trigonometric identity	Corresponding hyperbolic identity
$\cos^2 x + \sin^2 x = 1$	$\text{ch}^2 x - \text{sh}^2 x = 1$
$1 + \tan^2 x = \sec^2 x$	$1 - \text{th}^2 x = \text{sech}^2 x$
$\cot^2 x + 1 = \text{cosec}^2 x$	$\text{coth}^2 x - 1 = \text{cosech}^2 x$
<i>Compound angle formulae</i>	
$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$	$\text{sh}(A \pm B) = \text{sh} A \text{ch} B \pm \text{ch} A \text{sh} B$
$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$	$\text{ch}(A \pm B) = \text{ch} A \text{ch} B \pm \text{sh} A \text{sh} B$
$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$	$\text{th}(A \pm B) = \frac{\text{th} A \pm \text{th} B}{1 \pm \text{th} A \text{th} B}$
<i>Double angles</i>	
$\sin 2x = 2 \sin x \cos x$	$\text{sh} 2x = 2 \text{sh} x \text{ch} x$
$\cos 2x = \cos^2 x - \sin^2 x$	$\text{ch} 2x = \text{ch}^2 x + \text{sh}^2 x$
$= 2 \cos^2 x - 1$	$= 2 \text{ch}^2 x - 1$
$= 1 - 2 \sin^2 x$	$= 1 + 2 \text{sh}^2 x$
$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$	$\text{th} 2x = \frac{2 \text{th} x}{1 + \text{th}^2 x}$

- (c) Dividing each term in equation (1) by $\text{sh}^2 x$ gives:

$$\frac{\text{ch}^2 x}{\text{sh}^2 x} - \frac{\text{sh}^2 x}{\text{sh}^2 x} = \frac{1}{\text{sh}^2 x}$$

$$\text{i.e. } \text{coth}^2 x - 1 = \text{cosech}^2 x$$

Problem 10. Prove, using Osborne's rule

(a) $\text{ch} 2A = \text{ch}^2 A + \text{sh}^2 A$

(b) $1 - \text{th}^2 x = \text{sech}^2 x$

- (a) From trigonometric ratios,

$$\cos 2A = \cos^2 A - \sin^2 A \quad (1)$$

Osborne's rule states that trigonometric ratios may be replaced by their corresponding hyperbolic functions but the sign of any product of two sines has to be changed. In this case, $\sin^2 A = (\sin A)(\sin A)$, i.e. a product of two sines, thus the sign of the corresponding hyperbolic function, $\text{sh}^2 A$, is changed from + to -. Hence, from (1), $\text{ch} 2A = \text{ch}^2 A + \text{sh}^2 A$

- (b) From trigonometric ratios,

$$1 + \tan^2 x = \sec^2 x \quad (2)$$

$$\text{and } \tan^2 x = \frac{\sin^2 x}{\cos^2 x} = \frac{(\sin x)(\sin x)}{\cos^2 x}$$

i.e. a product of two sines.

Hence, in equation (2), the trigonometric ratios are changed to their equivalent hyperbolic function and the sign of $\text{th}^2 x$ changed + to -, i.e. $1 - \text{th}^2 x = \text{sech}^2 x$

Problem 11. Prove that $1 + 2 \text{sh}^2 x = \text{ch} 2x$

Left hand side (LHS)

$$\begin{aligned}
 &= 1 + 2 \text{sh}^2 x = 1 + 2 \left(\frac{e^x - e^{-x}}{2} \right)^2 \\
 &= 1 + 2 \left(\frac{e^{2x} - 2e^x e^{-x} + e^{-2x}}{4} \right) \\
 &= 1 + \frac{e^{2x} - 2 + e^{-2x}}{2} \\
 &= 1 + \left(\frac{e^{2x} + e^{-2x}}{2} \right) - \frac{2}{2} \\
 &= \frac{e^{2x} + e^{-2x}}{2} = \text{ch} 2x = \text{RHS}
 \end{aligned}$$

Problem 12. Show that $\text{th}^2 x + \text{sech}^2 x = 1$

$$\begin{aligned}\text{LHS} &= \text{th}^2 x + \text{sech}^2 x = \frac{\text{sh}^2 x}{\text{ch}^2 x} + \frac{1}{\text{ch}^2 x} \\ &= \frac{\text{sh}^2 x + 1}{\text{ch}^2 x}\end{aligned}$$

Since $\text{ch}^2 x - \text{sh}^2 x = 1$ then $1 + \text{sh}^2 x = \text{ch}^2 x$

$$\text{Thus } \frac{\text{sh}^2 x + 1}{\text{ch}^2 x} = \frac{\text{ch}^2 x}{\text{ch}^2 x} = 1 = \text{RHS}$$

Problem 13. Given $Ae^x + Be^{-x} \equiv 4\text{ch} x - 5\text{sh} x$, determine the values of A and B .

$$\begin{aligned}Ae^x + Be^{-x} &\equiv 4\text{ch} x - 5\text{sh} x \\ &= 4\left(\frac{e^x + e^{-x}}{2}\right) - 5\left(\frac{e^x - e^{-x}}{2}\right) \\ &= 2e^x + 2e^{-x} - \frac{5}{2}e^x + \frac{5}{2}e^{-x} \\ &= -\frac{1}{2}e^x + \frac{9}{2}e^{-x}\end{aligned}$$

Equating coefficients gives: $A = -\frac{1}{2}$ and $B = \frac{9}{2}$

Problem 14. If $4e^x - 3e^{-x} \equiv P\text{sh} x + Q\text{ch} x$, determine the values of P and Q

$$\begin{aligned}4e^x - 3e^{-x} &\equiv P\text{sh} x + Q\text{ch} x \\ &= P\left(\frac{e^x - e^{-x}}{2}\right) + Q\left(\frac{e^x + e^{-x}}{2}\right) \\ &= \frac{P}{2}e^x - \frac{P}{2}e^{-x} + \frac{Q}{2}e^x + \frac{Q}{2}e^{-x} \\ &= \left(\frac{P+Q}{2}\right)e^x + \left(\frac{Q-P}{2}\right)e^{-x}\end{aligned}$$

Equating coefficients gives:

$$4 = \frac{P+Q}{2} \text{ and } -3 = \frac{Q-P}{2}$$

$$\text{i.e. } P+Q=8 \quad (1)$$

$$-P+Q=-6 \quad (2)$$

Adding equations (1) and (2) gives: $2Q=2$, i.e. $Q=1$

Substituting in equation (1) gives: $P=7$

Now try the following Practice Exercise

Practice Exercise 73 Hyperbolic identities (Answers on page 840)

In Problems 1 to 4, prove the given identities.

- (a) $\text{ch}(P-Q) \equiv \text{ch} P \text{ch} Q - \text{sh} P \text{sh} Q$
(b) $\text{ch} 2x \equiv \text{ch}^2 x + \text{sh}^2 x$
- (a) $\text{coth} x \equiv 2 \text{cosech} 2x + \text{th} x$
(b) $\text{ch} 2\theta - 1 \equiv 2 \text{sh}^2 \theta$
- (a) $\text{th}(A-B) \equiv \frac{\text{th} A - \text{th} B}{1 - \text{th} A \text{th} B}$
(b) $\text{sh} 2A \equiv 2 \text{sh} A \text{ch} A$
- (a) $\text{sh}(A+B) \equiv \text{sh} A \text{ch} B + \text{ch} A \text{sh} B$
(b) $\frac{\text{sh}^2 x + \text{ch}^2 x - 1}{2\text{ch}^2 x \text{coth}^2 x} \equiv \tanh^4 x$
- Given $Pe^x - Qe^{-x} \equiv 6\text{ch} x - 2\text{sh} x$, find P and Q
- If $5e^x - 4e^{-x} \equiv A\text{sh} x + B\text{ch} x$, find A and B .

16.4 Solving equations involving hyperbolic functions

Equations such as $\sinh x = 3.25$ or $\cosh x = 3.478$ may be determined using a calculator. This is demonstrated in Problems 15 to 21.

Problem 15. Solve the equation $\text{sh} x = 3$, correct to 4 significant figures.

If $\sinh x = 3$, then $x = \sinh^{-1} 3$

This can be determined by calculator.

- Press hyp
- Choose 4, which is $\sinh^{-1}$
- Type in 3
- Close bracket)
- Press = and the answer is 1.818448459

i.e. the solution of $\text{sh} x = 3$ is: $x = 1.818$, correct to 4 significant figures.

Problem 16. Solve the equation $\text{ch} x = 1.52$, correct to 3 decimal places.

Using a calculator with a similar procedure as in Problem 15, check that:

$$x = 0.980, \text{ correct to 3 decimal places.}$$

With reference to Fig. 16.2, it can be seen that there will be two values corresponding to $y = \cosh x = 1.52$. Hence, $x = \pm 0.980$

Problem 17. Solve the equation $\tanh \theta = 0.256$, correct to 4 significant figures.

Using a calculator with a similar procedure as in Problem 15, check that

$$\theta = 0.2618, \text{ correct to 4 significant figures.}$$

Problem 18. Solve the equation $\operatorname{sech} x = 0.4562$, correct to 3 decimal places.

$$\text{If } \operatorname{sech} x = 0.4562, \text{ then } x = \operatorname{sech}^{-1} 0.4562 = \cosh^{-1} \left(\frac{1}{0.4562} \right) \text{ since } \cosh = \frac{1}{\operatorname{sech}}$$

$$\text{i.e. } x = 1.421, \text{ correct to 3 decimal places.}$$

With reference to the graph of $y = \operatorname{sech} x$ in Fig. 16.4, it can be seen that there will be two values corresponding to $y = \operatorname{sech} x = 0.4562$

$$\text{Hence, } x = \pm 1.421$$

Problem 19. Solve the equation $\operatorname{cosech} y = -0.4458$, correct to 4 significant figures.

$$\text{If } \operatorname{cosech} y = -0.4458, \text{ then } y = \operatorname{cosech}^{-1}(-0.4458) = \sinh^{-1} \left(\frac{1}{-0.4458} \right) \text{ since } \sinh = \frac{1}{\operatorname{cosech}}$$

$$\text{i.e. } y = -1.547, \text{ correct to 4 significant figures.}$$

Problem 20. Solve the equation $\coth A = 2.431$, correct to 3 decimal places.

$$\text{If } \coth A = 2.431, \text{ then } A = \coth^{-1} 2.431 = \tanh^{-1} \left(\frac{1}{2.431} \right) \text{ since } \tanh = \frac{1}{\coth}$$

$$\text{i.e. } A = 0.437, \text{ correct to 3 decimal places.}$$

Problem 21. A chain hangs in the form given by $y = 40 \operatorname{ch} \frac{x}{40}$. Determine, correct to 4 significant figures, (a) the value of y when x is 25, and (b) the value of x when $y = 54.30$

$$(a) \quad y = 40 \operatorname{ch} \frac{x}{40}, \text{ and when } x = 25,$$

$$\begin{aligned} y &= 40 \operatorname{ch} \frac{25}{40} = 40 \operatorname{ch} 0.625 \\ &= 40(1.2017536 \dots) = \mathbf{48.07} \end{aligned}$$

$$(b) \quad \text{When } y = 54.30, 54.30 = 40 \operatorname{ch} \frac{x}{40}, \text{ from which}$$

$$\operatorname{ch} \frac{x}{40} = \frac{54.30}{40} = 1.3575$$

$$\text{Hence, } \frac{x}{40} = \cosh^{-1} 1.3575 = \pm 0.822219 \dots$$

(see Fig. 16.2 for the reason as to why the answer is $\pm$) from which, $x = 40(\pm 0.822219 \dots) = \mathbf{\pm 32.89}$

Equations of the form $a \operatorname{ch} x + b \operatorname{sh} x = c$, where a , b and c are constants may be solved either by:

- plotting graphs of $y = a \operatorname{ch} x + b \operatorname{sh} x$ and $y = c$ and noting the points of intersection, or more accurately,
- by adopting the following procedure:
 - Change $\operatorname{sh} x$ to $\left(\frac{e^x - e^{-x}}{2} \right)$ and $\operatorname{ch} x$ to $\left(\frac{e^x + e^{-x}}{2} \right)$
 - Rearrange the equation into the form $pe^x + qe^{-x} + r = 0$, where p , q and r are constants.
 - Multiply each term by e^x , which produces an equation of the form $p(e^x)^2 + re^x + q = 0$ (since $(e^{-x})(e^x) = e^0 = 1$)
 - Solve the quadratic equation $p(e^x)^2 + re^x + q = 0$ for e^x by factorising or by using the quadratic formula.
 - Given $e^x = a$ constant (obtained by solving the equation in (iv)), take Napierian logarithms of both sides to give $x = \ln(\text{constant})$

This procedure is demonstrated in Problem 22.

Problem 22. Solve the equation $2.6 \operatorname{ch} x + 5.1 \operatorname{sh} x = 8.73$, correct to 4 decimal places.

Following the above procedure:

(i) $2.6 \operatorname{ch} x + 5.1 \operatorname{sh} x = 8.73$

$$\text{i.e. } 2.6 \left(\frac{e^x + e^{-x}}{2} \right) + 5.1 \left(\frac{e^x - e^{-x}}{2} \right) = 8.73$$

(ii) $1.3e^x + 1.3e^{-x} + 2.55e^x - 2.55e^{-x} = 8.73$

$$\text{i.e. } 3.85e^x - 1.25e^{-x} - 8.73 = 0$$

(iii) $3.85(e^x)^2 - 8.73e^x - 1.25 = 0$

(iv) e^x

$$= \frac{-(-8.73) \pm \sqrt{(-8.73)^2 - 4(3.85)(-1.25)}}{2(3.85)}$$

$$= \frac{8.73 \pm \sqrt{95.463}}{7.70} = \frac{8.73 \pm 9.7705}{7.70}$$

Hence $e^x = 2.4027$ or $e^x = -0.1351$

(v) $x = \ln 2.4027$ or $x = \ln(-0.1351)$ which has no real solution.

Hence $x = 0.8766$, correct to 4 decimal places.

Now try the following Practice Exercise

Practice Exercise 74 Hyperbolic equations (Answers on page 840)

In Problems 1 to 8, solve the given equations correct to 4 decimal places.

1. (a) $\sinh x = 1$ (b) $\operatorname{sh} A = -2.43$

2. (a) $\cosh B = 1.87$ (b) $2 \operatorname{ch} x = 3$

3. (a) $\tanh y = -0.76$ (b) $3 \operatorname{th} x = 2.4$

4. (a) $\operatorname{sech} B = 0.235$ (b) $\operatorname{sech} Z = 0.889$

5. (a) $\operatorname{cosech} \theta = 1.45$ (b) $5 \operatorname{cosech} x = 4.35$

6. (a) $\coth x = 2.54$ (b) $2 \coth y = -3.64$

7. $3.5 \operatorname{sh} x + 2.5 \operatorname{ch} x = 0$

8. $2 \operatorname{sh} x + 3 \operatorname{ch} x = 5$

9. $4 \operatorname{th} x - 1 = 0$

10. A chain hangs so that its shape is of the form $y = 56 \cosh \left(\frac{x}{56} \right)$. Determine, correct to 4 significant figures, (a) the value of y when x is 35, and (b) the value of x when y is 62.35

16.5 Series expansions for $\cosh x$ and $\sinh x$

By definition,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

from Chapter 4.

Replacing x by $-x$ gives:

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!} + \dots$$

$$\cosh x = \frac{1}{2}(e^x + e^{-x})$$

$$= \frac{1}{2} \left[\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots \right) + \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!} + \dots \right) \right]$$

$$= \frac{1}{2} \left[\left(2 + \frac{2x^2}{2!} + \frac{2x^4}{4!} + \dots \right) \right]$$

i.e. $\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$ (which is valid for all values of x). $\cosh x$ is an even function and contains only even powers of x in its expansion.

$$\sinh x = \frac{1}{2}(e^x - e^{-x})$$

$$= \frac{1}{2} \left[\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots \right) - \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!} + \dots \right) \right]$$

$$= \frac{1}{2} \left[2x + \frac{2x^3}{3!} + \frac{2x^5}{5!} + \dots \right]$$

i.e. $\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$ (which is valid for all values of x). $\sinh x$ is an odd function and contains only odd powers of x in its series expansion.

Problem 23. Using the series expansion for $\operatorname{ch} x$, evaluate $\operatorname{ch} 1$ correct to 4 decimal places.

$$\operatorname{ch} x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \text{ from above}$$

Let $x = 1$,

$$\begin{aligned} \text{then } \operatorname{ch} 1 &= 1 + \frac{1^2}{2 \times 1} + \frac{1^4}{4 \times 3 \times 2 \times 1} \\ &\quad + \frac{1^6}{6 \times 5 \times 4 \times 3 \times 2 \times 1} + \dots \\ &= 1 + 0.5 + 0.04167 + 0.001389 + \dots \end{aligned}$$

i.e. **ch 1 = 1.5431**, correct to 4 decimal places, which may be checked by using a calculator.

Problem 24. Determine, correct to 3 decimal places, the value of $\operatorname{sh} 3$ using the series expansion for $\operatorname{sh} x$.

$$\operatorname{sh} x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \text{ from above}$$

Let $x = 3$, then

$$\begin{aligned} \operatorname{sh} 3 &= 3 + \frac{3^3}{3!} + \frac{3^5}{5!} + \frac{3^7}{7!} + \frac{3^9}{9!} + \frac{3^{11}}{11!} + \dots \\ &= 3 + 4.5 + 2.025 + 0.43393 + 0.05424 \\ &\quad + 0.00444 + \dots \end{aligned}$$

i.e. **sh 3 = 10.018**, correct to 3 decimal places.

Problem 25. Determine the power series for $2 \operatorname{ch} \left(\frac{\theta}{2} \right) - \operatorname{sh} 2\theta$ as far as the term in θ^5

In the series expansion for $\operatorname{ch} x$, let $x = \frac{\theta}{2}$ then:

$$\begin{aligned} 2 \operatorname{ch} \left(\frac{\theta}{2} \right) &= 2 \left[1 + \frac{(\theta/2)^2}{2!} + \frac{(\theta/2)^4}{4!} + \dots \right] \\ &= 2 + \frac{\theta^2}{4} + \frac{\theta^4}{192} + \dots \end{aligned}$$

In the series expansion for $\operatorname{sh} x$, let $x = 2\theta$, then:

$$\begin{aligned} \operatorname{sh} 2\theta &= 2\theta + \frac{(2\theta)^3}{3!} + \frac{(2\theta)^5}{5!} + \dots \\ &= 2\theta + \frac{4}{3}\theta^3 + \frac{4}{15}\theta^5 + \dots \end{aligned}$$

Hence

$$\begin{aligned} \operatorname{ch} \left(\frac{\theta}{2} \right) - \operatorname{sh} 2\theta &= \left(2 + \frac{\theta^2}{4} + \frac{\theta^4}{192} + \dots \right) \\ &\quad - \left(2\theta + \frac{4}{3}\theta^3 + \frac{4}{15}\theta^5 + \dots \right) \\ &= 2 - 2\theta + \frac{\theta^2}{4} - \frac{4}{3}\theta^3 + \frac{\theta^4}{192} \\ &\quad - \frac{4}{15}\theta^5 + \dots \text{ as far the term in } \theta^5 \end{aligned}$$

Now try the following Practice Exercise

Practice Exercise 75 Series expansions for $\cosh x$ and $\sinh x$ (Answers on page 840)

- Use the series expansion for $\operatorname{ch} x$ to evaluate, correct to 4 decimal places: (a) $\operatorname{ch} 1.5$ (b) $\operatorname{ch} 0.8$
- Use the series expansion for $\operatorname{sh} x$ to evaluate, correct to 4 decimal places: (a) $\operatorname{sh} 0.5$ (b) $\operatorname{sh} 2$
- Expand the following as a power series as far as the term in x^5 : (a) $\operatorname{sh} 3x$ (b) $\operatorname{ch} 2x$

In Problems 4 and 5, prove the given identities, the series being taken as far as the term in θ^5 only.

- $\operatorname{sh} 2\theta - \operatorname{sh} \theta \equiv \theta + \frac{7}{6}\theta^3 + \frac{31}{120}\theta^5$
- $2 \operatorname{sh} \frac{\theta}{2} - \operatorname{ch} \frac{\theta}{2} \equiv -1 + \theta - \frac{\theta^2}{8} + \frac{\theta^3}{24} - \frac{\theta^4}{384} + \frac{\theta^5}{1920}$

For fully worked solutions to each of the problems in Practice Exercises 72 to 75 in this chapter, go to the website:

www.routledge.com/cw/bird

Chapter 17

Trigonometric identities and equations

Why it is important to understand: Trigonometric identities and equations

In engineering, trigonometric identities occur often, examples being in the more advanced areas of calculus to generate derivatives and integrals, with tensors/vectors, and with differential equations and partial differential equations. One of the skills required for more advanced work in mathematics, especially in calculus, is the ability to use identities to write expressions in alternative forms. In software engineering, working, say, on the next big blockbuster film, trigonometric identities are needed for computer graphics; an RF engineer working on the next-generation mobile phone will also need trigonometric identities. In addition, identities are needed in electrical engineering when dealing with a.c. power, and wave addition/subtraction and the solutions of trigonometric equations often require knowledge of trigonometric identities.

At the end of this chapter, you should be able to:

- state simple trigonometric identities
- prove simple identities
- solve equations of the form $b \sin A + c = 0$
- solve equations of the form $a \sin^2 A + c = 0$
- solve equations of the form $a \sin^2 A + b \sin A + c = 0$
- solve equations requiring trigonometric identities

17.1 Trigonometric identities

A **trigonometric identity** is a relationship that is true for all values of the unknown variable.

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \cot \theta = \frac{\cos \theta}{\sin \theta}, \sec \theta = \frac{1}{\cos \theta}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} \text{ and } \cot \theta = \frac{1}{\tan \theta}$$

are examples of trigonometric identities from Chapter 12.

Applying Pythagoras' theorem to the right-angled triangle shown in Fig. 17.1 gives:

$$a^2 + b^2 = c^2 \quad (1)$$

Dividing each term of equation (1) by c^2 gives:

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{c^2}{c^2}$$

$$\text{i.e. } \left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 = 1$$

$$(\cos\theta)^2 + (\sin\theta)^2 = 1$$

$$\text{Hence } \cos^2\theta + \sin^2\theta = 1 \quad (2)$$

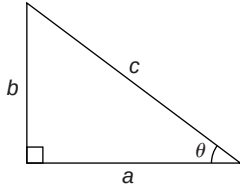


Figure 17.1

Dividing each term of equation (1) by a^2 gives:

$$\frac{a^2}{a^2} + \frac{b^2}{a^2} = \frac{c^2}{a^2}$$

$$\text{i.e. } 1 + \left(\frac{b}{a}\right)^2 = \left(\frac{c}{a}\right)^2$$

$$\text{Hence } 1 + \tan^2\theta = \sec^2\theta \quad (3)$$

Dividing each term of equation (1) by b^2 gives:

$$\frac{a^2}{b^2} + \frac{b^2}{b^2} = \frac{c^2}{b^2}$$

$$\text{i.e. } \left(\frac{a}{b}\right)^2 + 1 = \left(\frac{c}{b}\right)^2$$

$$\text{Hence } \cot^2\theta + 1 = \operatorname{cosec}^2\theta \quad (4)$$

Equations (2), (3) and (4) are three further examples of trigonometric identities. For the proof of further trigonometric identities, see Section 17.2.

17.2 Worked problems on trigonometric identities

Problem 1. Prove the identity
 $\sin^2\theta \cot\theta \sec\theta = \sin\theta$

With trigonometric identities it is necessary to start with the left-hand side (LHS) and attempt to make it equal to the right-hand side (RHS) or vice-versa. It is often useful

to change all of the trigonometric ratios into sines and cosines where possible. Thus,

$$\begin{aligned} \text{LHS} &= \sin^2\theta \cot\theta \sec\theta \\ &= \sin^2\theta \left(\frac{\cos\theta}{\sin\theta}\right) \left(\frac{1}{\cos\theta}\right) \\ &= \sin\theta \text{ (by cancelling) } = \text{RHS} \end{aligned}$$

Problem 2. Prove that

$$\frac{\tan x + \sec x}{\sec x \left(1 + \frac{\tan x}{\sec x}\right)} = 1$$

$$\begin{aligned} \text{LHS} &= \frac{\tan x + \sec x}{\sec x \left(1 + \frac{\tan x}{\sec x}\right)} \\ &= \frac{\frac{\sin x}{\cos x} + \frac{1}{\cos x}}{\left(\frac{1}{\cos x}\right) \left(1 + \frac{\frac{\sin x}{\cos x}}{\frac{1}{\cos x}}\right)} \\ &= \frac{\frac{\sin x + 1}{\cos x}}{\left(\frac{1}{\cos x}\right) \left[1 + \left(\frac{\sin x}{\cos x}\right) \left(\frac{\cos x}{1}\right)\right]} \\ &= \frac{\frac{\sin x + 1}{\cos x}}{\left(\frac{1}{\cos x}\right) [1 + \sin x]} \\ &= \left(\frac{\sin x + 1}{\cos x}\right) \left(\frac{\cos x}{1 + \sin x}\right) \\ &= 1 \text{ (by cancelling) } = \text{RHS} \end{aligned}$$

Problem 3. Prove that $\frac{1 + \cot\theta}{1 + \tan\theta} = \cot\theta$

$$\begin{aligned} \text{LHS} &= \frac{1 + \cot\theta}{1 + \tan\theta} \\ &= \frac{1 + \frac{\cos\theta}{\sin\theta}}{1 + \frac{\sin\theta}{\cos\theta}} = \frac{\frac{\sin\theta + \cos\theta}{\sin\theta}}{\frac{\cos\theta + \sin\theta}{\cos\theta}} \\ &= \frac{\sin\theta + \cos\theta}{\sin\theta} \cdot \frac{\cos\theta}{\cos\theta + \sin\theta} = \cot\theta \end{aligned}$$

$$\begin{aligned}
 &= \left(\frac{\sin \theta + \cos \theta}{\sin \theta} \right) \left(\frac{\cos \theta}{\cos \theta + \sin \theta} \right) \\
 &= \frac{\cos \theta}{\sin \theta} = \cot \theta = \text{RHS}
 \end{aligned}$$

Problem 4. Show that
 $\cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta$

From equation (2), $\cos^2 \theta + \sin^2 \theta = 1$, from which,
 $\cos^2 \theta = 1 - \sin^2 \theta$

Hence, LHS

$$\begin{aligned}
 &= \cos^2 \theta - \sin^2 \theta = (1 - \sin^2 \theta) - \sin^2 \theta \\
 &= 1 - \sin^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta = \text{RHS}
 \end{aligned}$$

Problem 5. Prove that

$$\sqrt{\left(\frac{1 - \sin x}{1 + \sin x} \right)} = \sec x - \tan x$$

$$\begin{aligned}
 \text{LHS} &= \sqrt{\left(\frac{1 - \sin x}{1 + \sin x} \right)} = \sqrt{\left\{ \frac{(1 - \sin x)(1 - \sin x)}{(1 + \sin x)(1 - \sin x)} \right\}} \\
 &= \sqrt{\left\{ \frac{(1 - \sin x)^2}{(1 - \sin^2 x)} \right\}}
 \end{aligned}$$

Since $\cos^2 x + \sin^2 x = 1$ then $1 - \sin^2 x = \cos^2 x$

$$\begin{aligned}
 \text{LHS} &= \sqrt{\left\{ \frac{(1 - \sin x)^2}{(1 - \sin^2 x)} \right\}} = \sqrt{\left\{ \frac{(1 - \sin x)^2}{\cos^2 x} \right\}} \\
 &= \frac{1 - \sin x}{\cos x} = \frac{1}{\cos x} - \frac{\sin x}{\cos x} \\
 &= \sec x - \tan x = \text{RHS}
 \end{aligned}$$

Now try the following Practice Exercise

Practice Exercise 76 Trigonometric identities (Answers on page 840)

In Problems 1 to 6 prove the trigonometric identities.

- $\sin x \cot x = \cos x$
- $\frac{1}{\sqrt{1 - \cos^2 \theta}} = \operatorname{cosec} \theta$

- $2 \cos^2 A - 1 = \cos^2 A - \sin^2 A$
- $\frac{\cos x - \cos^3 x}{\sin x} = \sin x \cos x$
- $(1 + \cot \theta)^2 + (1 - \cot \theta)^2 = 2 \operatorname{cosec}^2 \theta$
- $\frac{\sin^2 x (\sec x + \operatorname{cosec} x)}{\cos x \tan x} = 1 + \tan x$

17.3 Trigonometric equations

Equations which contain trigonometric ratios are called **trigonometric equations**. There are usually an infinite number of solutions to such equations; however, solutions are often restricted to those between 0° and 360° .

A knowledge of angles of any magnitude is essential in the solution of trigonometric equations and calculators cannot be relied upon to give all the solutions (as shown in Chapter 15). Fig. 17.2 shows a summary for angles of any magnitude.

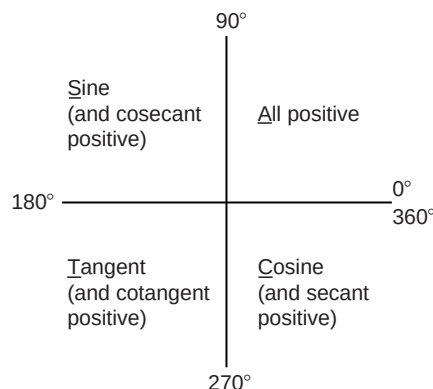


Figure 17.2

Equations of the type $a \sin^2 A + b \sin A + c = 0$

- When $a = 0$,** $b \sin A + c = 0$, hence
 $\sin A = -\frac{c}{b}$ and $A = \sin^{-1} \left(-\frac{c}{b} \right)$
 There are two values of A between 0° and 360° which satisfy such an equation, provided $-1 \leq \frac{c}{b} \leq 1$ (see Problems 6 to 8).
- When $b = 0$,** $a \sin^2 A + c = 0$, hence
 $\sin^2 A = -\frac{c}{a}$, $\sin A = \sqrt{\left(-\frac{c}{a} \right)}$
 and $A = \sin^{-1} \sqrt{\left(-\frac{c}{a} \right)}$

If either a or c is a negative number, then the value within the square root sign is positive. Since when a square root is taken there is a positive and negative answer there are four values of A between 0° and 360° which satisfy such an equation, provided $-1 \leq \frac{c}{a} \leq 1$ (see Problems 9 and 10).

(iii) **When a , b and c are all non-zero:**

$a \sin^2 A + b \sin A + c = 0$ is a quadratic equation in which the unknown is $\sin A$. The solution of a quadratic equation is obtained either by factoring (if possible) or by using the quadratic formula:

$$\sin A = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

(see Problems 11 and 12).

(iv) Often the trigonometric identities $\cos^2 A + \sin^2 A = 1$, $1 + \tan^2 A = \sec^2 A$ and $\cot^2 A + 1 = \operatorname{cosec}^2 A$ need to be used to reduce equations to one of the above forms (see Problems 13 to 15).

17.4 Worked problems (i) on trigonometric equations

Problem 6. Solve the trigonometric equation $5 \sin \theta + 3 = 0$ for values of θ from 0° to 360°

$$5 \sin \theta + 3 = 0, \text{ from which } \sin \theta = -\frac{3}{5} = -0.6000$$

Hence $\theta = \sin^{-1}(-0.6000)$. Sine is negative in the third and fourth quadrants (see Fig. 17.3). The acute angle $\sin^{-1}(0.6000) = 36.87^\circ$ (shown as α in Fig. 17.3(b)). Hence,

$$\theta = 180^\circ + 36.87^\circ, \text{ i.e. } 216.87^\circ \text{ or}$$

$$\theta = 360^\circ - 36.87^\circ, \text{ i.e. } 323.13^\circ$$

Problem 7. Solve $1.5 \tan x - 1.8 = 0$ for $0^\circ \leq x \leq 360^\circ$

$$1.5 \tan x - 1.8 = 0, \text{ from which}$$

$$\tan x = \frac{1.8}{1.5} = 1.2000$$

$$\text{Hence } x = \tan^{-1} 1.2000$$

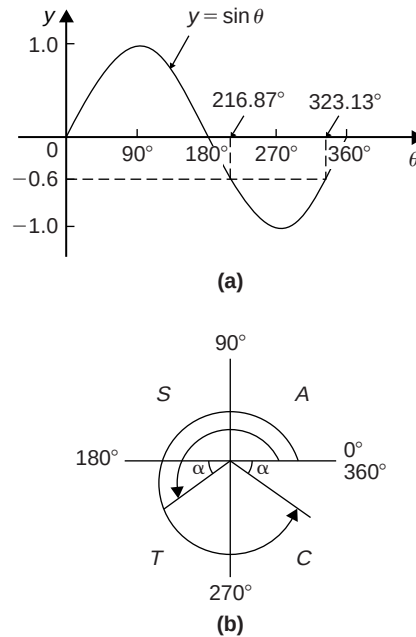


Figure 17.3

Tangent is positive in the first and third quadrants (see Fig. 17.4).

The acute angle $\tan^{-1} 1.2000 = 50.19^\circ$. Hence,

$$x = 50.19^\circ \text{ or } 180^\circ + 50.19^\circ = 230.19^\circ$$

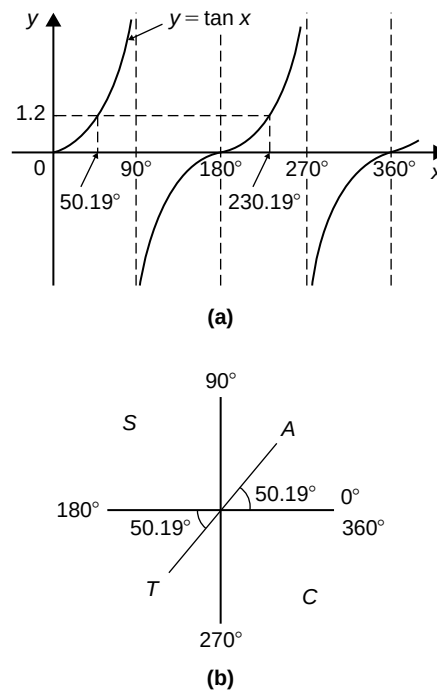


Figure 17.4

Problem 8. Solve for θ in the range $0^\circ \leq \theta \leq 360^\circ$ for $2 \sin \theta = \cos \theta$

Dividing both sides by $\cos \theta$ gives: $\frac{2 \sin \theta}{\cos \theta} = 1$

From Section 17.1, $\tan \theta = \frac{\sin \theta}{\cos \theta}$

hence $2 \tan \theta = 1$

Dividing by 2 gives: $\tan \theta = \frac{1}{2}$

from which, $\theta = \tan^{-1} \frac{1}{2}$

Since tangent is positive in the first and third quadrants, $\theta = 26.57^\circ$ and 206.57°

Problem 9. Solve $4 \sec t = 5$ for values of t between 0° and 360°

$4 \sec t = 5$, from which $\sec t = \frac{5}{4} = 1.2500$

Hence $t = \sec^{-1} 1.2500$

Secant = (1/cosine) is positive in the first and fourth quadrants (see Fig. 17.5) The acute angle $\sec^{-1} 1.2500 = 36.87^\circ$. Hence,

$$t = 36.87^\circ \text{ or } 360^\circ - 36.87^\circ = 323.13^\circ$$

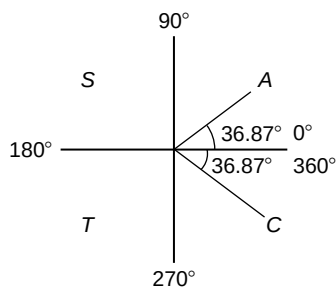


Figure 17.5

Now try the following Practice Exercise

Practice Exercise 77 Trigonometric equations (Answers on page 840)

In Problems 1 to 3 solve the equations for angles between 0° and 360° .

- $4 - 7 \sin \theta = 0$
- $3 \operatorname{cosec} A + 5.5 = 0$
- $4(2.32 - 5.4 \cot t) = 0$

In Problems 4 to 6, solve for θ in the range $0^\circ \leq \theta \leq 360^\circ$.

- $\sec \theta = 2$
- $\cot \theta = 0.6$
- $\operatorname{cosec} \theta = 1.5$

In Problems 7 to 9, solve for x in the range $-180^\circ \leq x \leq 180^\circ$.

- $\sec x = -1.5$
- $\cot x = 1.2$
- $\operatorname{cosec} x = -2$

In Problem 10 and 11, solve for θ in the range $0^\circ \leq \theta \leq 360^\circ$.

- $3 \sin \theta = 2 \cos \theta$
- $5 \cos \theta = -\sin \theta$

17.5 Worked problems (ii) on trigonometric equations

Problem 10. Solve $2 - 4 \cos^2 A = 0$ for values of A in the range $0^\circ < A < 360^\circ$.

$2 - 4 \cos^2 A = 0$, from which $\cos^2 A = \frac{2}{4} = 0.5000$

Hence $\cos A = \sqrt{(0.5000)} = \pm 0.7071$ and

$A = \cos^{-1}(\pm 0.7071)$

Cosine is positive in quadrants one and four and negative in quadrants two and three. Thus in this case there are four solutions, one in each quadrant (see Fig. 17.6). The acute angle $\cos^{-1} 0.7071 = 45^\circ$.

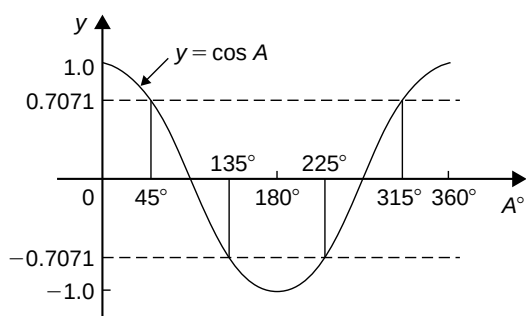
Hence, $A = 45^\circ, 135^\circ, 225^\circ$ or 315°

Problem 11. Solve $\frac{1}{2} \cot^2 y = 1.3$ for $0^\circ < y < 360^\circ$

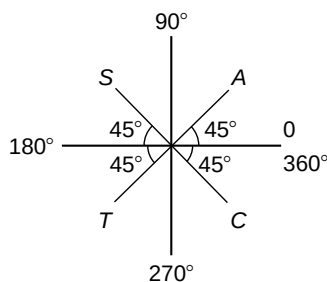
$\frac{1}{2} \cot^2 y = 1.3$, from which, $\cot^2 y = 2(1.3) = 2.6$

Hence $\cot y = \sqrt{2.6} = \pm 1.6125$, and $y = \cot^{-1}(\pm 1.6125)$. There are four solutions, one in each quadrant. The acute angle $\cot^{-1} 1.6125 = 31.81^\circ$

Hence $y = 31.81^\circ, 148.19^\circ, 211.81^\circ$ or 328.19°



(a)



(b)

Figure 17.6

Now try the following Practice Exercise

Practice Exercise 78 Trigonometric equations (Answers on page 840)

In Problems 1 to 5 solve the equations for angles between 0° and 360° .

- $5 \sin^2 y = 3$
- $\cos^2 \theta = 0.25$
- $\tan^2 x = 3$
- $5 + 3 \operatorname{cosec}^2 D = 8$
- $2 \cot^2 \theta = 5$

17.6 Worked problems (iii) on trigonometric equations

Problem 12. Solve the equation

$$8 \sin^2 \theta + 2 \sin \theta - 1 = 0,$$

for all values of θ between 0° and 360° .

Factorising $8 \sin^2 \theta + 2 \sin \theta - 1 = 0$ gives

$$(4 \sin \theta - 1)(2 \sin \theta + 1) = 0$$

Hence $4 \sin \theta - 1 = 0$, from which, $\sin \theta = \frac{1}{4} = 0.2500$

or $2 \sin \theta + 1 = 0$, from which, $\sin \theta = -\frac{1}{2} = -0.5000$

(Instead of factorising, the quadratic formula can, of course, be used.)

$\theta = \sin^{-1} 0.2500 = 14.48^\circ$ or 165.52° , since sine is positive in the first and second quadrants, or $\theta = \sin^{-1}(-0.5000) = 210^\circ$ or 330° , since sine is negative in the third and fourth quadrants.

Hence $\theta = 14.48^\circ, 165.52^\circ, 210^\circ$ or 330°

Problem 13. Solve $6 \cos^2 \theta + 5 \cos \theta - 6 = 0$ for values of θ from 0° to 360°

Factorising $6 \cos^2 \theta + 5 \cos \theta - 6 = 0$ gives

$$(3 \cos \theta - 2)(2 \cos \theta + 3) = 0$$

Hence $3 \cos \theta - 2 = 0$, from which, $\cos \theta = \frac{2}{3} = 0.6667$

or $2 \cos \theta + 3 = 0$, from which, $\cos \theta = -\frac{3}{2} = -1.5000$

The minimum value of a cosine is -1 , hence the latter expression has no solution and is thus neglected. Hence,

$$\theta = \cos^{-1} 0.6667 = 48.18^\circ \text{ or } 311.82^\circ$$

since cosine is positive in the first and fourth quadrants.

Now try the following Practice Exercise

Practice Exercise 79 Trigonometric equations (Answers on page 840)

In Problems 1 to 4 solve the equations for angles between 0° and 360° .

- $15 \sin^2 A + \sin A - 2 = 0$
- $8 \tan^2 \theta + 2 \tan \theta = 15$
- $2 \operatorname{cosec}^2 t - 5 \operatorname{cosec} t = 12$
- $2 \cos^2 \theta + 9 \cos \theta - 5 = 0$

17.7 Worked problems (iv) on trigonometric equations

Problem 14. Solve $5 \cos^2 t + 3 \sin t - 3 = 0$ for values of t from 0° to 360° .

Since $\cos^2 t + \sin^2 t = 1$, $\cos^2 t = 1 - \sin^2 t$. Substituting for $\cos^2 t$ in $5\cos^2 t + 3\sin t - 3 = 0$ gives:

$$5(1 - \sin^2 t) + 3\sin t - 3 = 0$$

$$5 - 5\sin^2 t + 3\sin t - 3 = 0$$

$$-5\sin^2 t + 3\sin t + 2 = 0$$

$$5\sin^2 t - 3\sin t - 2 = 0$$

Factorising gives $(5\sin t + 2)(\sin t - 1) = 0$. Hence $5\sin t + 2 = 0$, from which, $\sin t = -\frac{2}{5} = -0.4000$, or $\sin t - 1 = 0$, from which, $\sin t = 1$
 $t = \sin^{-1}(-0.4000) = 203.58^\circ$ or 336.42° , since sine is negative in the third and fourth quadrants, or $t = \sin^{-1} 1 = 90^\circ$. Hence $t = 90^\circ, 203.58^\circ$ or 336.42° as shown in Fig. 17.7.

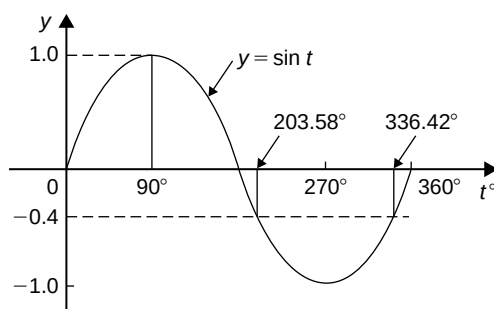


Figure 17.7

Problem 15. Solve $18\sec^2 A - 3\tan A = 21$ for values of A between 0° and 360° .

$1 + \tan^2 A = \sec^2 A$. Substituting for $\sec^2 A$ in $18\sec^2 A - 3\tan A = 21$ gives
 $18(1 + \tan^2 A) - 3\tan A = 21$,

$$\text{i.e. } 18 + 18\tan^2 A - 3\tan A - 21 = 0$$

$$18\tan^2 A - 3\tan A - 3 = 0$$

Factorising gives $(6\tan A - 3)(\tan A + 1) = 0$

Hence $6\tan A - 3 = 0$, from which, $\tan A = \frac{3}{6} = 0.5000$ or $3\tan A + 1 = 0$, from which, $\tan A = -\frac{1}{3} = -0.3333$
 Thus $A = \tan^{-1}(0.5000) = 26.57^\circ$ or 206.57° , since tangent is positive in the first and third quadrants, or

$A = \tan^{-1}(-0.3333) = 161.57^\circ$ or 341.57° , since tangent is negative in the second and fourth quadrants.

Hence, $A = 26.57^\circ, 161.57^\circ, 206.57^\circ$ or 341.57°

Problem 16. Solve $3\operatorname{cosec}^2 \theta - 5 = 4\cot \theta$ in the range $0 < \theta < 360^\circ$.

$\cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$. Substituting for $\operatorname{cosec}^2 \theta$ in $3\operatorname{cosec}^2 \theta - 5 = 4\cot \theta$ gives:

$$3(\cot^2 \theta + 1) - 5 = 4\cot \theta$$

$$3\cot^2 \theta + 3 - 5 = 4\cot \theta$$

$$3\cot^2 \theta - 4\cot \theta - 2 = 0$$

Since the LHS does not factorise the quadratic formula is used. Thus,

$$\begin{aligned} \cot \theta &= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(3)(-2)}}{2(3)} \\ &= \frac{4 \pm \sqrt{(16 + 24)}}{6} = \frac{4 \pm \sqrt{40}}{6} \\ &= \frac{10.3246}{6} \text{ or } -\frac{2.3246}{6} \end{aligned}$$

Hence $\cot \theta = 1.7208$ or -0.3874 , $\theta = \cot^{-1} 1.7208 = 30.17^\circ$ or 210.17° , since cotangent is positive in the first and third quadrants, or $\theta = \cot^{-1}(-0.3874) = 111.18^\circ$ or 291.18° , since cotangent is negative in the second and fourth quadrants.

Hence, $\theta = 30.17^\circ, 111.18^\circ, 210.17^\circ$ or 291.18°

Now try the following Practice Exercise

Practice Exercise 80 Trigonometric equations (Answers on page 840)

In Problems 1 to 12 solve the equations for angles between 0° and 360° .

- $2\cos^2 \theta + \sin \theta = 1$
- $4\cos^2 t + 5\sin t = 3$

3. $2\cos\theta - 4\sin^2\theta = 0$

4. $3\cos\theta + 2\sin^2\theta = 3$

5. $12\sin^2\theta - 6 = \cos\theta$

6. $16\sec x - 2 = 14\tan^2 x$

7. $4\cot^2 A - 6\operatorname{cosec} A + 6 = 0$

8. $5\sec t + 2\tan^2 t = 3$

9. $2.9\cos^2 a - 7\sin a + 1 = 0$

10. $3\operatorname{cosec}^2 \beta = 8 - 7\cot \beta$

11. $\cot\theta = \sin\theta$

12. $\tan\theta + 3\cot\theta = 5\sec\theta$

For fully worked solutions to each of the problems in Practice Exercises 76 to 80 in this chapter,
go to the website:

www.routledge.com/cw/bird



The relationship between trigonometric and hyperbolic functions

Why it is important to understand: The relationship between trigonometric and hyperbolic functions

There are similarities between notations used for hyperbolic and trigonometric functions. Both trigonometric and hyperbolic functions have applications in many areas of engineering. For example, the shape of a chain hanging under gravity (known as a catenary curve) is described by the hyperbolic cosine, $\cosh x$, and the deformation of uniform beams can be expressed in terms of hyperbolic tangents. Hyperbolic functions are also used in electrical engineering applications such as transmission line theory. Einstein's special theory of relativity used hyperbolic functions and both trigonometric and hyperbolic functions are needed for certain areas of more advanced integral calculus. There are many identities showing relationships between hyperbolic and trigonometric functions; these may be used to evaluate trigonometric and hyperbolic functions of complex numbers.

At the end of this chapter, you should be able to:

- identify the relationship between trigonometric and hyperbolic functions
- state hyperbolic identities

18.1 The relationship between trigonometric and hyperbolic functions

In Chapter 23 it is shown that

$$\cos \theta + j \sin \theta = e^{j\theta} \quad (1)$$

$$\text{and } \cos \theta - j \sin \theta = e^{-j\theta} \quad (2)$$

Adding equations (1) and (2) gives:

$$\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta}) \quad (3)$$

Subtracting equation (2) from equation (1) gives:

$$\sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta}) \quad (4)$$

Substituting $j\theta$ for θ in equations (3) and (4) gives:

$$\cos j\theta = \frac{1}{2}(e^{j(j\theta)} + e^{-j(j\theta)})$$

$$\text{and } \sin j\theta = \frac{1}{2j}(e^{j(j\theta)} - e^{-j(j\theta)})$$

$$\text{Since } j^2 = -1, \cos j\theta = \frac{1}{2}(e^{-\theta} + e^{\theta}) = \frac{1}{2}(e^{\theta} + e^{-\theta})$$

$$\text{Hence from Chapter 16, } \cos j\theta = \cosh \theta \quad (5)$$

$$\text{Similarly, } \sin j\theta = \frac{1}{2j}(e^{-\theta} - e^{\theta}) = -\frac{1}{2j}(e^{\theta} - e^{-\theta})$$

$$= \frac{-1}{j} \left[\frac{1}{2}(e^{\theta} - e^{-\theta}) \right]$$

$$= -\frac{1}{j} \sinh \theta \quad (\text{see Chapter 16})$$

$$\text{But } -\frac{1}{j} = -\frac{1}{j} \times \frac{j}{j} = -\frac{j}{j^2} = j,$$

$$\text{hence } \sin j\theta = j \sinh \theta \quad (6)$$

Equations (5) and (6) may be used to verify that in all standard trigonometric identities, $j\theta$ may be written for θ and the identity still remains true.

Problem 1. Verify that $\cos^2 j\theta + \sin^2 j\theta = 1$

From equation (5), $\cos j\theta = \cosh \theta$, and from equation (6), $\sin j\theta = j \sinh \theta$

Thus, $\cos^2 j\theta + \sin^2 j\theta = \cosh^2 \theta + j^2 \sinh^2 \theta$, and since $j^2 = -1$,

$$\cos^2 j\theta + \sin^2 j\theta = \cosh^2 \theta - \sinh^2 \theta$$

But from Chapter 16, Problem 6,

$$\cosh^2 \theta - \sinh^2 \theta = 1,$$

$$\text{hence } \cos^2 j\theta + \sin^2 j\theta = 1$$

Problem 2. Verify that $\sin j2A = 2 \sin jA \cos jA$

From equation (6), writing $2A$ for θ , $\sin j2A = j \sinh 2A$, and from Chapter 16, Table 16.1, page 183, $\sinh 2A = 2 \sinh A \cosh A$

$$\text{Hence, } \sin j2A = j(2 \sinh A \cosh A)$$

$$\text{But, } \sinh A = \frac{1}{2}(e^A - e^{-A}) \text{ and } \cosh A = \frac{1}{2}(e^A + e^{-A})$$

$$\text{Hence, } \sin j2A = j2 \left(\frac{e^A - e^{-A}}{2} \right) \left(\frac{e^A + e^{-A}}{2} \right)$$

$$= -\frac{2}{j} \left(\frac{e^A - e^{-A}}{2} \right) \left(\frac{e^A + e^{-A}}{2} \right)$$

$$= -\frac{2}{j} \left(\frac{\sin j\theta}{j} \right) (\cos j\theta)$$

$$= 2 \sin jA \cos jA \text{ since } j^2 = -1$$

$$\text{i.e. } \sin j2A = 2 \sin jA \cos jA$$

Now try the following Practice Exercise

Practice Exercise 81 The relationship between trigonometric and hyperbolic functions (Answers on page 840)

Verify the following identities by expressing in exponential form.

- $\sin j(A+B) = \sin jA \cos jB + \cos jA \sin jB$
- $\cos j(A-B) = \cos jA \cos jB + \sin jA \sin jB$
- $\cos j2A = 1 - 2 \sin^2 jA$
- $\sin jA \cos jB = \frac{1}{2}[\sin j(A+B) + \sin j(A-B)]$
- $\sin jA - \sin jB$

$$= 2 \cos j \left(\frac{A+B}{2} \right) \sin j \left(\frac{A-B}{2} \right)$$

18.2 Hyperbolic identities

$$\text{From Chapter 16, } \cosh \theta = \frac{1}{2}(e^{\theta} + e^{-\theta})$$

Substituting $j\theta$ for θ gives:

$$\cosh j\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta}) = \cos \theta, \text{ from equation (3),}$$

$$\text{i.e. } \cosh j\theta = \cos \theta \quad (7)$$

Similarly, from Chapter 16,

$$\sinh \theta = \frac{1}{2}(e^{\theta} - e^{-\theta})$$

Substituting $j\theta$ for θ gives:

$$\sinh j\theta = \frac{1}{2}(e^{j\theta} - e^{-j\theta}) = j \sin \theta, \text{ from equation (4).}$$

Hence $\sinh j\theta = j \sin \theta$ (8)

$$\tan j\theta = \frac{\sin j\theta}{\cosh j\theta}$$

From equations (5) and (6),

$$\frac{\sin j\theta}{\cos j\theta} = \frac{j \sinh \theta}{\cosh \theta} = j \tanh \theta$$

Hence $\tan j\theta = j \tanh \theta$ (9)

Similarly, $\tanh j\theta = \frac{\sinh j\theta}{\cosh j\theta}$

From equations (7) and (8),

$$\frac{\sinh j\theta}{\cosh j\theta} = \frac{j \sin \theta}{\cos \theta} = j \tan \theta$$

Hence $\tanh j\theta = j \tan \theta$ (10)

Two methods are commonly used to verify hyperbolic identities. These are (a) by substituting $j\theta$ (and $j\phi$) in the corresponding trigonometric identity and using the relationships given in equations (5) to (10) (see Problems 3 to 5) and (b) by applying Osborne's rule given in Chapter 16, page 182.

Problem 3. By writing jA for θ in $\cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$, determine the corresponding hyperbolic identity.

Substituting jA for θ gives:

$$\cot^2 jA + 1 = \operatorname{cosec}^2 jA,$$

i.e. $\frac{\cos^2 jA}{\sin^2 jA} + 1 = \frac{1}{\sin^2 jA}$

But from equation (5), $\cos jA = \cosh A$

and from equation (6), $\sin jA = j \sinh A$

Hence $\frac{\cosh^2 A}{j^2 \sinh^2 A} + 1 = \frac{1}{j^2 \sinh^2 A}$

and since $j^2 = -1$, $-\frac{\cosh^2 A}{\sinh^2 A} + 1 = -\frac{1}{\sinh^2 A}$

Multiplying throughout by -1 , gives:

$$\frac{\cosh^2 A}{\sinh^2 A} - 1 = \frac{1}{\sinh^2 A}$$

i.e. $\coth^2 A - 1 = \operatorname{cosech}^2 A$

Problem 4. By substituting jA and jB for θ and ϕ , respectively, in the trigonometric identity for $\cos \theta - \cos \phi$, show that

$$\begin{aligned} \cosh A - \cosh B \\ = 2 \sinh \left(\frac{A+B}{2} \right) \sinh \left(\frac{A-B}{2} \right) \end{aligned}$$

$$\cos \theta - \cos \phi = -2 \sin \left(\frac{\theta + \phi}{2} \right) \sin \left(\frac{\theta - \phi}{2} \right)$$

(see Chapter 19, page 209)

thus $\cos jA - \cos jB$

$$= -2 \sin j \left(\frac{A+B}{2} \right) \sin j \left(\frac{A-B}{2} \right)$$

But from equation (5), $\cos jA = \cosh A$

and from equation (6), $\sin jA = j \sinh A$

Hence, $\cosh A - \cosh B$

$$= -2 j \sinh \left(\frac{A+B}{2} \right) j \sinh \left(\frac{A-B}{2} \right)$$

$$= -2 j^2 \sinh \left(\frac{A+B}{2} \right) \sinh \left(\frac{A-B}{2} \right)$$

But $j^2 = -1$, hence

$$\cosh A - \cosh B = 2 \sinh \left(\frac{A+B}{2} \right) \sinh \left(\frac{A-B}{2} \right)$$

Problem 5. Develop the hyperbolic identity corresponding to $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ by writing jA for θ

Substituting jA for θ gives:

$$\sin 3jA = 3 \sin jA - 4 \sin^3 jA$$

and since from equation (6),

$$\sin jA = j \sinh A,$$

$$j \sinh 3A = 3 j \sinh A - 4 j^3 \sinh^3 A$$

Dividing throughout by j gives:

$$\sinh 3A = 3 \sinh A - j^2 4 \sinh^3 A$$

But $j^2 = -1$, hence

$$\sinh 3A = 3 \sinh A + 4 \sinh^3 A$$

[An examination of Problems 3 to 5 shows that whenever the trigonometric identity contains a term which is the product of two sines, or the implied product of two sines (e.g. $\tan^2 \theta = \sin^2 \theta / \cos^2 \theta$, thus $\tan^2 \theta$ is the implied product of two sines), the sign of the corresponding term in the hyperbolic function changes. This relationship between trigonometric and hyperbolic functions is known as Osborne's rule, as discussed in Chapter 16, page 182].

Now try the following Practice Exercise

Practice Exercise 82 Hyperbolic identities
(Answers on page 840)

In Problems 1 to 9, use the substitution $A = j\theta$ (and $B = j\phi$) to obtain the hyperbolic identities corresponding to the trigonometric identities given.

1. $1 + \tan^2 A = \sec^2 A$
2. $\cos(A + B) = \cos A \cos B - \sin A \sin B$
3. $\sin(A - B) = \sin A \cos B - \cos A \sin B$
4. $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
5. $\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$
6. $\sin^3 A = \frac{3}{4} \sin A - \frac{1}{4} \sin 3A$
7. $\cot^2 A (\sec^2 A - 1) = 1$

For fully worked solutions to each of the problems in Practice Exercises 81 and 82 in this chapter, go to the website:

www.routledge.com/cw/bird

Chapter 19

Compound angles

Why it is important to understand: Compound angles

It is often necessary to rewrite expressions involving sines, cosines and tangents in alternative forms. To do this formulae known as trigonometric identities are used as explained previously. Compound angle (or sum and difference) formulae, and double angles are further commonly used identities. Compound angles are required, for example, in the analysis of acoustics (where a beat is an interference between two sounds of slightly different frequencies), and with phase detectors (which is a frequency mixer, analogue multiplier, or logic circuit that generates a voltage signal which represents the difference in phase between two signal inputs). Many rational functions of sine and cosine are difficult to integrate without compound angle formulae.

At the end of this chapter, you should be able to:

- state compound angle formulae for $\sin(A \pm B)$, $\cos(A \pm B)$ and $\tan(A \pm B)$
- convert a $\sin \omega t + b \cos \omega t$ into $R \sin(\omega t + \alpha)$
- derive double angle formulae
- change products of sines and cosines into sums or differences
- change sums or differences of sines and cosines into products
- develop expressions for power in a.c. circuits – purely resistive, inductive and capacitive circuits, R–L and R–C circuits

19.1 Compound angle formulae

An electric current i may be expressed as $i = 5 \sin(\omega t - 0.33)$ amperes. Similarly, the displacement x of a body from a fixed point can be expressed as $x = 10 \sin(2t + 0.67)$ metres. The angles $(\omega t - 0.33)$ and $(2t + 0.67)$ are called **compound angles** because they are the sum or difference of two angles. The **compound angle formulae** for sines and cosines of the sum and difference of two angles A and B are:

$$\begin{aligned}\sin(A + B) &= \sin A \cos B + \cos A \sin B \\ \sin(A - B) &= \sin A \cos B - \cos A \sin B \\ \cos(A + B) &= \cos A \cos B - \sin A \sin B \\ \cos(A - B) &= \cos A \cos B + \sin A \sin B\end{aligned}$$

(Note, $\sin(A + B)$ is **not** equal to $(\sin A + \sin B)$, and so on.)

The formulae stated above may be used to derive two further compound angle formulae:

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

The compound angle formulae are true for all values of A and B , and by substituting values of A and B into the formulae they may be shown to be true.

Problem 1. Expand and simplify the following expressions:

(a) $\sin(\pi + \alpha)$ (b) $-\cos(90^\circ + \beta)$

(c) $\sin(A - B) - \sin(A + B)$

(a) $\sin(\pi + \alpha) = \sin \pi \cos \alpha + \cos \pi \sin \alpha$ (from the formula for $\sin(A + B)$)

$$= (0)(\cos \alpha) + (-1) \sin \alpha = -\sin \alpha$$

(b) $-\cos(90^\circ + \beta)$

$$= -[\cos 90^\circ \cos \beta - \sin 90^\circ \sin \beta]$$

$$= -[(0)(\cos \beta) - (1) \sin \beta] = \sin \beta$$

(c) $\sin(A - B) - \sin(A + B)$

$$= [\sin A \cos B - \cos A \sin B]$$

$$- [\sin A \cos B + \cos A \sin B]$$

$$= -2\cos A \sin B$$

Problem 2. Prove that

$$\cos(y - \pi) + \sin\left(y + \frac{\pi}{2}\right) = 0$$

$$\cos(y - \pi) = \cos y \cos \pi + \sin y \sin \pi$$

$$= (\cos y)(-1) + (\sin y)(0)$$

$$= -\cos y$$

$$\sin\left(y + \frac{\pi}{2}\right) = \sin y \cos \frac{\pi}{2} + \cos y \sin \frac{\pi}{2}$$

$$= (\sin y)(0) + (\cos y)(1) = \cos y$$

Hence $\cos(y - \pi) + \sin\left(y + \frac{\pi}{2}\right)$

$$= (-\cos y) + (\cos y) = 0$$

Problem 3. Show that

$$\tan\left(x + \frac{\pi}{4}\right) \tan\left(x - \frac{\pi}{4}\right) = -1$$

$$\tan\left(x + \frac{\pi}{4}\right) = \frac{\tan x + \tan \frac{\pi}{4}}{1 - \tan x \tan \frac{\pi}{4}}$$

from the formula for $\tan(A + B)$

$$= \frac{\tan x + 1}{1 - (\tan x)(1)} = \left(\frac{1 + \tan x}{1 - \tan x}\right)$$

since $\tan \frac{\pi}{4} = 1$

$$\tan\left(x - \frac{\pi}{4}\right) = \frac{\tan x - \tan \frac{\pi}{4}}{1 + \tan x \tan \frac{\pi}{4}} = \left(\frac{\tan x - 1}{1 + \tan x}\right)$$

Hence $\tan\left(x + \frac{\pi}{4}\right) \tan\left(x - \frac{\pi}{4}\right)$

$$= \left(\frac{1 + \tan x}{1 - \tan x}\right) \left(\frac{\tan x - 1}{1 + \tan x}\right)$$

$$= \frac{\tan x - 1}{1 - \tan x} = \frac{-(1 - \tan x)}{1 - \tan x} = -1$$

Problem 4. If $\sin P = 0.8142$ and $\cos Q = 0.4432$ evaluate, correct to 3 decimal places:

(a) $\sin(P - Q)$ (b) $\cos(P + Q)$

(c) $\tan(P + Q)$, using the compound angle formulae.

Since $\sin P = 0.8142$ then

$$P = \sin^{-1} 0.8142 = 54.51^\circ$$

Thus $\cos P = \cos 54.51^\circ = 0.5806$ and

$$\tan P = \tan 54.51^\circ = 1.4025$$

Since $\cos Q = 0.4432$, $Q = \cos^{-1} 0.4432 = 63.69^\circ$.

Thus $\sin Q = \sin 63.69^\circ = 0.8964$ and

$$\tan Q = \tan 63.69^\circ = 2.0225$$

(a) $\sin(P - Q)$

$$= \sin P \cos Q - \cos P \sin Q$$

$$= (0.8142)(0.4432) - (0.5806)(0.8964)$$

$$= 0.3609 - 0.5204 = -0.160$$

(b) $\cos(P + Q)$

$$= \cos P \cos Q - \sin P \sin Q$$

$$= (0.5806)(0.4432) - (0.8142)(0.8964)$$

$$= 0.2573 - 0.7298 = -0.473$$

(c) $\tan(P + Q)$

$$= \frac{\tan P + \tan Q}{1 - \tan P \tan Q} = \frac{(1.4025) + (2.0225)}{1 - (1.4025)(2.0225)}$$

$$= \frac{3.4250}{-1.8366} = -1.865$$

Problem 5. Solve the equation

$$4 \sin(x - 20^\circ) = 5 \cos x$$

for values of x between 0° and 90°

$$4 \sin(x - 20^\circ) = 4[\sin x \cos 20^\circ - \cos x \sin 20^\circ]$$

from the formula for $\sin(A - B)$

$$= 4[\sin x (0.9397) - \cos x (0.3420)]$$

$$= 3.7588 \sin x - 1.3680 \cos x$$

Since $4 \sin(x - 20^\circ) = 5 \cos x$ then

$$3.7588 \sin x - 1.3680 \cos x = 5 \cos x$$

Rearranging gives:

$$3.7588 \sin x = 5 \cos x + 1.3680 \cos x$$

$$= 6.3680 \cos x$$

$$\text{and } \frac{\sin x}{\cos x} = \frac{6.3680}{3.7588} = 1.6942$$

i.e. $\tan x = 1.6942$, and $x = \tan^{-1} 1.6942 = 59.449^\circ$ or $59^\circ 27'$ [Check: LHS = $4 \sin(59.449^\circ - 20^\circ)$

$$= 4 \sin 39.449^\circ = 2.542$$

$$\text{RHS} = 5 \cos x = 5 \cos 59.449^\circ = 2.542]$$

Now try the following Practice Exercise**Practice Exercise 83 Compound angle formulae (Answers on page 841)**

1. Reduce the following to the sine of one angle:

$$(a) \sin 37^\circ \cos 21^\circ + \cos 37^\circ \sin 21^\circ$$

$$(b) \sin 7t \cos 3t - \cos 7t \sin 3t$$

2. Reduce the following to the cosine of one angle:

$$(a) \cos 71^\circ \cos 33^\circ - \sin 71^\circ \sin 33^\circ$$

$$(b) \cos \frac{\pi}{3} \cos \frac{\pi}{4} + \sin \frac{\pi}{3} \sin \frac{\pi}{4}$$

3. Show that:

$$(a) \sin\left(x + \frac{\pi}{3}\right) + \sin\left(x + \frac{2\pi}{3}\right) = \sqrt{3} \cos x$$

$$(b) -\sin\left(\frac{3\pi}{2} - \phi\right) = \cos \phi$$

4. Prove that:

$$(a) \sin\left(\theta + \frac{\pi}{4}\right) - \sin\left(\theta - \frac{3\pi}{4}\right)$$

$$= \sqrt{2}(\sin \theta + \cos \theta)$$

$$(b) \frac{\cos(270^\circ + \theta)}{\cos(360^\circ - \theta)} = \tan \theta$$

5. Given $\cos A = 0.42$ and $\sin B = 0.73$, evaluate

(a) $\sin(A - B)$ (b) $\cos(A - B)$ (c) $\tan(A + B)$, correct to 4 decimal places.In Problems 6 and 7, solve the equations for values of θ between 0° and 360° .

$$6. \quad 3 \sin(\theta + 30^\circ) = 7 \cos \theta$$

$$7. \quad 4 \sin(\theta - 40^\circ) = 2 \sin \theta$$

19.2 Conversion of $a \sin \omega t + b \cos \omega t$ into $R \sin(\omega t + \alpha)$

- (i) $R \sin(\omega t + \alpha)$ represents a sine wave of maximum value R , periodic time $2\pi/\omega$, frequency $\omega/2\pi$ and leading $R \sin \omega t$ by angle α (see Chapter 15).
- (ii) $R \sin(\omega t + \alpha)$ may be expanded using the compound angle formula for $\sin(A + B)$, where $A = \omega t$ and $B = \alpha$. Hence,

$$R \sin(\omega t + \alpha)$$

$$= R[\sin \omega t \cos \alpha + \cos \omega t \sin \alpha]$$

$$= R \sin \omega t \cos \alpha + R \cos \omega t \sin \alpha$$

$$= (R \cos \alpha) \sin \omega t + (R \sin \alpha) \cos \omega t$$

- (iii) If $a = R \cos \alpha$ and $b = R \sin \alpha$, where a and b are constants, then $R \sin(\omega t + \alpha) = a \sin \omega t + b \cos \omega t$, i.e. a sine and cosine function of the same frequency when added produce a sine wave of the

same frequency (which is further demonstrated in Chapter 27).

- (iv) Since $a = R \cos \alpha$, then $\cos \alpha = a/R$, and since $b = R \sin \alpha$, then $\sin \alpha = b/R$

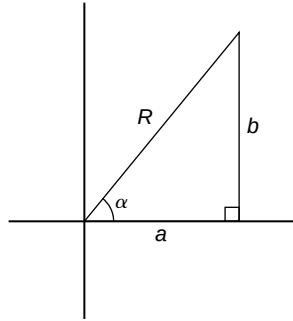


Figure 19.1

If the values of a and b are known then the values of R and α may be calculated. The relationship between constants a , b , R and α are shown in Fig. 19.1.

From Fig. 19.1, by Pythagoras' theorem:

$$R = \sqrt{a^2 + b^2}$$

and from trigonometric ratios:

$$\alpha = \tan^{-1} b/a$$

Problem 6. Find an expression for $3 \sin \omega t + 4 \cos \omega t$ in the form $R \sin(\omega t + \alpha)$ and sketch graphs of $3 \sin \omega t$, $4 \cos \omega t$ and $R \sin(\omega t + \alpha)$ on the same axes.

Let $3 \sin \omega t + 4 \cos \omega t = R \sin(\omega t + \alpha)$

then $3 \sin \omega t + 4 \cos \omega t$

$$\begin{aligned} &= R[\sin \omega t \cos \alpha + \cos \omega t \sin \alpha] \\ &= (R \cos \alpha) \sin \omega t + (R \sin \alpha) \cos \omega t \end{aligned}$$

Equating coefficients of $\sin \omega t$ gives:

$$3 = R \cos \alpha, \text{ from which, } \cos \alpha = \frac{3}{R}$$

Equating coefficients of $\cos \omega t$ gives:

$$4 = R \sin \alpha, \text{ from which, } \sin \alpha = \frac{4}{R}$$

There is only one quadrant where both $\sin \alpha$ and $\cos \alpha$ are positive, and this is the first, as shown in Fig. 19.2. From Fig. 19.2, by Pythagoras' theorem:

$$R = \sqrt{(3^2 + 4^2)} = 5$$

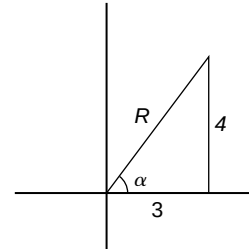


Figure 19.2

From trigonometric ratios: $\alpha = \tan^{-1} \frac{4}{3} = 53.13^\circ$ or 0.927 radians.

Hence $3 \sin \omega t + 4 \cos \omega t = 5 \sin(\omega t + 0.927)$

A sketch of $3 \sin \omega t$, $4 \cos \omega t$ and $5 \sin(\omega t + 0.927)$ is shown in Fig. 19.3 on page 204.

Two periodic functions of the same frequency may be combined by

- plotting the functions graphically and combining ordinates at intervals, or
- resolution of phasors by drawing or calculation.

Problem 6, together with Problems 7 and 8 following, demonstrate a third method of combining waveforms.

Problem 7. Express $4.6 \sin \omega t - 7.3 \cos \omega t$ in the form $R \sin(\omega t + \alpha)$

Let $4.6 \sin \omega t - 7.3 \cos \omega t = R \sin(\omega t + \alpha)$

then $4.6 \sin \omega t - 7.3 \cos \omega t$

$$\begin{aligned} &= R[\sin \omega t \cos \alpha + \cos \omega t \sin \alpha] \\ &= (R \cos \alpha) \sin \omega t + (R \sin \alpha) \cos \omega t \end{aligned}$$

Equating coefficients of $\sin \omega t$ gives:

$$4.6 = R \cos \alpha, \text{ from which, } \cos \alpha = \frac{4.6}{R}$$

Equating coefficients of $\cos \omega t$ gives:

$$-7.3 = R \sin \alpha, \text{ from which, } \sin \alpha = \frac{-7.3}{R}$$

There is only one quadrant where cosine is positive and sine is negative, i.e. the fourth quadrant, as shown in Fig. 19.4. By Pythagoras' theorem:

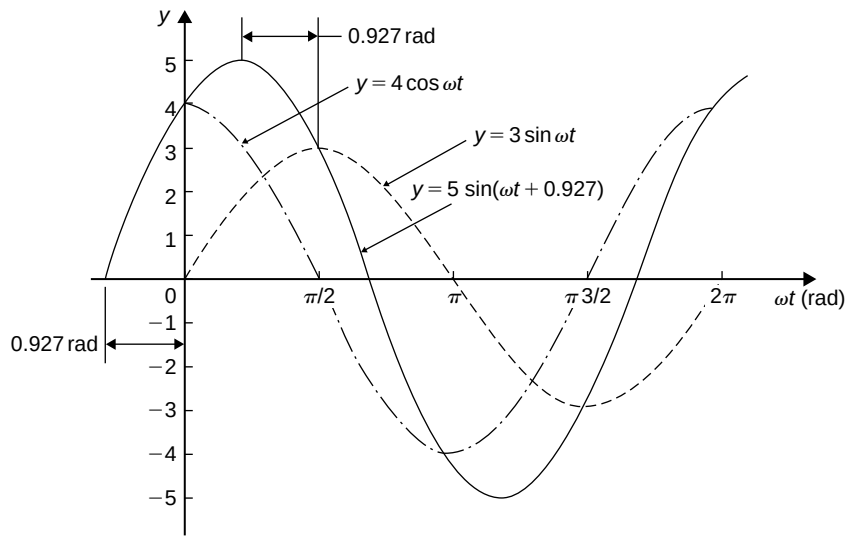


Figure 19.3

$$R = \sqrt{[(4.6)^2 + (-7.3)^2]} = 8.628$$

By trigonometric ratios:

$$\begin{aligned}\alpha &= \tan^{-1}\left(\frac{-7.3}{4.6}\right) \\ &= -57.78^\circ \text{ or } -1.008 \text{ radians.}\end{aligned}$$

Hence

$$4.6 \sin \omega t - 7.3 \cos \omega t = 8.628 \sin(\omega t - 1.008)$$

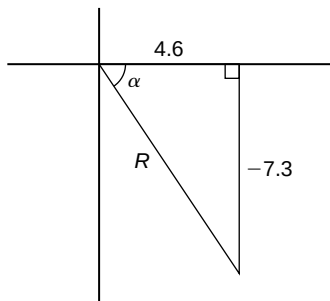


Figure 19.4

Problem 8. Express $-2.7 \sin \omega t - 4.1 \cos \omega t$ in the form $R \sin(\omega t + \alpha)$

$$\text{Let } -2.7 \sin \omega t - 4.1 \cos \omega t = R \sin(\omega t + \alpha)$$

$$\begin{aligned}&= R[\sin \omega t \cos \alpha + \cos \omega t \sin \alpha] \\ &= (R \cos \alpha) \sin \omega t + (R \sin \alpha) \cos \omega t\end{aligned}$$

Equating coefficients gives:

$$-2.7 = R \cos \alpha, \text{ from which, } \cos \alpha = \frac{-2.7}{R}$$

$$\text{and } -4.1 = R \sin \alpha, \text{ from which, } \sin \alpha = \frac{-4.1}{R}$$

There is only one quadrant in which both cosine **and** sine are negative, i.e. the third quadrant, as shown in Fig. 19.5. From Fig. 19.5,

$$R = \sqrt{[(-2.7)^2 + (-4.1)^2]} = 4.909$$

$$\text{and } \theta = \tan^{-1} \frac{4.1}{2.7} = 56.63^\circ$$

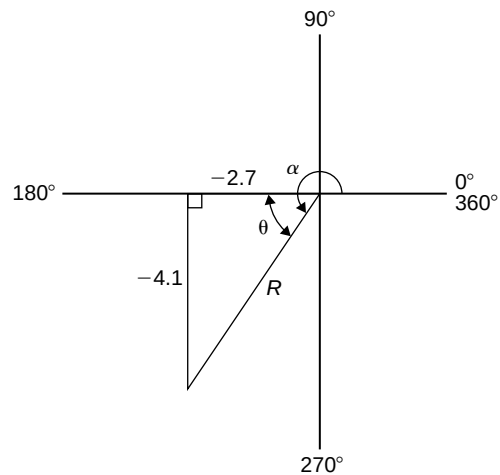


Figure 19.5

Hence $\alpha = 180^\circ + 56.63^\circ = 236.63^\circ$ or 4.130 radians.
Thus,

$$-2.7 \sin \omega t - 4.1 \cos \omega t = 4.909 \sin(\omega t + 4.130)$$

An angle of 236.63° is the same as -123.37° or -2.153 radians. Hence $-2.7 \sin \omega t - 4.1 \cos \omega t$ may be expressed also as $4.909 \sin(\omega t - 2.153)$, which is preferred since it is the **principal value** (i.e. $-\pi \leq \alpha \leq \pi$)

Problem 9. Express $3 \sin \theta + 5 \cos \theta$ in the form $R \sin(\theta + \alpha)$, and hence solve the equation $3 \sin \theta + 5 \cos \theta = 4$, for values of θ between 0° and 360°

Let $3 \sin \theta + 5 \cos \theta = R \sin(\theta + \alpha)$

$$\begin{aligned} &= R[\sin \theta \cos \alpha + \cos \theta \sin \alpha] \\ &= (R \cos \alpha) \sin \theta + (R \sin \alpha) \cos \theta \end{aligned}$$

Equating coefficients gives:

$$3 = R \cos \alpha, \text{ from which, } \cos \alpha = \frac{3}{R}$$

$$\text{and } 5 = R \sin \alpha, \text{ from which, } \sin \alpha = \frac{5}{R}$$

Since both $\sin \alpha$ and $\cos \alpha$ are positive, R lies in the first quadrant, as shown in Fig. 19.6.

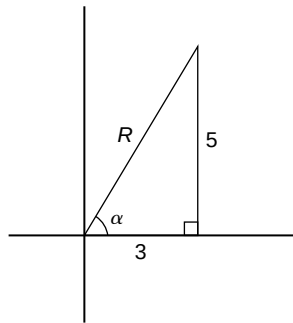


Figure 19.6

From Fig. 19.6, $R = \sqrt{(3^2 + 5^2)} = 5.831$ and
 $\alpha = \tan^{-1} \frac{5}{3} = 59.03^\circ$
Hence $3 \sin \theta + 5 \cos \theta = 5.831 \sin(\theta + 59.03^\circ)$

$$\text{However } 3 \sin \theta + 5 \cos \theta = 4$$

Thus $5.831 \sin(\theta + 59.03^\circ) = 4$, from which

$$(\theta + 59.03^\circ) = \sin^{-1} \left(\frac{4}{5.831} \right)$$

$$\text{i.e. } \theta + 59.03^\circ = 43.32^\circ \text{ or } 136.68^\circ$$

$$\text{Hence } \theta = 43.32^\circ - 59.03^\circ = -15.71^\circ$$

$$\text{or } \theta = 136.68^\circ - 59.03^\circ = 77.65^\circ$$

Since -15.71° is the same as $-15.71^\circ + 360^\circ$, i.e. 344.29° , then the solutions are $\theta = 77.65^\circ$ or 344.29° , which may be checked by substituting into the original equation.

Problem 10. Solve the equation
 $3.5 \cos A - 5.8 \sin A = 6.5$ for $0^\circ \leq A \leq 360^\circ$

Let $3.5 \cos A - 5.8 \sin A = R \sin(A + \alpha)$

$$\begin{aligned} &= R[\sin A \cos \alpha + \cos A \sin \alpha] \\ &= (R \cos \alpha) \sin A + (R \sin \alpha) \cos A \end{aligned}$$

Equating coefficients gives:

$$3.5 = R \sin \alpha, \text{ from which, } \sin \alpha = \frac{3.5}{R}$$

$$\text{and } -5.8 = R \cos \alpha, \text{ from which, } \cos \alpha = \frac{-5.8}{R}$$

There is only one quadrant in which both sine is positive **and** cosine is negative, i.e. the second, as shown in Fig. 19.7.

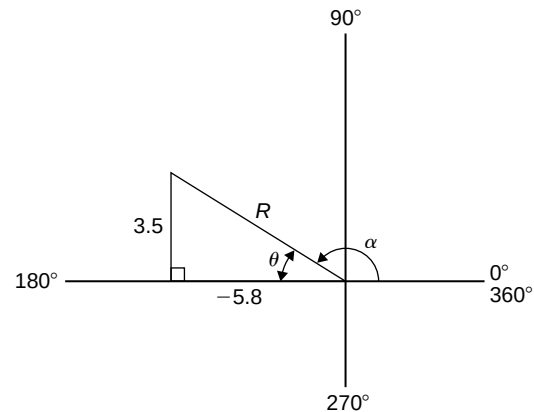


Figure 19.7

From Fig. 19.7, $R = \sqrt{[(3.5)^2 + (-5.8)^2]} = 6.774$ and
 $\theta = \tan^{-1} \frac{3.5}{-5.8} = 31.12^\circ$

$$\text{Hence } \alpha = 180^\circ - 31.12^\circ = 148.88^\circ$$

Thus

$$3.5 \cos A - 5.8 \sin A = 6.774 \sin(A + 148.88^\circ) = 6.5$$

Hence $\sin(A + 148.88^\circ) = \frac{6.5}{6.774}$, from which,

$$\begin{aligned}(A + 148.88^\circ) &= \sin^{-1} \frac{6.5}{6.774} \\ &= 73.65^\circ \text{ or } 106.35^\circ\end{aligned}$$

Thus $A = 73.65^\circ - 148.88^\circ = -75.23^\circ$

$$\equiv (-75.23^\circ + 360^\circ) = 284.77^\circ$$

or $A = 106.35^\circ - 148.88^\circ = -42.53^\circ$

$$\equiv (-42.53^\circ + 360^\circ) = 317.47^\circ$$

The solutions are thus **$A = 284.77^\circ$ or 317.47°** , which may be checked in the original equation.

Now try the following Practice Exercise

Practice Exercise 84 The conversion of $a \sin \omega t + b \cos \omega t$ into $R \sin(\omega t \pm \alpha)$ (Answers on page 841)

In Problems 1 to 4, change the functions into the form $R \sin(\omega t \pm \alpha)$.

- $5 \sin \omega t + 8 \cos \omega t$
- $4 \sin \omega t - 3 \cos \omega t$
- $-7 \sin \omega t + 4 \cos \omega t$
- $-3 \sin \omega t - 6 \cos \omega t$
- Solve the following equations for values of θ between 0° and 360° : (a) $2 \sin \theta + 4 \cos \theta = 3$
(b) $12 \sin \theta - 9 \cos \theta = 7$
- Solve the following equations for $0^\circ < A < 360^\circ$: (a) $3 \cos A + 2 \sin A = 2.8$
(b) $12 \cos A - 4 \sin A = 11$
- Solve the following equations for values of θ between 0° and 360° : (a) $3 \sin \theta + 4 \cos \theta = 3$
(b) $2 \cos \theta + \sin \theta = 2$
- Solve the following equations for values of θ between 0° and 360° : (a) $6 \cos \theta + \sin \theta = \sqrt{3}$
(b) $2 \sin 3\theta + 8 \cos 3\theta = 1$

- The third harmonic of a wave motion is given by $4.3 \cos 3\theta - 6.9 \sin 3\theta$. Express this in the form $R \sin(3\theta \pm \alpha)$
- The displacement x metres of a mass from a fixed point about which it is oscillating is given by $x = 2.4 \sin \omega t + 3.2 \cos \omega t$, where t is the time in seconds. Express x in the form $R \sin(\omega t + \alpha)$.
- Two voltages, $v_1 = 5 \cos \omega t$ and $v_2 = -8 \sin \omega t$ are inputs to an analogue circuit. Determine an expression for the output voltage if this is given by $(v_1 + v_2)$.
- The motion of a piston moving in a cylinder can be described by:
 $x = (5 \cos 2t + 5 \sin 2t) \text{ cm}$.
Express x in the form $R \sin(\omega t + \alpha)$

19.3 Double angles

- (i) If, in the compound angle formula for $\sin(A + B)$, we let $B = A$ then

$$\sin 2A = 2 \sin A \cos A$$

Also, for example,

$$\sin 4A = 2 \sin 2A \cos 2A$$

and $\sin 8A = 2 \sin 4A \cos 4A$, and so on.

- (ii) If, in the compound angle formula for $\cos(A + B)$, we let $B = A$ then

$$\cos 2A = \cos^2 A - \sin^2 A$$

Since $\cos^2 A + \sin^2 A = 1$, then $\cos^2 A = 1 - \sin^2 A$, and $\sin^2 A = 1 - \cos^2 A$, and two further formula for $\cos 2A$ can be produced.

$$\begin{aligned}\text{Thus } \cos 2A &= \cos^2 A - \sin^2 A \\ &= (1 - \sin^2 A) - \sin^2 A\end{aligned}$$

$$\text{i.e. } \cos 2A = 1 - 2 \sin^2 A$$

$$\begin{aligned}\text{and } \cos 2A &= \cos^2 A - \sin^2 A \\ &= \cos^2 A - (1 - \cos^2 A)\end{aligned}$$

$$\text{i.e. } \cos 2A = 2 \cos^2 A - 1$$

Also, for example,

$$\cos 4A = \cos^2 2A - \sin^2 2A \text{ or}$$

$$1 - 2\sin^2 2A \text{ or}$$

$$2\cos^2 2A - 1$$

$$\text{and } \cos 6A = \cos^2 3A - \sin^2 3A \text{ or}$$

$$1 - 2\sin^2 3A \text{ or}$$

$$2\cos^2 3A - 1,$$

and so on.

- (iii) If, in the compound angle formula for $\tan(A+B)$, we let $B=A$ then

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Also, for example,

$$\tan 4A = \frac{2 \tan 2A}{1 - \tan^2 2A}$$

$$\text{and } \tan 5A = \frac{2 \tan \frac{5}{2}A}{1 - \tan^2 \frac{5}{2}A} \text{ and so on.}$$

Problem 11. $I_3 \sin 3\theta$ is the third harmonic of a waveform. Express the third harmonic in terms of the first harmonic $\sin \theta$, when $I_3 = 1$

When $I_3 = 1$,

$$I_3 \sin 3\theta = \sin 3\theta = \sin(2\theta + \theta)$$

$$= \sin 2\theta \cos \theta + \cos 2\theta \sin \theta,$$

from the $\sin(A+B)$ formula

$$= (2 \sin \theta \cos \theta) \cos \theta + (1 - 2 \sin^2 \theta) \sin \theta,$$

from the double angle expansions

$$= 2 \sin \theta \cos^2 \theta + \sin \theta - 2 \sin^3 \theta$$

$$= 2 \sin \theta (1 - \sin^2 \theta) + \sin \theta - 2 \sin^3 \theta,$$

(since $\cos^2 \theta = 1 - \sin^2 \theta$)

$$= 2 \sin \theta - 2 \sin^3 \theta + \sin \theta - 2 \sin^3 \theta$$

$$\text{i.e. } \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

Problem 12. Prove that $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$

$$\begin{aligned} \text{LHS} &= \frac{1 - \cos 2\theta}{\sin 2\theta} = \frac{1 - (1 - 2 \sin^2 \theta)}{2 \sin \theta \cos \theta} \\ &= \frac{2 \sin^2 \theta}{2 \sin \theta \cos \theta} = \frac{\sin \theta}{\cos \theta} \\ &= \tan \theta = \text{RHS} \end{aligned}$$

Problem 13. Prove that

$$\cot 2x + \operatorname{cosec} 2x = \cot x$$

$$\begin{aligned} \text{LHS} &= \cot 2x + \operatorname{cosec} 2x = \frac{\cos 2x}{\sin 2x} + \frac{1}{\sin 2x} \\ &= \frac{\cos 2x + 1}{\sin 2x} \\ &= \frac{(2 \cos^2 x - 1) + 1}{\sin 2x} \\ &= \frac{2 \cos^2 x}{\sin 2x} = \frac{2 \cos^2 x}{2 \sin x \cos x} \\ &= \frac{\cos x}{\sin x} = \cot x = \text{RHS} \end{aligned}$$

Problem 14. Solve the equation

$$\cos 2\theta + 3 \sin \theta = 2 \text{ for } \theta \text{ in the range } 0^\circ \leq \theta \leq 360^\circ$$

Replacing the double angle term with the relationship $\cos 2\theta = 1 - 2 \sin^2 \theta$ gives:

$$1 - 2 \sin^2 \theta + 3 \sin \theta = 2$$

$$\text{Rearranging gives: } -2 \sin^2 \theta + 3 \sin \theta - 1 = 0$$

$$\text{or } 2 \sin^2 \theta - 3 \sin \theta + 1 = 0$$

which is a quadratic in $\sin \theta$

Using the quadratic formula or by factorising gives:

$$(2 \sin \theta - 1)(\sin \theta - 1) = 0$$

$$\text{from which, } 2 \sin \theta - 1 = 0 \text{ or } \sin \theta - 1 = 0$$

$$\text{and } \sin \theta = \frac{1}{2} \text{ or } \sin \theta = 1$$

$$\text{from which, } \theta = 30^\circ \text{ or } 150^\circ \text{ or } 90^\circ$$

Now try the following Practice Exercise

Practice Exercise 85 Double angles (Answers on page 841)

1. The power p in an electrical circuit is given by

$$p = \frac{v^2}{R}. \text{ Determine the power in terms of } V, R \text{ and } \cos 2t \text{ when } v = V \cos t.$$

2. Prove the following identities:

$$(a) \quad 1 - \frac{\cos 2\phi}{\cos^2 \phi} = \tan^2 \phi$$

$$(b) \quad \frac{1 + \cos 2t}{\sin^2 t} = 2 \cot^2 t$$

$$(c) \quad \frac{(\tan 2x)(1 + \tan x)}{\tan x} = \frac{2}{1 - \tan x}$$

$$(d) \quad 2 \operatorname{cosec} 2\theta \cos 2\theta = \cot \theta - \tan \theta$$

3. If the third harmonic of a waveform is given by $V_3 \cos 3\theta$, express the third harmonic in terms of the first harmonic $\cos \theta$, when $V_3 = 1$

In Problems 4 to 8, solve for θ in the range $-180^\circ \leq \theta \leq 180^\circ$

$$4. \quad \cos 2\theta = \sin \theta$$

$$5. \quad 3 \sin 2\theta + 2 \cos \theta = 0$$

$$6. \quad \sin 2\theta + \cos \theta = 0$$

$$7. \quad \cos 2\theta + 2 \sin \theta = -3$$

$$8. \quad \tan \theta + \cot \theta = 2$$

19.4 Changing products of sines and cosines into sums or differences

(i) $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$ (from the formulae in Section 19.1)

$$\begin{aligned} \text{i.e. } \sin A \cos B \\ = \frac{1}{2} [\sin(A + B) + \sin(A - B)] \end{aligned} \quad (1)$$

(ii) $\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$

$$\begin{aligned} \text{i.e. } \cos A \sin B \\ = \frac{1}{2} [\sin(A + B) - \sin(A - B)] \end{aligned} \quad (2)$$

(iii) $\cos(A + B) + \cos(A - B) = 2 \cos A \cos B$

$$\begin{aligned} \text{i.e. } \cos A \cos B \\ = \frac{1}{2} [\cos(A + B) + \cos(A - B)] \end{aligned} \quad (3)$$

(iv) $\cos(A + B) - \cos(A - B) = -2 \sin A \sin B$

$$\begin{aligned} \text{i.e. } \sin A \sin B \\ = -\frac{1}{2} [\cos(A + B) - \cos(A - B)] \end{aligned} \quad (4)$$

Problem 15. Express $\sin 4x \cos 3x$ as a sum or difference of sines and cosines.

From equation (1),

$$\begin{aligned} \sin 4x \cos 3x &= \frac{1}{2} [\sin(4x + 3x) + \sin(4x - 3x)] \\ &= \frac{1}{2} (\sin 7x + \sin x) \end{aligned}$$

Problem 16. Express $2 \cos 5\theta \sin 2\theta$ as a sum or difference of sines or cosines.

From equation (2),

$$\begin{aligned} 2 \cos 5\theta \sin 2\theta &= 2 \left\{ \frac{1}{2} [\sin(5\theta + 2\theta) - \sin(5\theta - 2\theta)] \right\} \\ &= \sin 7\theta - \sin 3\theta \end{aligned}$$

Problem 17. Express $3 \cos 4t \cos t$ as a sum or difference of sines or cosines.

From equation (3),

$$\begin{aligned} 3 \cos 4t \cos t &= 3 \left\{ \frac{1}{2} [\cos(4t + t) + \cos(4t - t)] \right\} \\ &= \frac{3}{2} (\cos 5t + \cos 3t) \end{aligned}$$

Thus, if the integral $\int 3 \cos 4t \cos t \, dt$ was required (for integration see Chapter 39), then

$$\begin{aligned} \int 3 \cos 4t \cos t \, dt &= \int \frac{3}{2} (\cos 5t + \cos 3t) \, dt \\ &= \frac{3}{2} \left[\frac{\sin 5t}{5} + \frac{\sin 3t}{3} \right] + c \end{aligned}$$

Problem 18. In an alternating current circuit, voltage $v = 5 \sin \omega t$ and current $i = 10 \sin(\omega t - \pi/6)$. Find an expression for the instantaneous power p at time t given that $p = vi$, expressing the answer as a sum or difference of sines and cosines.

$$\begin{aligned} p = vi &= (5 \sin \omega t) [10 \sin(\omega t - \pi/6)] \\ &= 50 \sin \omega t \sin(\omega t - \pi/6) \end{aligned}$$

From equation (4),

$$\begin{aligned} & 50 \sin \omega t \sin(\omega t - \pi/6) \\ &= (50) \left[-\frac{1}{2} \left\{ \cos(\omega t + \omega t - \pi/6) \right. \right. \\ &\quad \left. \left. - \cos[\omega t - (\omega t - \pi/6)] \right\} \right] \\ &= -25 \{ \cos(2\omega t - \pi/6) - \cos \pi/6 \} \end{aligned}$$

i.e. instantaneous power,

$$p = 25[\cos \pi/6 - \cos(2\omega t - \pi/6)]$$

Now try the following Practice Exercise

Practice Exercise 86 Changing products of sines and cosines into sums or differences (Answers on page 841)

In Problems 1 to 5, express as sums or differences:

1. $\sin 7t \cos 2t$
2. $\cos 8x \sin 2x$
3. $2 \sin 7t \sin 3t$
4. $4 \cos 3\theta \cos \theta$
5. $3 \sin \frac{\pi}{3} \cos \frac{\pi}{6}$
6. Determine $\int 2 \sin 3t \cos t \, dt$
7. Evaluate $\int_0^{\frac{\pi}{2}} 4 \cos 5x \cos 2x \, dx$
8. Solve the equation: $2 \sin 2\phi \sin \phi = \cos \phi$ in the range $\phi = 0$ to $\phi = 180^\circ$

19.5 Changing sums or differences of sines and cosines into products

In the compound angle formula let,

$$(A + B) = X$$

and

$$(A - B) = Y$$

Solving the simultaneous equations gives:

$$A = \frac{X + Y}{2} \text{ and } B = \frac{X - Y}{2}$$

Thus $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$ becomes,

$$\sin X + \sin Y = 2 \sin \left(\frac{X + Y}{2} \right) \cos \left(\frac{X - Y}{2} \right) \quad (5)$$

Similarly,

$$\sin X - \sin Y = 2 \cos \left(\frac{X + Y}{2} \right) \sin \left(\frac{X - Y}{2} \right) \quad (6)$$

$$\cos X + \cos Y = 2 \cos \left(\frac{X + Y}{2} \right) \cos \left(\frac{X - Y}{2} \right) \quad (7)$$

$$\cos X - \cos Y = -2 \sin \left(\frac{X + Y}{2} \right) \sin \left(\frac{X - Y}{2} \right) \quad (8)$$

Problem 19. Express $\sin 5\theta + \sin 3\theta$ as a product.

From equation (5),

$$\begin{aligned} \sin 5\theta + \sin 3\theta &= 2 \sin \left(\frac{5\theta + 3\theta}{2} \right) \cos \left(\frac{5\theta - 3\theta}{2} \right) \\ &= 2 \sin 4\theta \cos \theta \end{aligned}$$

Problem 20. Express $\sin 7x - \sin x$ as a product.

From equation (6),

$$\begin{aligned} \sin 7x - \sin x &= 2 \cos \left(\frac{7x + x}{2} \right) \sin \left(\frac{7x - x}{2} \right) \\ &= 2 \cos 4x \sin 3x \end{aligned}$$

Problem 21. Express $\cos 2t - \cos 5t$ as a product.

From equation (8),

$$\begin{aligned} \cos 2t - \cos 5t &= -2 \sin \left(\frac{2t + 5t}{2} \right) \sin \left(\frac{2t - 5t}{2} \right) \\ &= -2 \sin \frac{7}{2}t \sin \left(-\frac{3}{2}t \right) = 2 \sin \frac{7}{2}t \sin \frac{3}{2}t \\ &\quad \left(\text{since } \sin \left(-\frac{3}{2}t \right) = -\sin \frac{3}{2}t \right) \end{aligned}$$

Problem 22. Show that $\frac{\cos 6x + \cos 2x}{\sin 6x + \sin 2x} = \cot 4x$

From equation (7),

$$\cos 6x + \cos 2x = 2 \cos 4x \cos 2x$$

From equation (5),

$$\sin 6x + \sin 2x = 2 \sin 4x \cos 2x$$

Hence

$$\begin{aligned} \frac{\cos 6x + \cos 2x}{\sin 6x + \sin 2x} &= \frac{2 \cos 4x \cos 2x}{2 \sin 4x \cos 2x} \\ &= \frac{\cos 4x}{\sin 4x} = \cot 4x \end{aligned}$$

Problem 23. Solve the equation $\cos 4\theta + \cos 2\theta = 0$ for θ in the range $0^\circ \leq \theta \leq 360^\circ$

From equation (7),

$$\cos 4\theta + \cos 2\theta = 2 \cos \left(\frac{4\theta + 2\theta}{2} \right) \cos \left(\frac{4\theta - 2\theta}{2} \right)$$

$$\text{Hence,} \quad 2 \cos 3\theta \cos \theta = 0$$

$$\text{Dividing by 2 gives:} \quad \cos 3\theta \cos \theta = 0$$

$$\text{Hence, either} \quad \cos 3\theta = 0 \quad \text{or} \quad \cos \theta = 0$$

$$\text{Thus,} \quad 3\theta = \cos^{-1} 0 \quad \text{or} \quad \theta = \cos^{-1} 0$$

from which, $3\theta = 90^\circ$ or 270° or 450° or 630° or 810° or 990°

and $\theta = 30^\circ, 90^\circ, 150^\circ, 210^\circ, 270^\circ$ or 330°

Now try the following Practice Exercise

Practice Exercise 87 Changing sums or differences of sines and cosines into products (Answers on page 841)

In Problems 1 to 5, express as products:

- $\sin 3x + \sin x$
- $\frac{1}{2}(\sin 9\theta - \sin 7\theta)$
- $\cos 5t + \cos 3t$
- $\frac{1}{8}(\cos 5t - \cos t)$
- $\frac{1}{2} \left(\cos \frac{\pi}{3} + \cos \frac{\pi}{4} \right)$
- Show that:

$$(a) \quad \frac{\sin 4x - \sin 2x}{\cos 4x + \cos 2x} = \tan x$$

$$(b) \quad \frac{1}{2} \{ \sin(5x - \alpha) - \sin(x + \alpha) \} \\ = \cos 3x \sin(2x - \alpha)$$

In Problems 7 and 8, solve for θ in the range $0^\circ \leq \theta \leq 180^\circ$

$$7. \quad \cos 6\theta + \cos 2\theta = 0$$

$$8. \quad \sin 3\theta - \sin \theta = 0$$

In Problems 9 and 10, solve in the range 0° to 360°

$$9. \quad \cos 2x = 2 \sin x$$

$$10. \quad \sin 4t + \sin 2t = 0$$

19.6 Power waveforms in a.c. circuits

(a) Purely resistive a.c. circuits

Let a voltage $v = V_m \sin \omega t$ be applied to a circuit comprising resistance only. The resulting current is $i = I_m \sin \omega t$, and the corresponding instantaneous power, p , is given by:

$$p = vi = (V_m \sin \omega t)(I_m \sin \omega t)$$

$$\text{i.e.} \quad p = V_m I_m \sin^2 \omega t$$

From double angle formulae of Section 19.3,

$$\cos 2A = 1 - 2 \sin^2 A, \text{ from which,}$$

$$\sin^2 A = \frac{1}{2}(1 - \cos 2A) \text{ thus}$$

$$\sin^2 \omega t = \frac{1}{2}(1 - \cos 2\omega t)$$

$$\text{Then power} \quad p = V_m I_m \left[\frac{1}{2}(1 - \cos 2\omega t) \right]$$

$$\text{i.e.} \quad p = \frac{1}{2} V_m I_m (1 - \cos 2\omega t)$$

The waveforms of v , i and p are shown in Fig. 19.8. The waveform of power repeats itself after π/ω seconds and hence the power has a frequency twice that of voltage and current. The power is always positive, having a maximum value of $V_m I_m$. The average or mean value of the power is $\frac{1}{2} V_m I_m$

The rms value of voltage $V = 0.707 V_m$, i.e. $V = \frac{V_m}{\sqrt{2}}$, from which, $V_m = \sqrt{2} V$

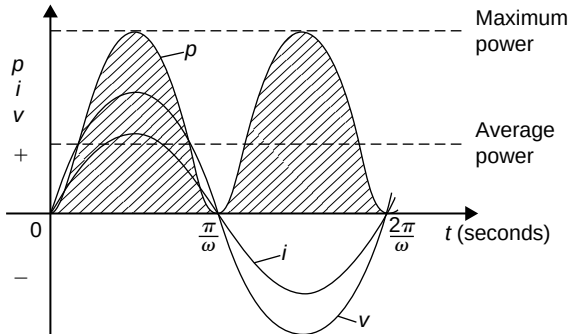


Figure 19.8

Similarly, the rms value of current, $I = \frac{I_m}{\sqrt{2}}$, from which, $I_m = \sqrt{2}I$. Hence the average power, P , developed in a purely resistive a.c. circuit is given by $P = \frac{1}{2}V_m I_m = \frac{1}{2}(\sqrt{2}V)(\sqrt{2}I) = VI$ watts. Also, power $P = I^2 R$ or V^2/R as for a d.c. circuit, since $V = IR$.

Summarising, the average power P in a purely resistive a.c. circuit given by

$$P = VI = I^2 R = \frac{V^2}{R}$$

where V and I are rms values.

(b) Purely inductive a.c. circuits

Let a voltage $v = V_m \sin \omega t$ be applied to a circuit containing pure inductance (theoretical case). The resulting

current is $i = I_m \sin\left(\omega t - \frac{\pi}{2}\right)$ since current lags voltage by $\frac{\pi}{2}$ radians or 90° in a purely inductive circuit, and the corresponding instantaneous power, p , is given by:

$$p = vi = (V_m \sin \omega t) I_m \sin\left(\omega t - \frac{\pi}{2}\right)$$

$$\text{i.e. } p = V_m I_m \sin \omega t \sin\left(\omega t - \frac{\pi}{2}\right)$$

However,

$$\sin\left(\omega t - \frac{\pi}{2}\right) = -\cos \omega t \text{ thus}$$

$$p = -V_m I_m \sin \omega t \cos \omega t$$

Rearranging gives:

$$p = -\frac{1}{2} V_m I_m (2 \sin \omega t \cos \omega t)$$

However, from double angle formulae,

$$2 \sin \omega t \cos \omega t = \sin 2\omega t.$$

Thus **power**, $p = -\frac{1}{2} V_m I_m \sin 2\omega t$

The waveforms of v , i and p are shown in Fig. 19.9. The frequency of power is twice that of voltage and current. For the power curve shown in Fig. 19.9, the area above the horizontal axis is equal to the area below, thus over a complete cycle the average power P is zero. It is noted that when v and i are both positive, power p is positive and energy is delivered from the source to the inductance; when v and i have opposite signs, power p is negative and energy is returned from the inductance to the source.

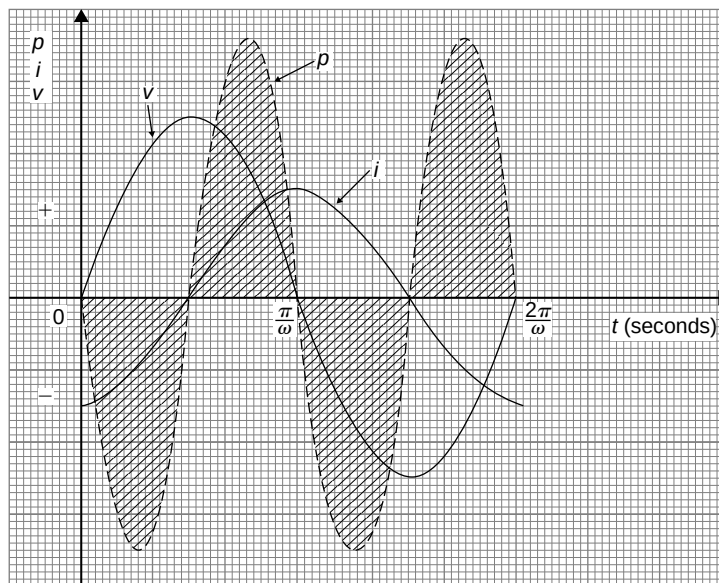


Figure 19.9

In general, when the current through an inductance is increasing, energy is transferred from the circuit to the magnetic field, but this energy is returned when the current is decreasing.

Summarising, **the average power P in a purely inductive a.c. circuit is zero.**

(c) Purely capacitive a.c. circuits

Let a voltage $v = V_m \sin \omega t$ be applied to a circuit containing pure capacitance. The resulting current is $i = I_m \sin(\omega t + \frac{\pi}{2})$, since current leads voltage by 90° in a purely capacitive circuit, and the corresponding instantaneous power, p , is given by:

$$p = vi = (V_m \sin \omega t) I_m \sin(\omega t + \frac{\pi}{2})$$

$$\text{i.e. } p = V_m I_m \sin \omega t \sin(\omega t + \frac{\pi}{2})$$

$$\text{However, } \sin(\omega t + \frac{\pi}{2}) = \cos \omega t$$

$$\text{thus } p = V_m I_m \sin \omega t \cos \omega t$$

Rearranging gives

$$p = \frac{1}{2} V_m I_m (2 \sin \omega t \cos \omega t)$$

$$\text{Thus power, } p = \frac{1}{2} V_m I_m \sin 2\omega t.$$

The waveforms of v , i and p are shown in Fig. 19.10. Over a complete cycle the average power P is zero. When the voltage across a capacitor is increasing, energy is transferred from the circuit to the electric

field, but this energy is returned when the voltage is decreasing.

Summarising, **the average power P in a purely capacitive a.c. circuit is zero.**

(d) R-L or R-C a.c. circuits

Let a voltage $v = V_m \sin \omega t$ be applied to a circuit containing resistance and inductance or resistance and capacitance. Let the resulting current be $i = I_m \sin(\omega t + \phi)$, where phase angle ϕ will be positive for an R-C circuit and negative for an R-L circuit. The corresponding instantaneous power, p , is given by:

$$p = vi = (V_m \sin \omega t) I_m \sin(\omega t + \phi)$$

$$\text{i.e. } p = V_m I_m \sin \omega t \sin(\omega t + \phi)$$

Products of sine functions may be changed into differences of cosine functions as shown in Section 19.4,

$$\text{i.e. } \sin A \sin B = -\frac{1}{2} [\cos(A+B) - \cos(A-B)]$$

Substituting $\omega t = A$ and $(\omega t + \phi) = B$ gives:

$$\text{power, } p = V_m I_m \left\{ -\frac{1}{2} [\cos(\omega t + \omega t + \phi) - \cos(\omega t - (\omega t + \phi))] \right\}$$

$$\text{i.e. } p = \frac{1}{2} V_m I_m [\cos(-\phi) - \cos(2\omega t + \phi)]$$

However, $\cos(-\phi) = \cos \phi$

$$\text{Thus } p = \frac{1}{2} V_m I_m [\cos \phi - \cos(2\omega t + \phi)]$$

The instantaneous power p thus consists of

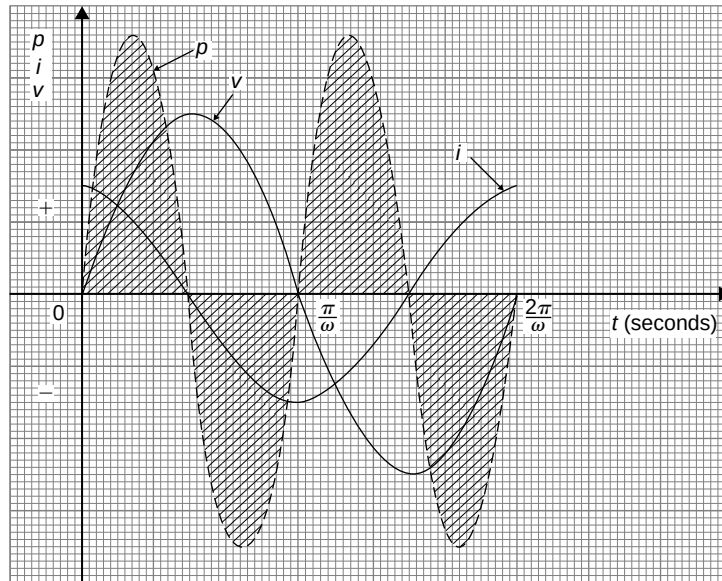


Figure 19.10

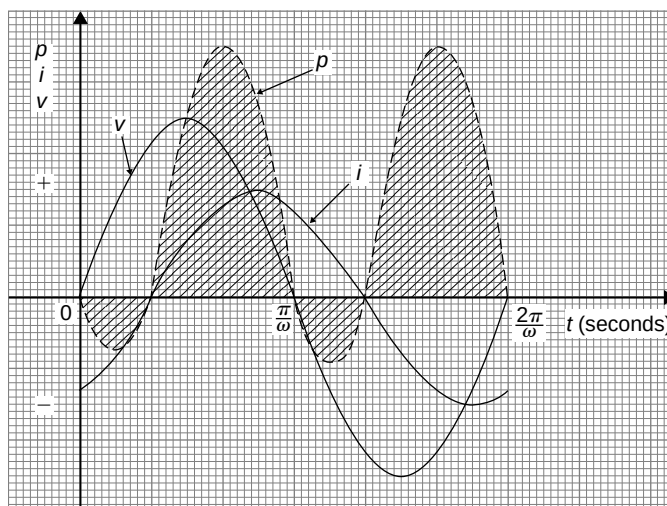


Figure 19.11

- (i) a sinusoidal term, $-\frac{1}{2}V_m I_m \cos(2\omega t + \phi)$ which has a mean value over a cycle of zero, and
- (ii) a constant term, $\frac{1}{2}V_m I_m \cos \phi$ (since ϕ is constant for a particular circuit).

Thus the average value of power $P = \frac{1}{2}V_m I_m \cos \phi$.

Since $V_m = \sqrt{2}V$ and $I_m = \sqrt{2}I$, average power

$$P = \frac{1}{2}(\sqrt{2}V)(\sqrt{2}I)\cos \phi$$

i.e. $P = VI \cos \phi$

The waveforms of v , i and p , are shown in Fig. 19.11 for an R - L circuit. The waveform of power is seen to pulsate at twice the supply frequency. The areas of the power curve (shown shaded) above the horizontal time axis represent power supplied to the load; the small

areas below the axis represent power being returned to the supply from the inductance as the magnetic field collapses.

A similar shape of power curve is obtained for an R - C circuit, the small areas below the horizontal axis representing power being returned to the supply from the charged capacitor. The difference between the areas above and below the horizontal axis represents the heat loss due to the circuit resistance. Since power is dissipated only in a pure resistance, the alternative equations for power, $P = I_R^2 R$, may be used, where I_R is the rms current flowing through the resistance.

Summarising, **the average power P in a circuit containing resistance and inductance and/or capacitance, whether in series or in parallel, is given by $P = VI \cos \phi$ or $P = I_R^2 R$** (V , I and I_R being rms values).

For fully worked solutions to each of the problems in Practice Exercises 83 to 87 in this chapter, go to the website:

www.routledge.com/cw/bird

Revision Test 5 Further trigonometry and hyperbolic functions

This Revision Test covers the material contained in Chapters 15 to 19. The marks for each question are shown in brackets at the end of each question.

1. Solve the following equations in the range 0° to 360° .
 - (a) $\sin^{-1}(-0.4161) = x$
 - (b) $\cot^{-1}(2.4198) = \theta$ (8)
2. Sketch the following curves, labelling relevant points:
 - (a) $y = 4\cos(\theta + 45^\circ)$
 - (b) $y = 5\sin(2t - 60^\circ)$ (8)
3. The current in an alternating current circuit at any time t seconds is given by:

$$i = 120\sin(100\pi t + 0.274) \text{ amperes.}$$

Determine

 - (a) the amplitude, periodic time, frequency and phase angle (with reference to $120\sin 100\pi t$)
 - (b) the value of current when $t = 0$
 - (c) the value of current when $t = 6\text{ ms}$
 - (d) the time when the current first reaches 80 A

Sketch one cycle of the oscillation. (19)
4. A complex voltage waveform v is comprised of a 141.4 V rms fundamental voltage at a frequency of 100 Hz , a 35% third harmonic component leading the fundamental voltage at zero time by $\pi/3$ radians, and a 20% fifth harmonic component lagging the fundamental at zero time by $\pi/4$ radians.
 - (a) Write down an expression to represent voltage v .
 - (b) Draw the complex voltage waveform using harmonic synthesis over one cycle of the fundamental waveform using scales of 12 cm for the time for one cycle horizontally and $1\text{ cm} = 20\text{ V}$ vertically. (15)
5. Evaluate correct to 4 significant figures:
 - (a) $\sinh 2.47$ (b) $\tanh 0.6439$
 - (c) $\operatorname{sech} 1.385$ (d) $\operatorname{cosech} 0.874$ (6)
6. The increase in resistance of strip conductors due to eddy currents at power frequencies is given by:

$$\lambda = \frac{\alpha t}{2} \left[\frac{\sinh \alpha t + \sin \alpha t}{\cosh \alpha t - \cos \alpha t} \right]$$

Calculate λ , correct to 5 significant figures, when $\alpha = 1.08$ and $t = 1$ (5)
7. If $A \operatorname{ch} x - B \operatorname{sh} x \equiv 4e^x - 3e^{-x}$ determine the values of A and B . (6)
8. Solve the following equation:

$$3.52 \operatorname{ch} x + 8.42 \operatorname{sh} x = 5.32$$

correct to 4 decimal places. (8)
9. Prove the following identities:
 - (a) $\sqrt{\frac{1 - \cos^2 \theta}{\cos^2 \theta}} = \tan \theta$
 - (b) $\cos\left(\frac{3\pi}{2} + \phi\right) = \sin \phi$
 - (c) $\frac{\sin^2 x}{1 + \cos 2x} = \frac{1}{2} \tan^2 x$ (9)
10. Solve the following trigonometric equations in the range $0^\circ \leq x \leq 360^\circ$:
 - (a) $4\cos x + 1 = 0$
 - (b) $3.25 \operatorname{cosec} x = 5.25$
 - (c) $5\sin^2 x + 3\sin x = 4$
 - (d) $2\sec^2 \theta + 5\tan \theta = 3$ (18)
11. Solve the equation $5\sin(\theta - \pi/6) = 8\cos \theta$ for values $0 \leq \theta \leq 2\pi$ (8)
12. Express $5.3\cos t - 7.2\sin t$ in the form $R\sin(t + \alpha)$. Hence solve the equation $5.3\cos t - 7.2\sin t = 4.5$ in the range $0 \leq t \leq 2\pi$ (12)
13. Determine $\int 2\cos 3t \sin t \, dt$ (3)

For lecturers/instructors/teachers, fully worked solutions to each of the problems in Revision Test 5, together with a full marking scheme, are available at the website:

www.routledge.com/cw/bird